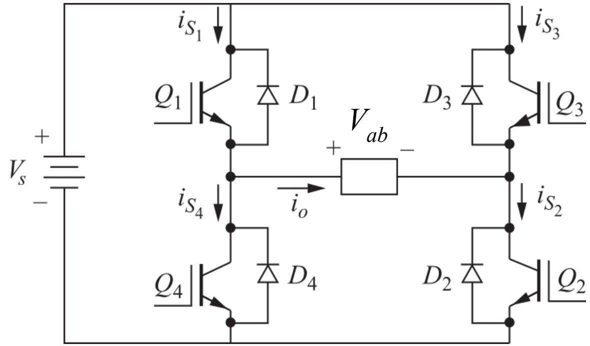
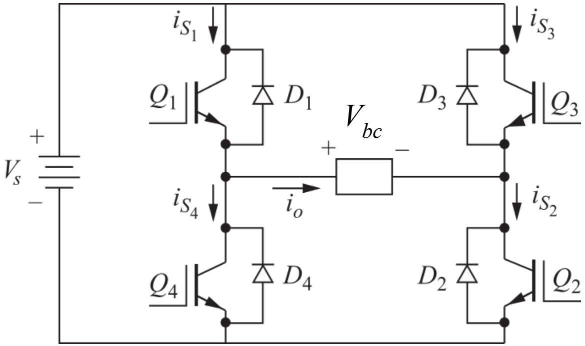
Conversión CC/CA

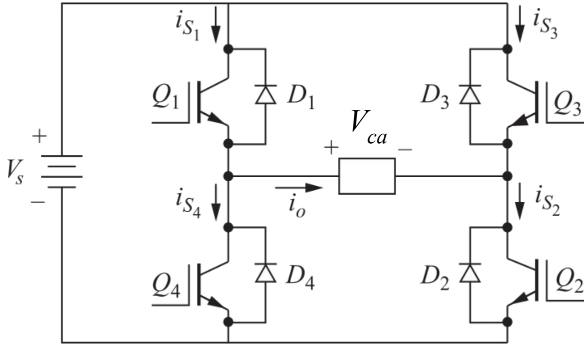
Inversores Puente Trifásicos:
Onda Cuadrada

Inversor trifásico: principio

Un inversor trifásico debe generar tres tensiones alternas, desfasadas 120° una respecto a la otra. ¿Podemos resolverlo con 3 inversores puente monofásicos?







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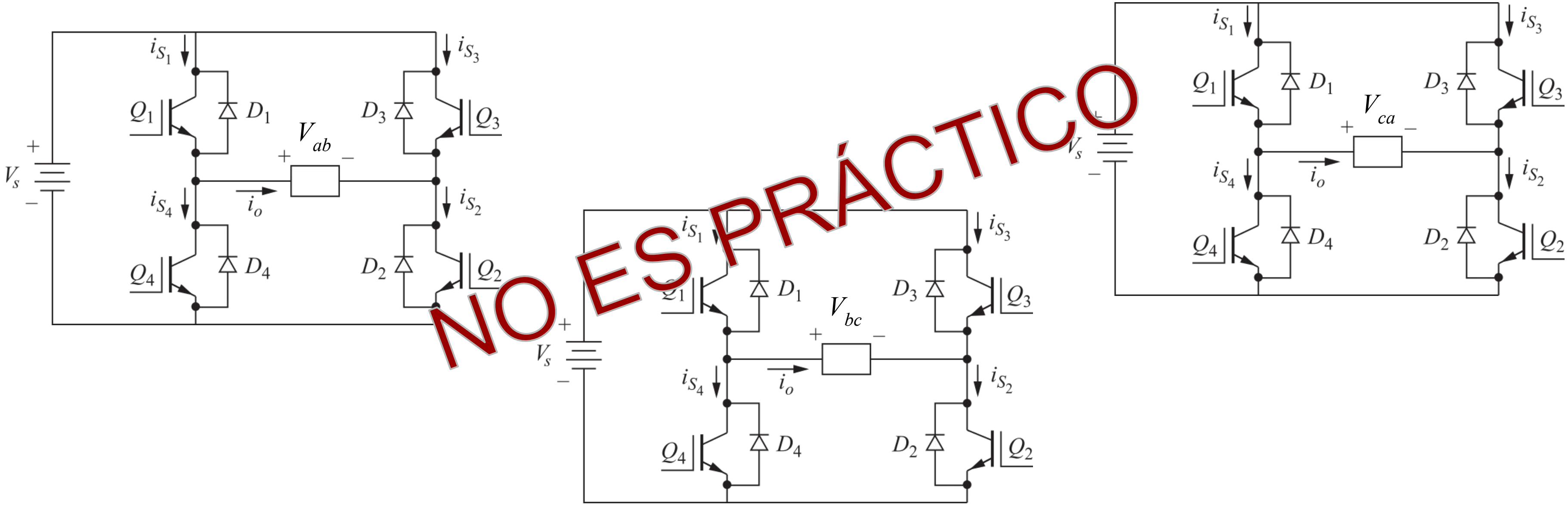
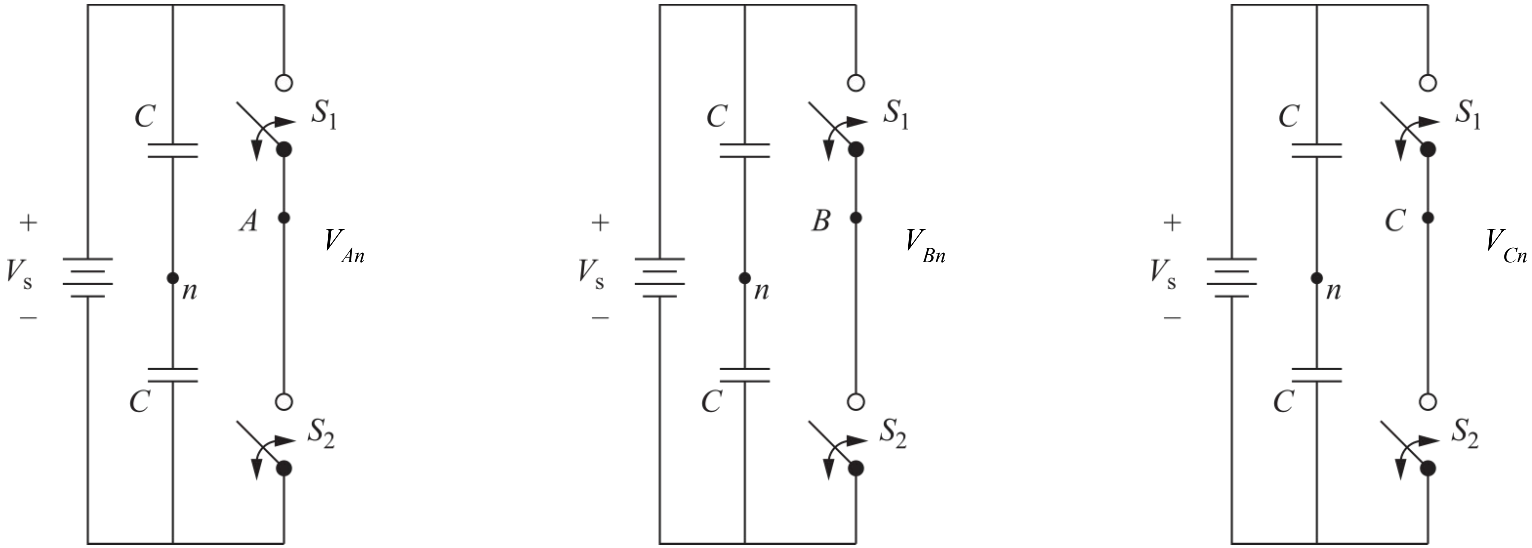


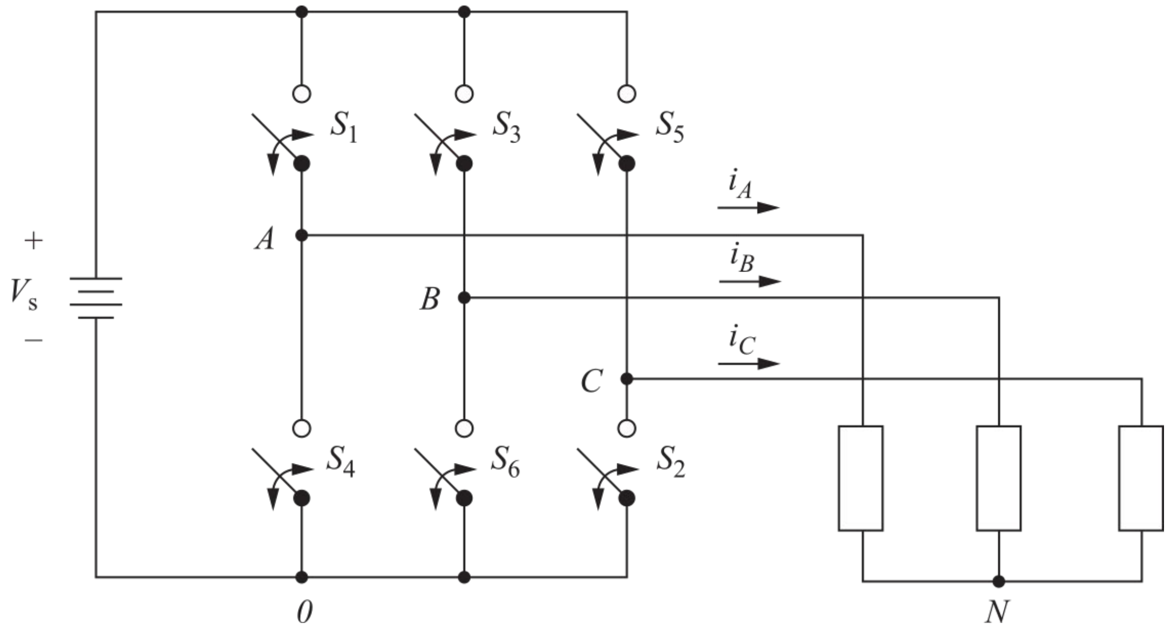
Inversor trifásico: principio

Si bien la combinación de tres inversores puente monofásicos no resulta práctico, si es viable conceptualmente utilizar tres semipuente.



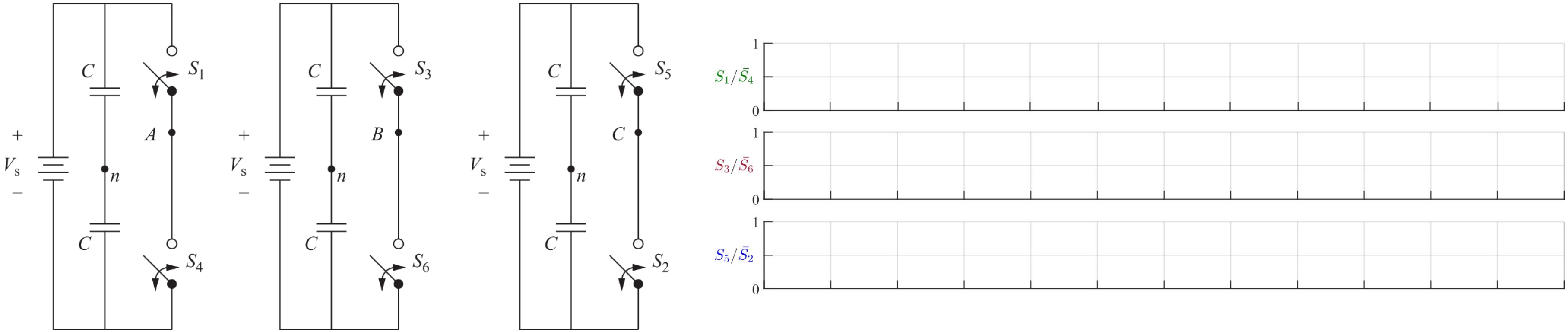
Inversor puente trifásico: esquema

La estructura puente para un inversor trifásico resulta de la siguiente forma. En este caso se muestra conectada una carga conectada en estrella.



Inversor puente trifásico: principio

Para su funcionamiento, debemos generar señales de disparo desfasadas 120° una respecto a la otra.



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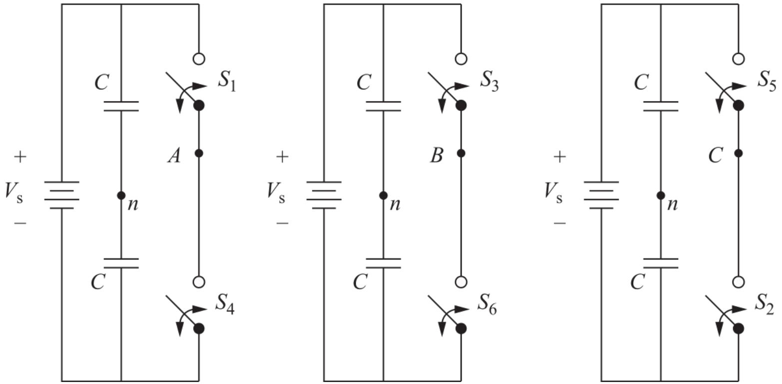

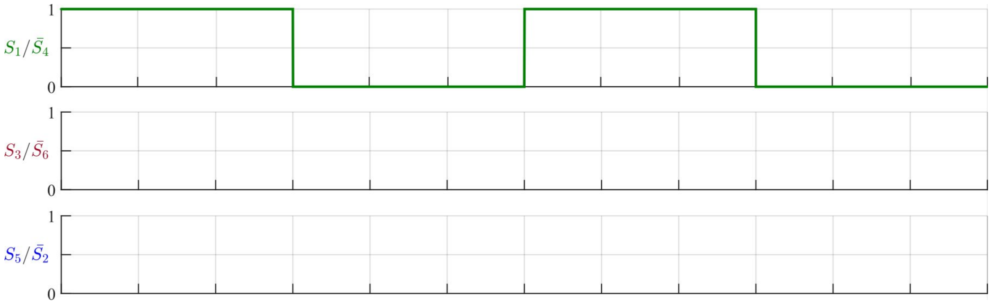

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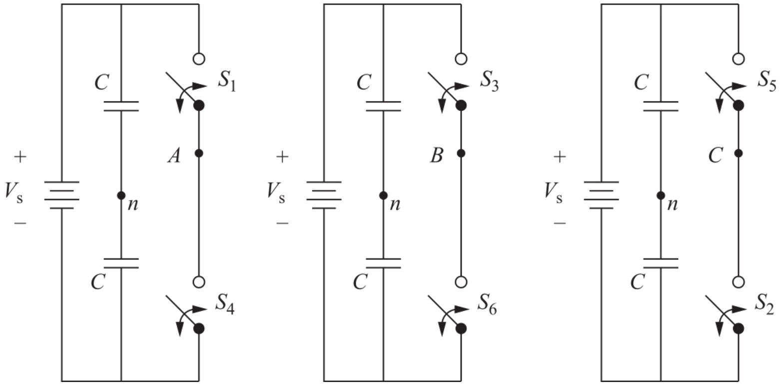

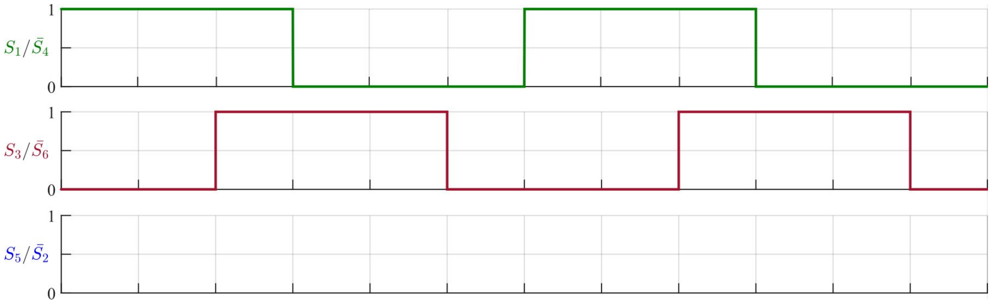

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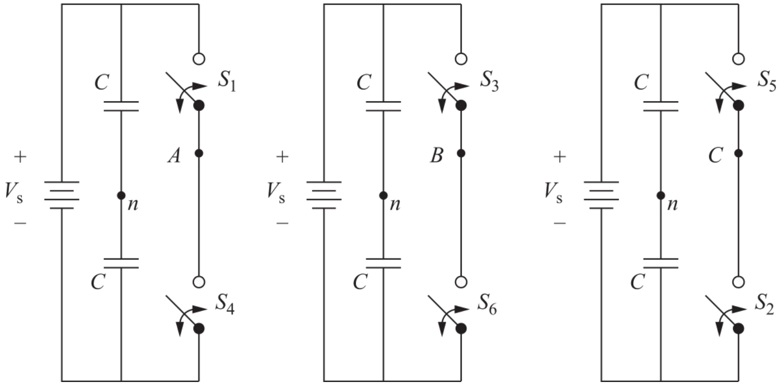

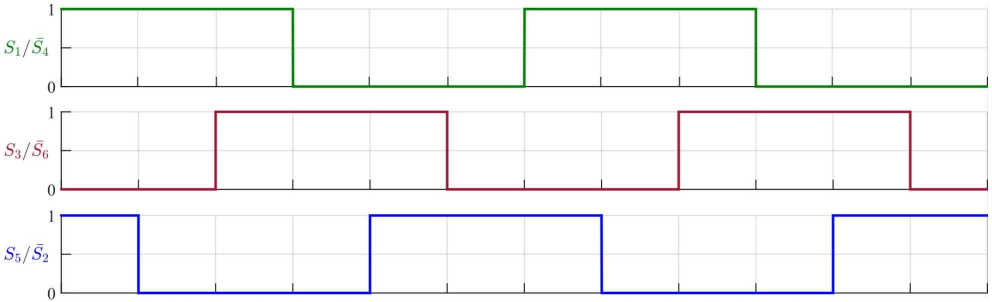

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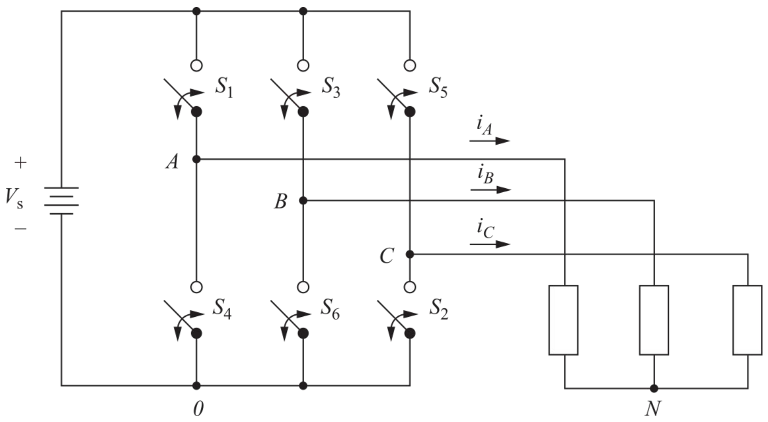
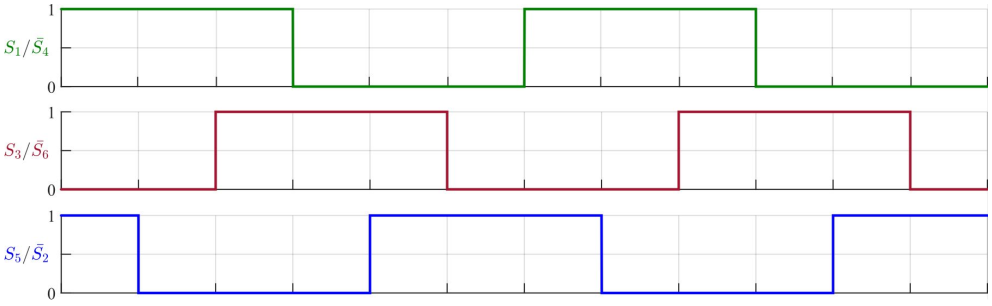
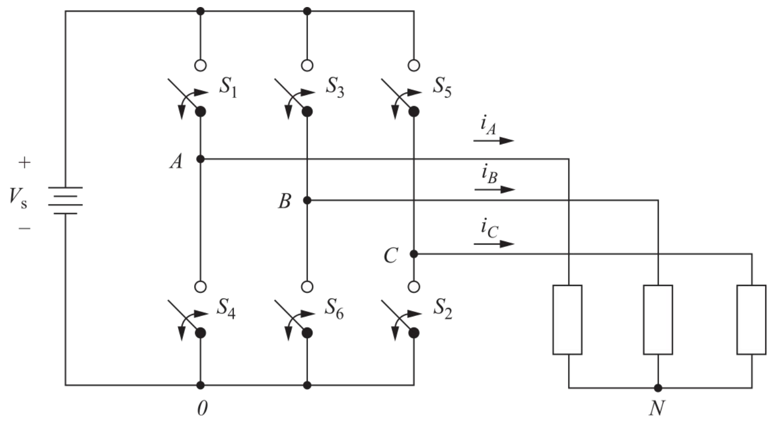
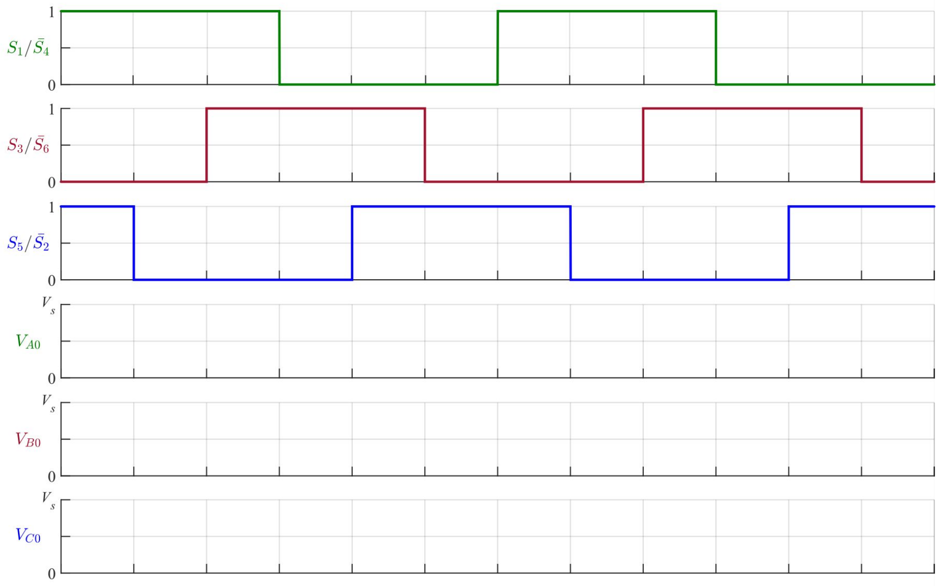




Inversor puente trifásico: formas de onda

Tensiones de columna





Inversor puente trifásico: formas de onda

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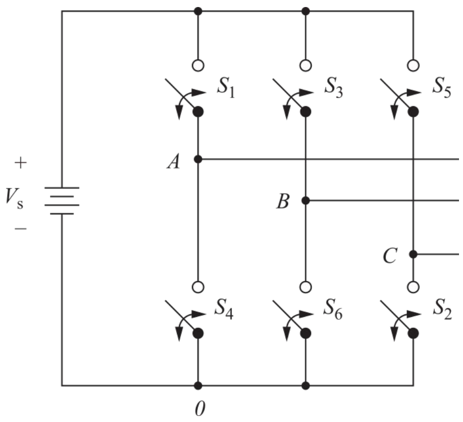

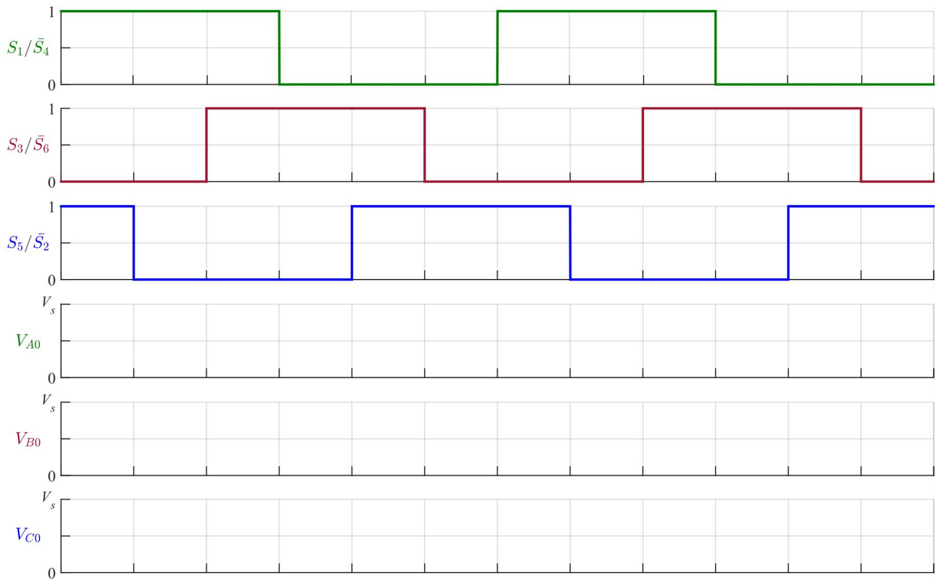

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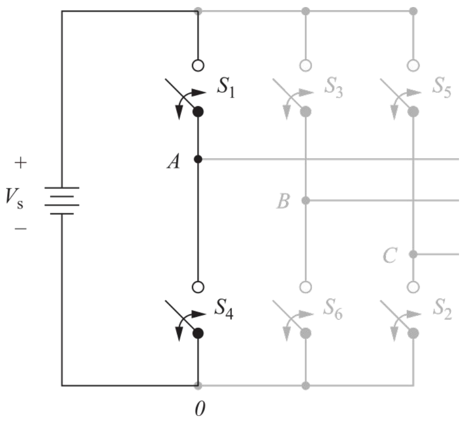

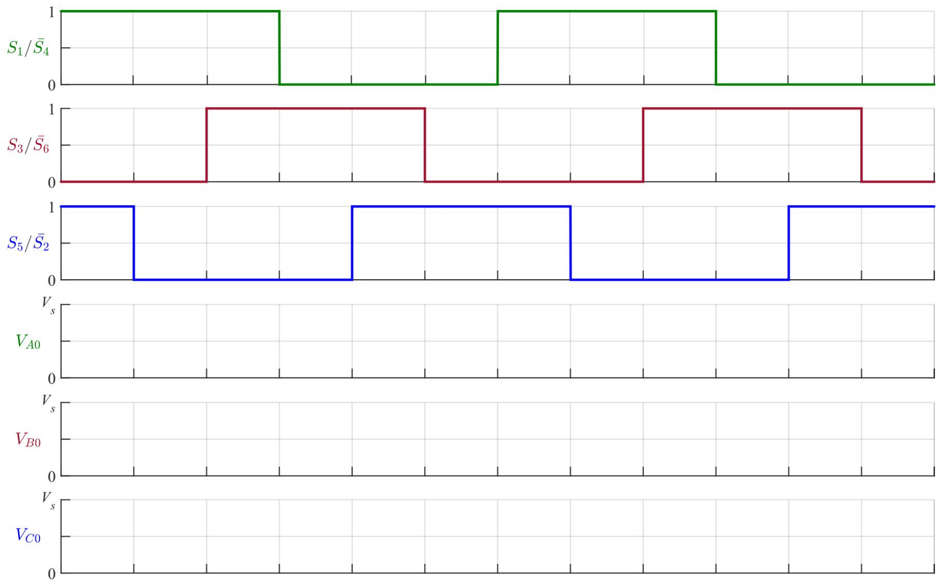

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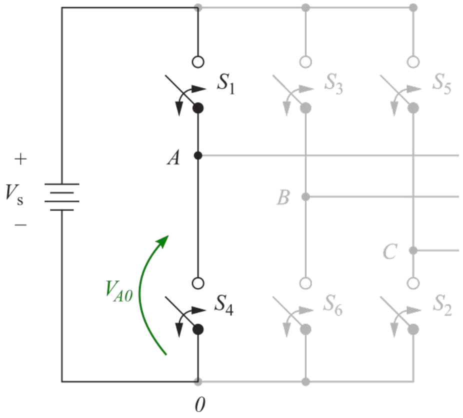

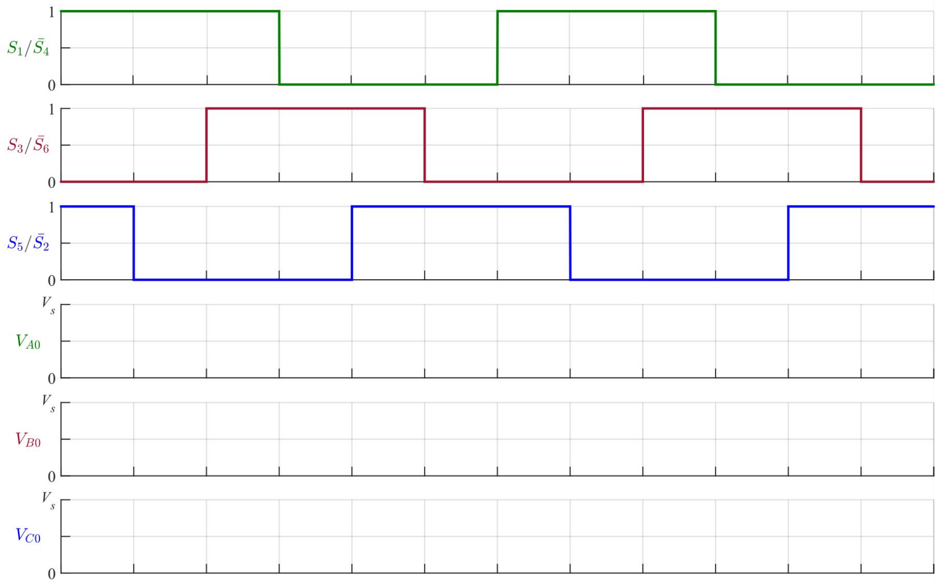

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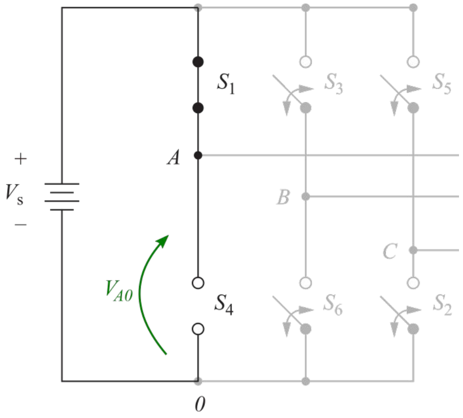

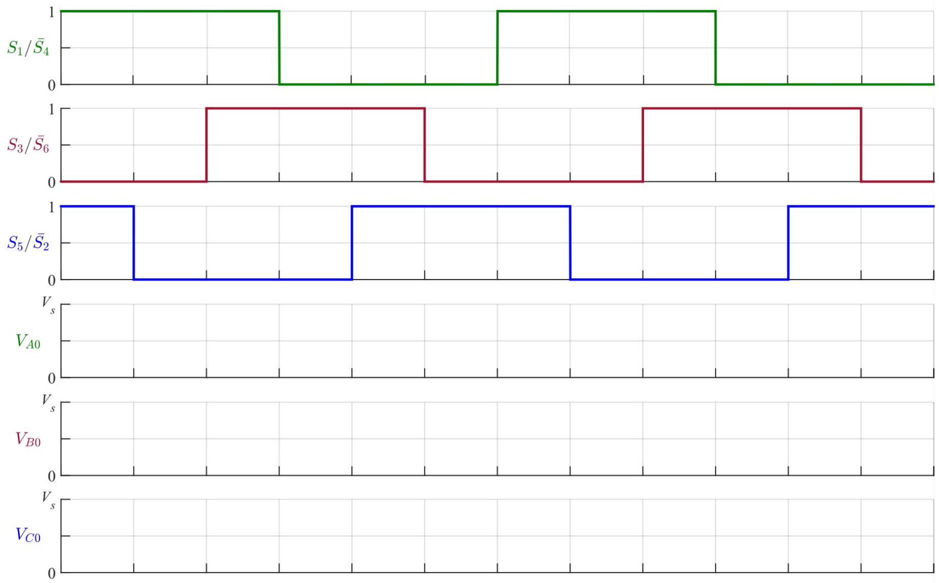

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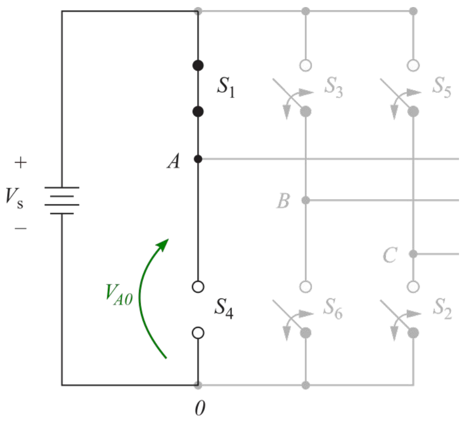

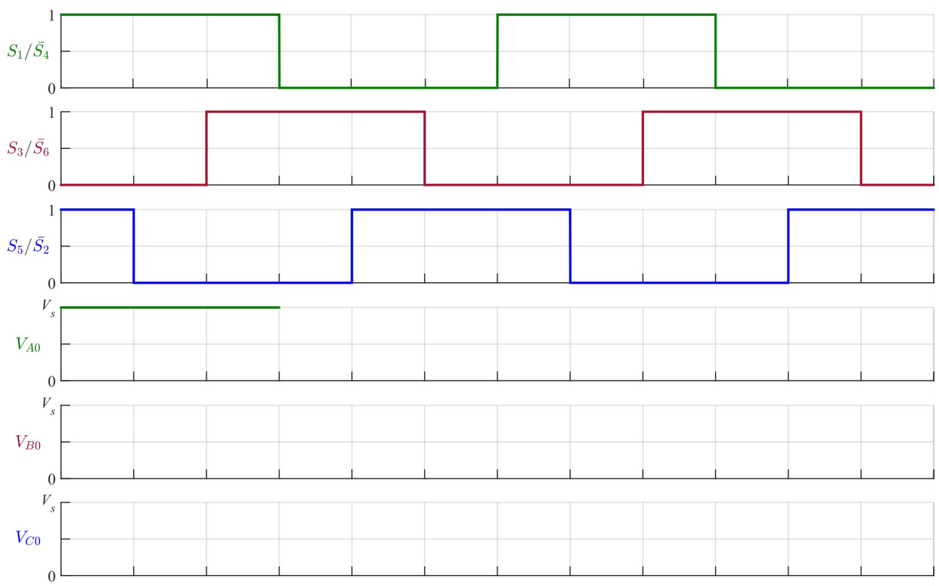

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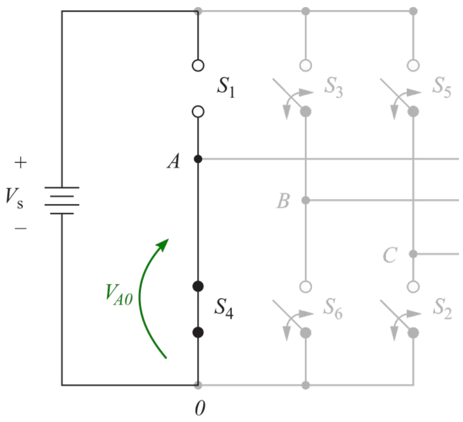

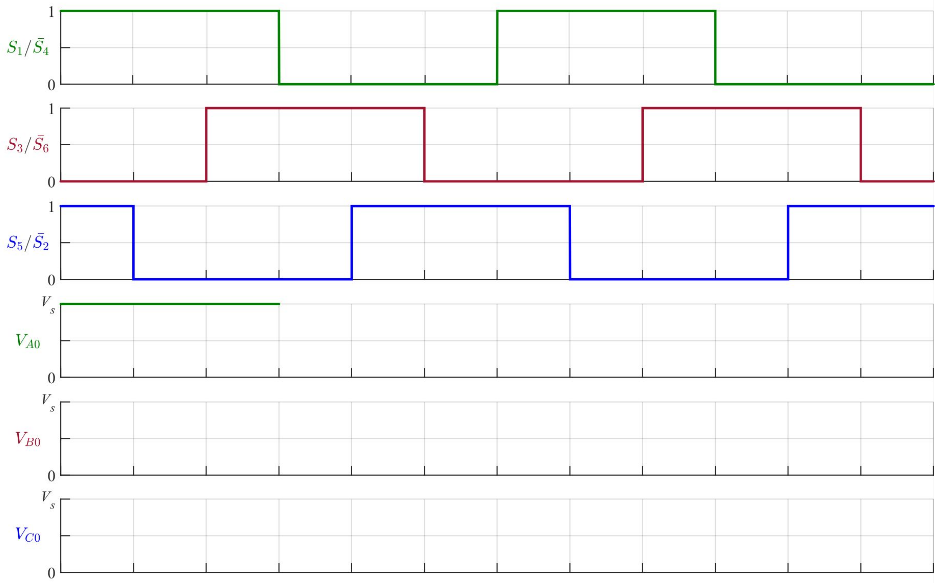

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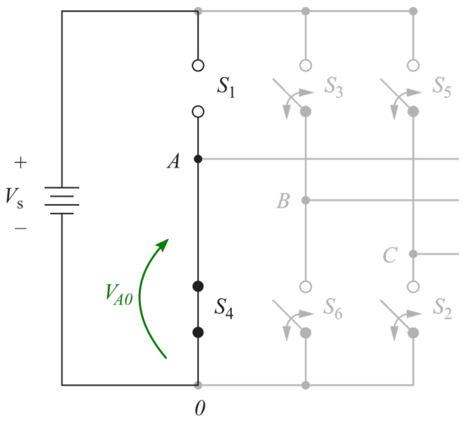

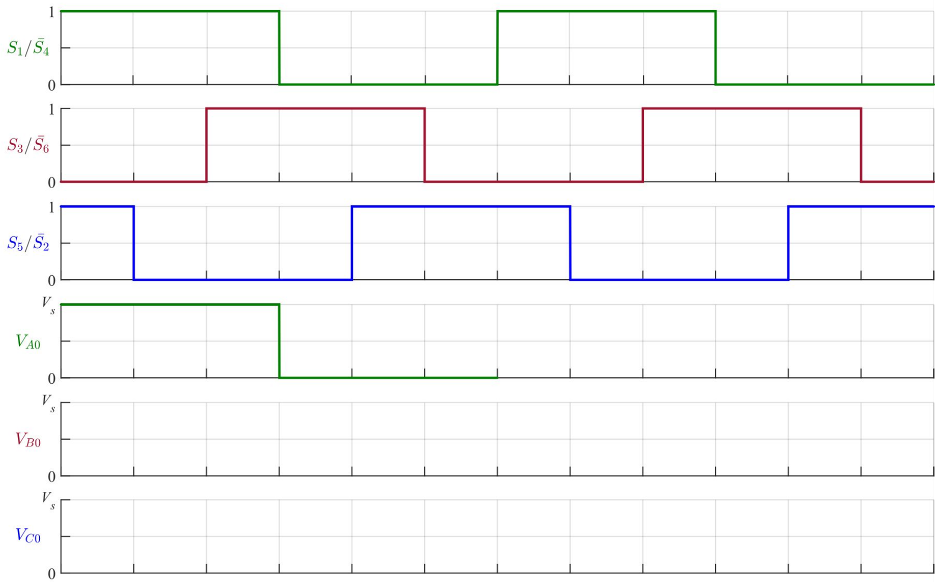

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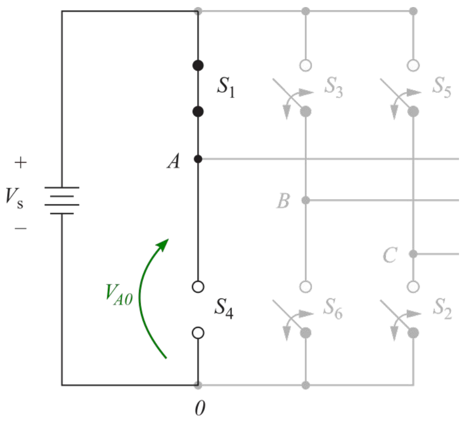

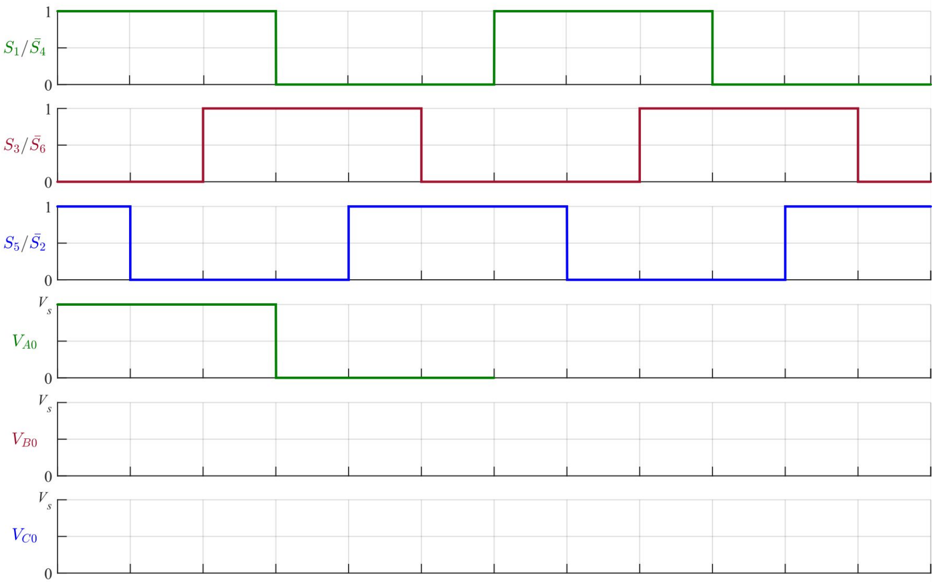

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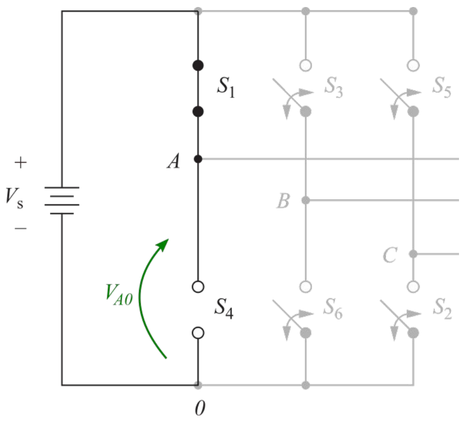

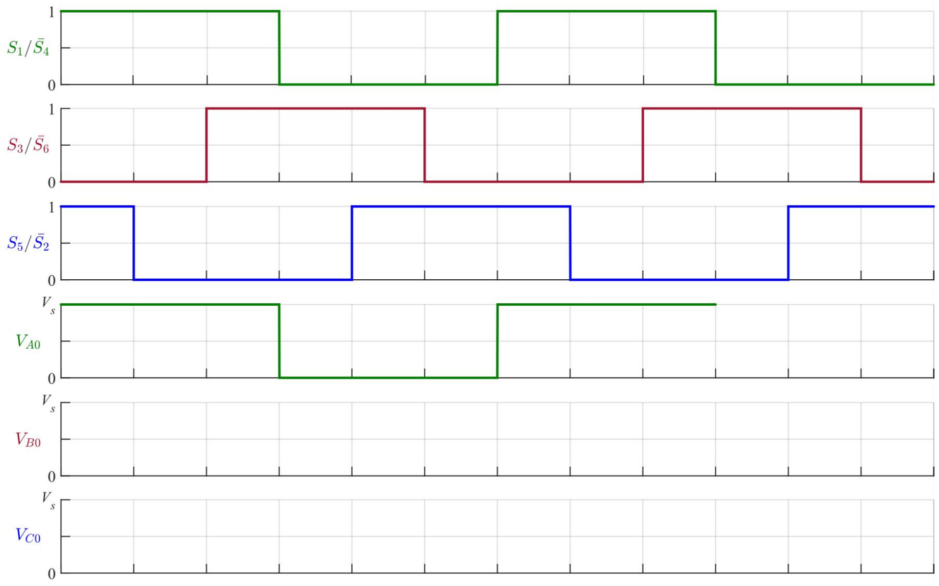

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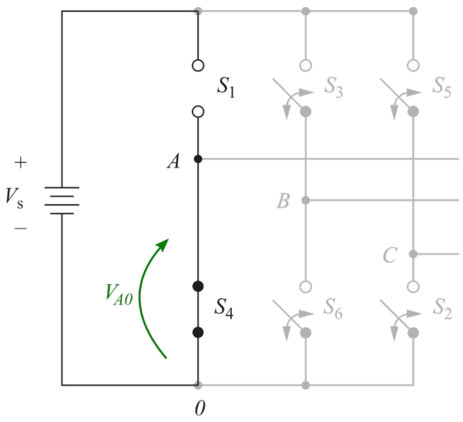

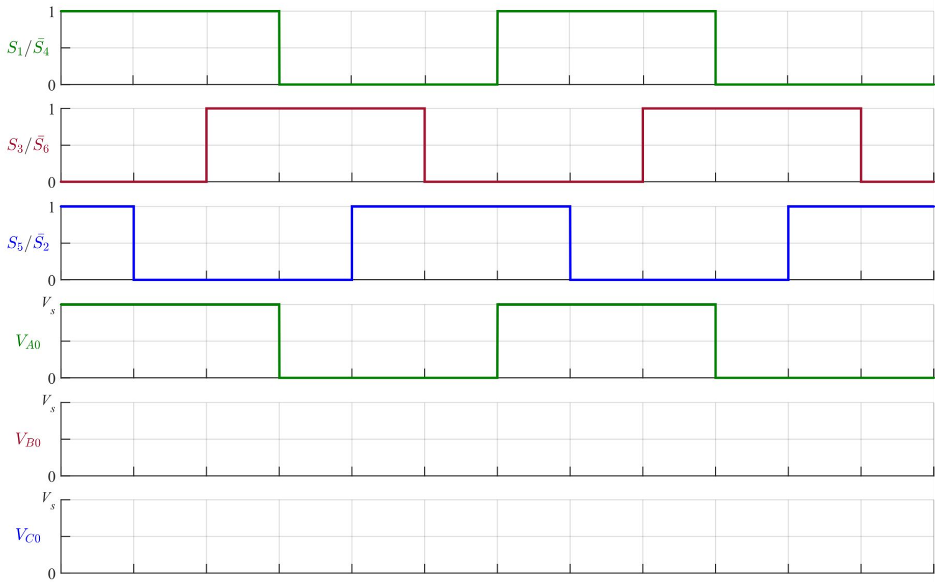

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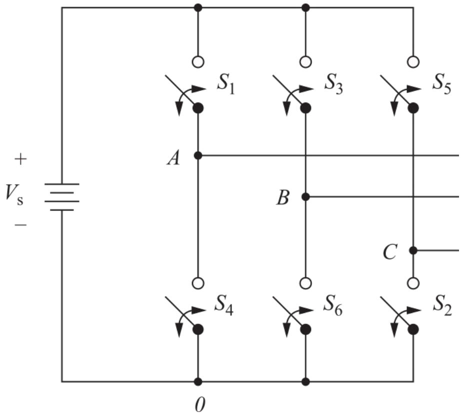

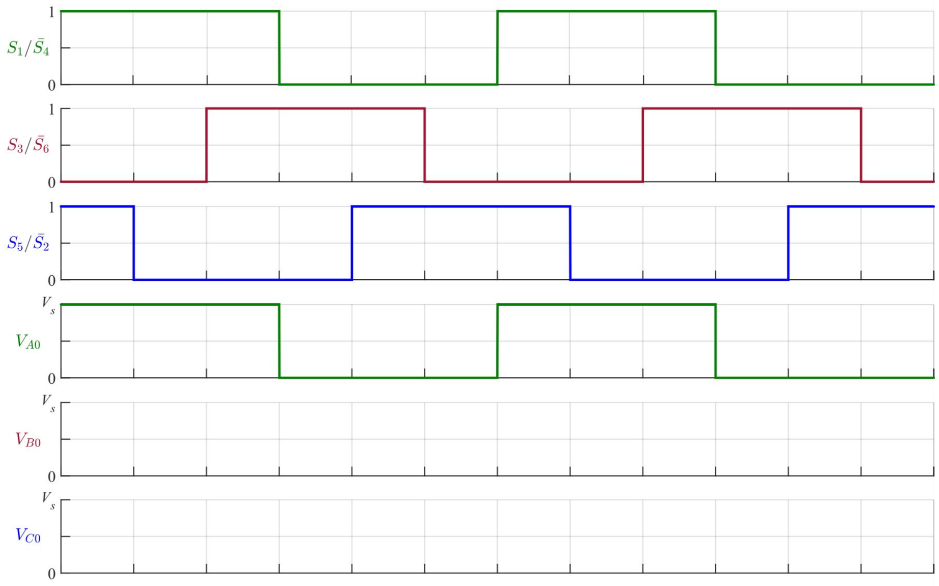

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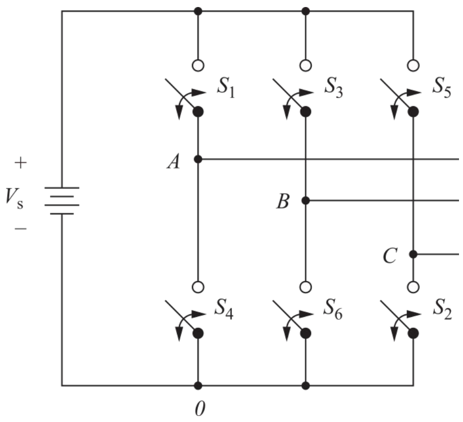

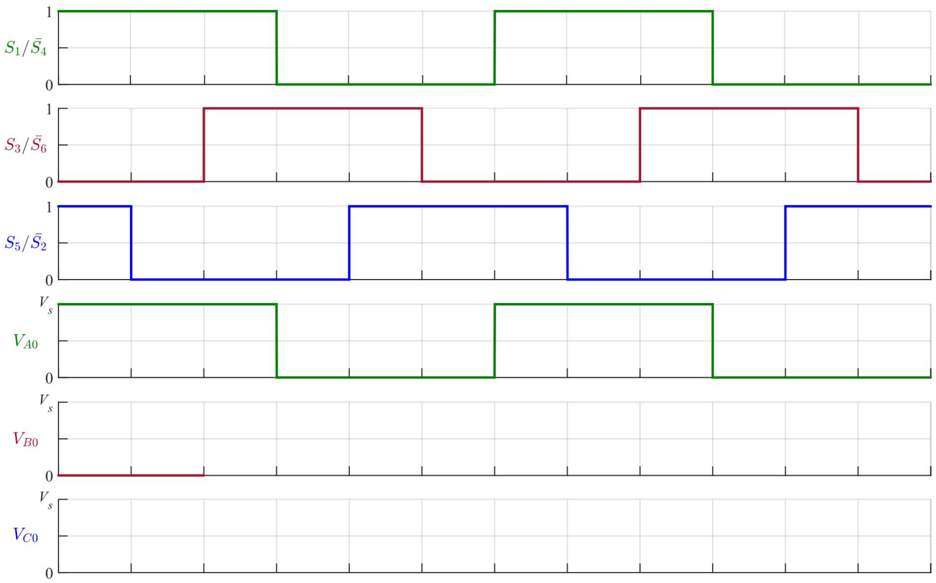

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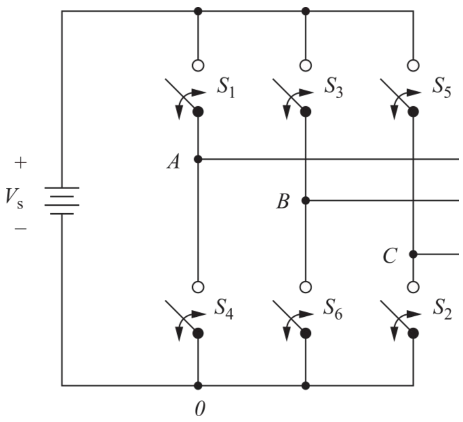

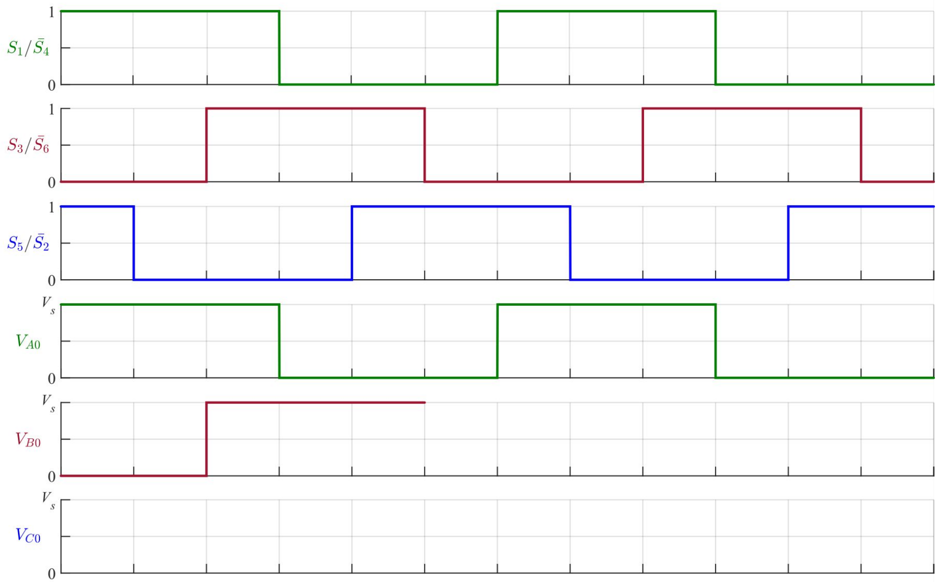

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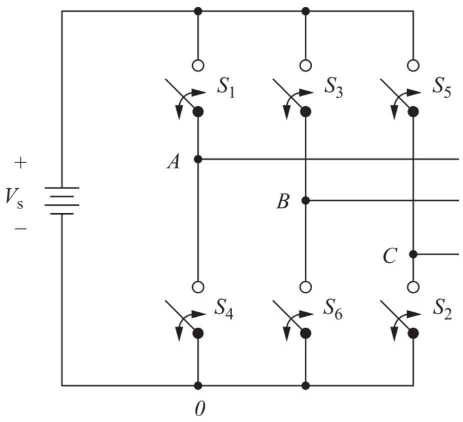

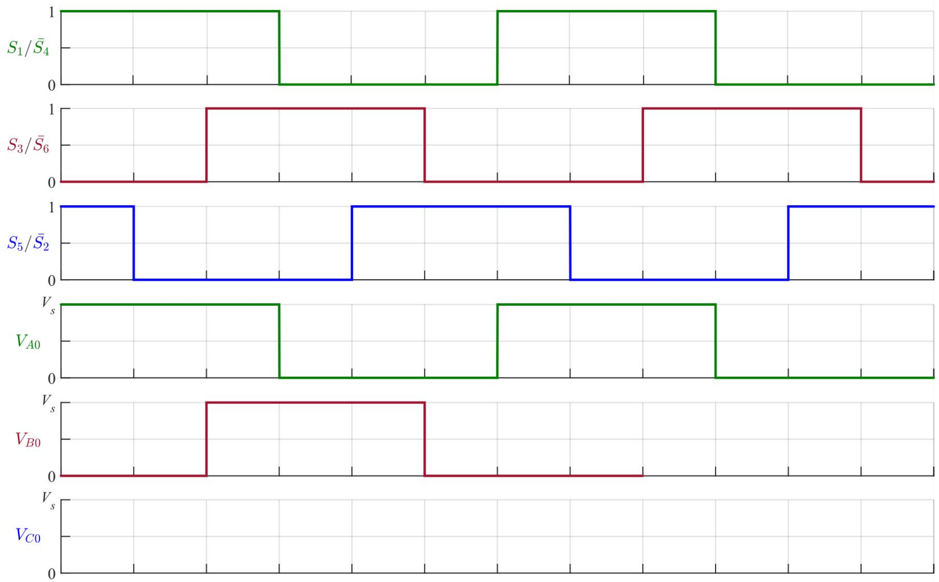

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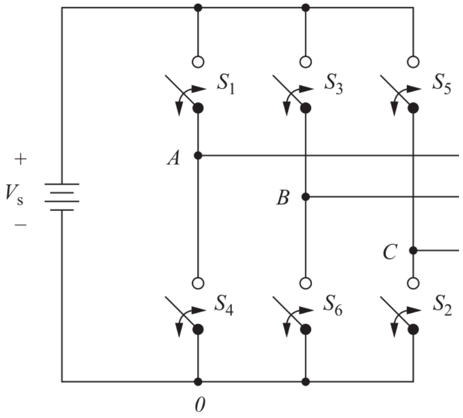

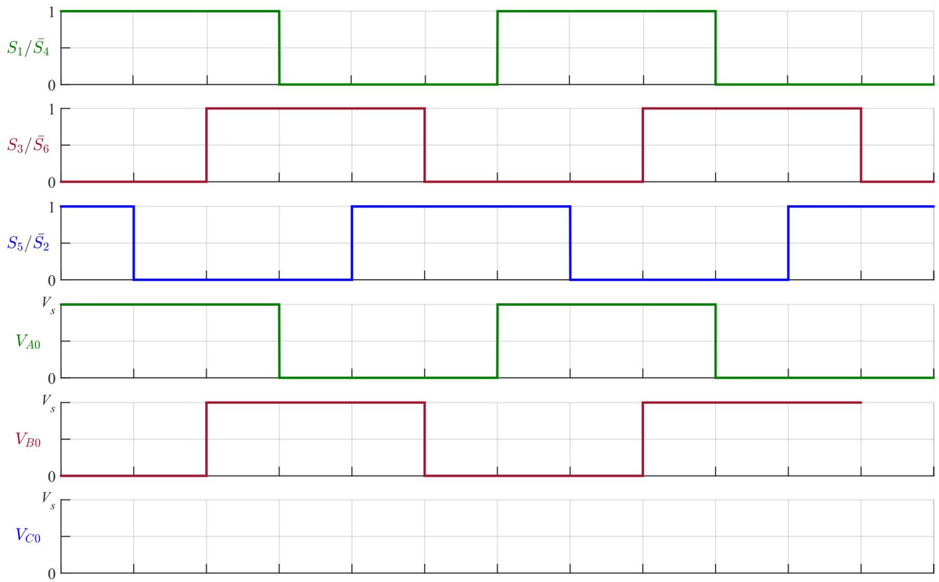

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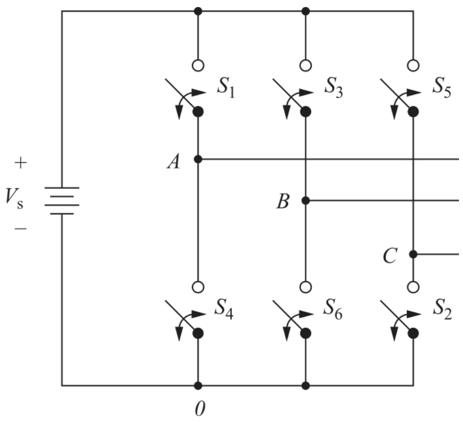

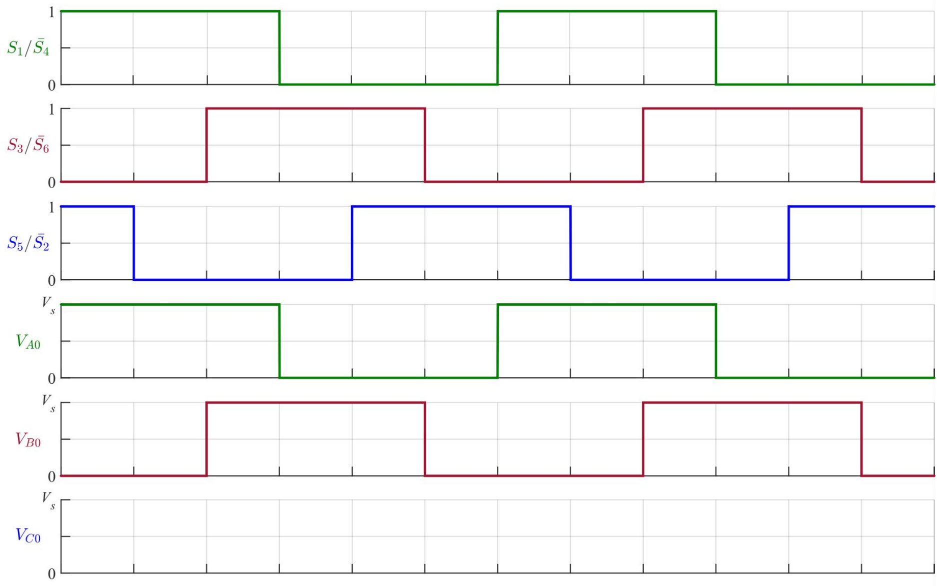

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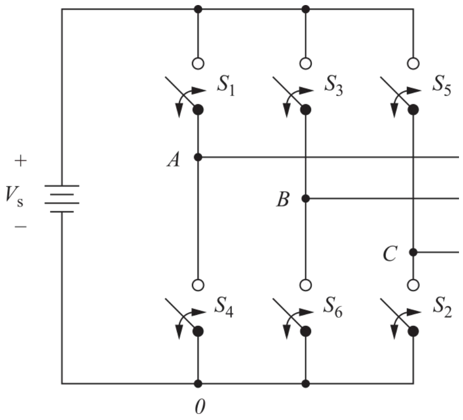

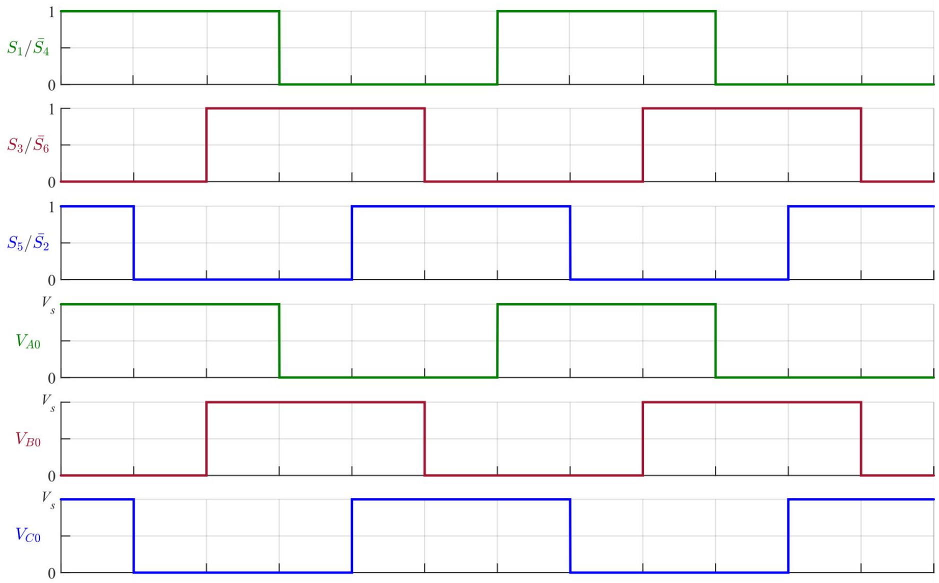

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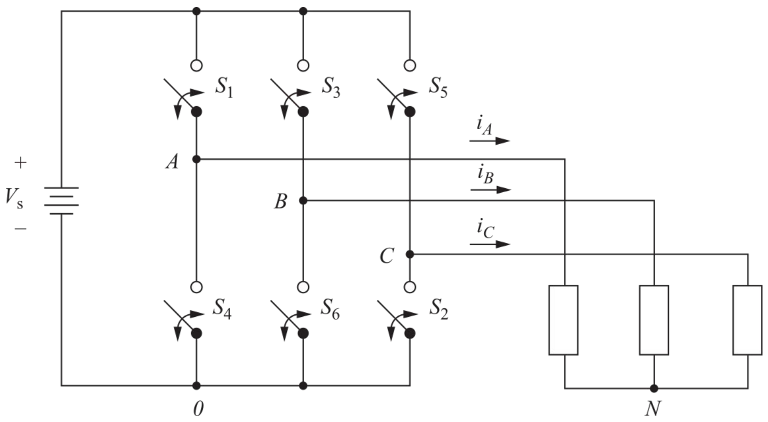
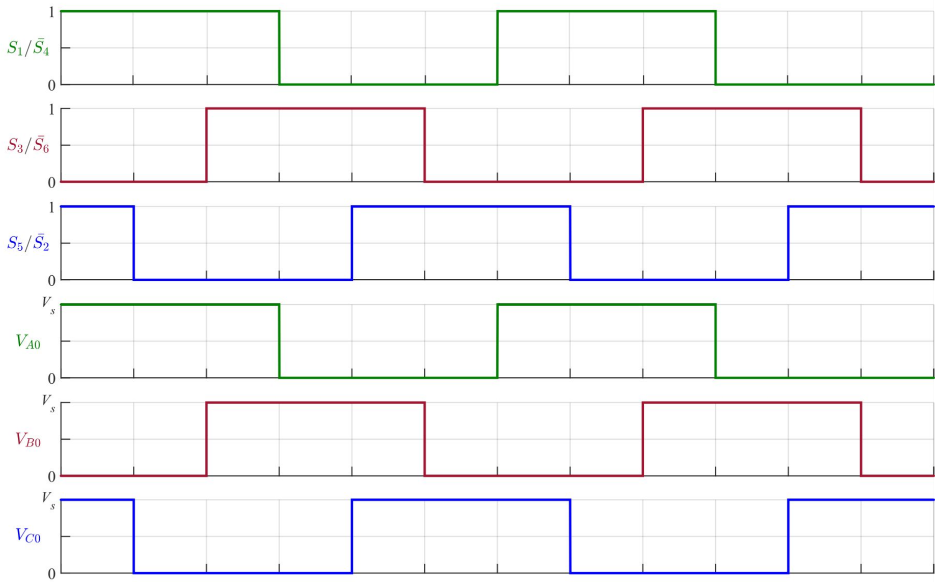
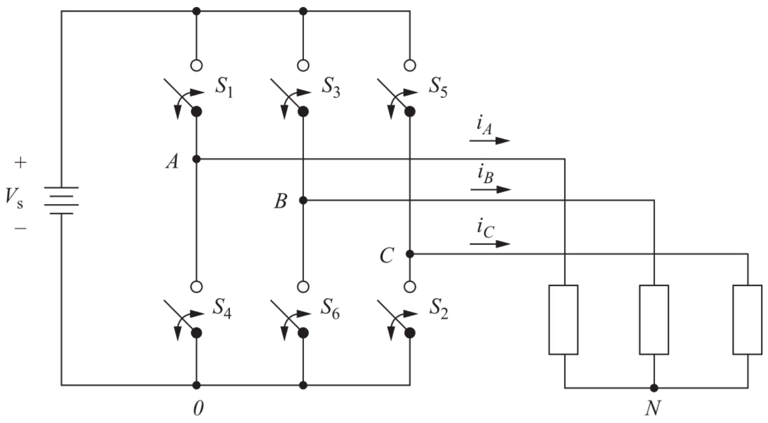
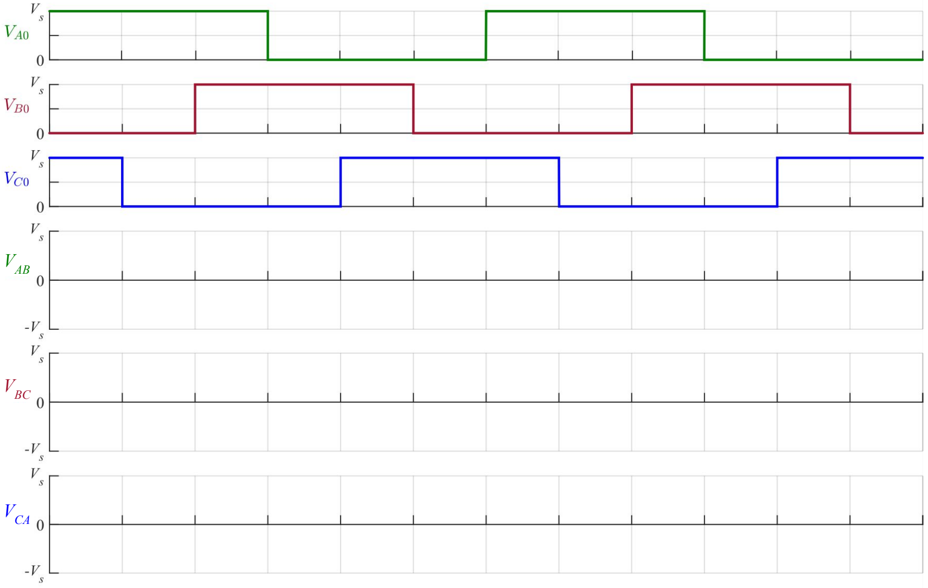




Inversor puente trifásico: formas de onda

Tensiones de línea

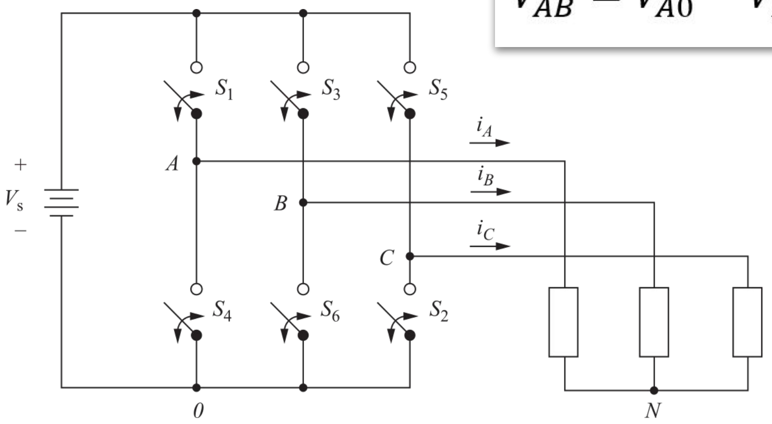


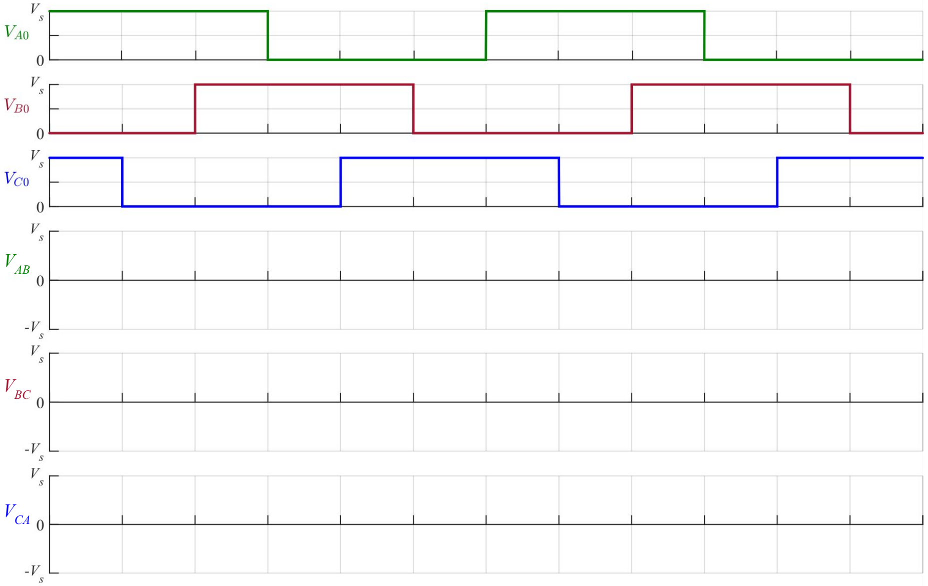


Inversor puente trifásico: formas de onda

Tensiones de línea

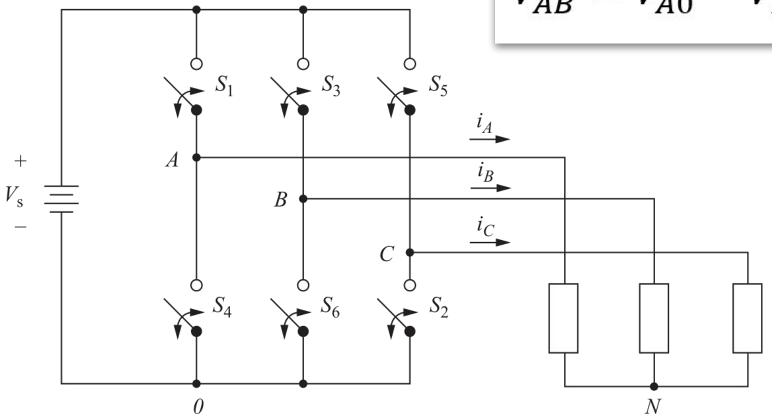
$$V_{AB} = V_{A0} - V_{B0}$$





Inversor puente trifásico: formas de onda

Tensiones de línea



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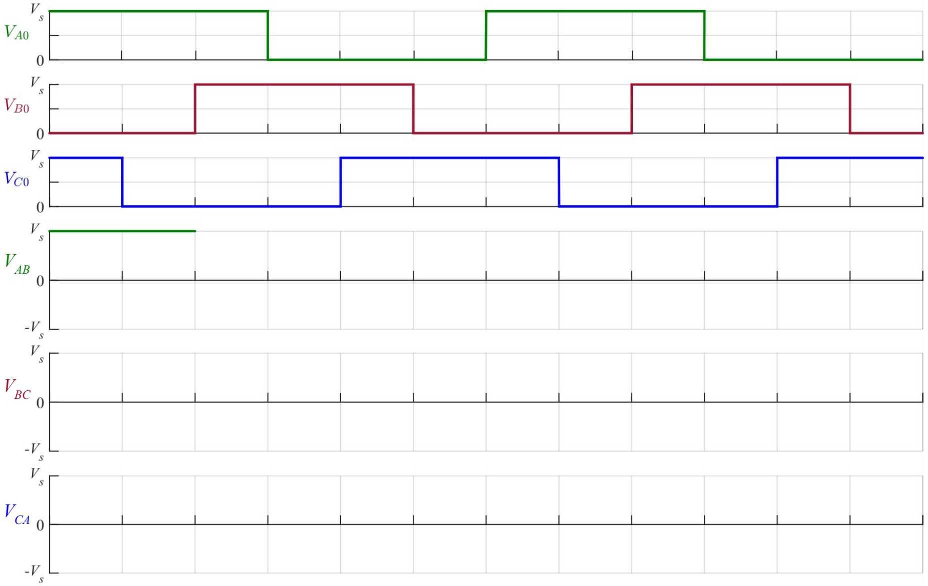

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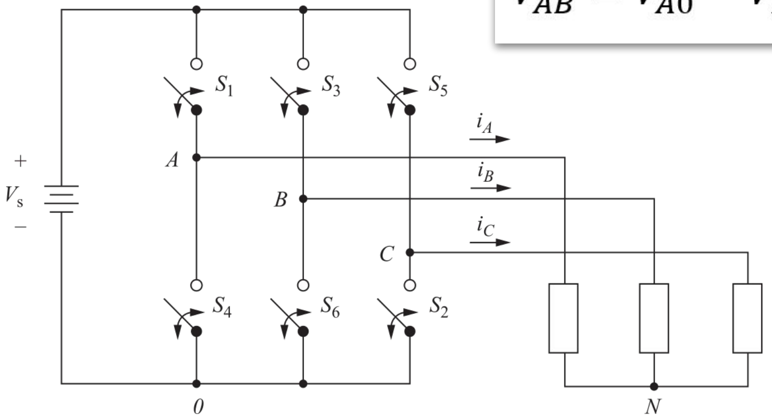
Tensiones de línea



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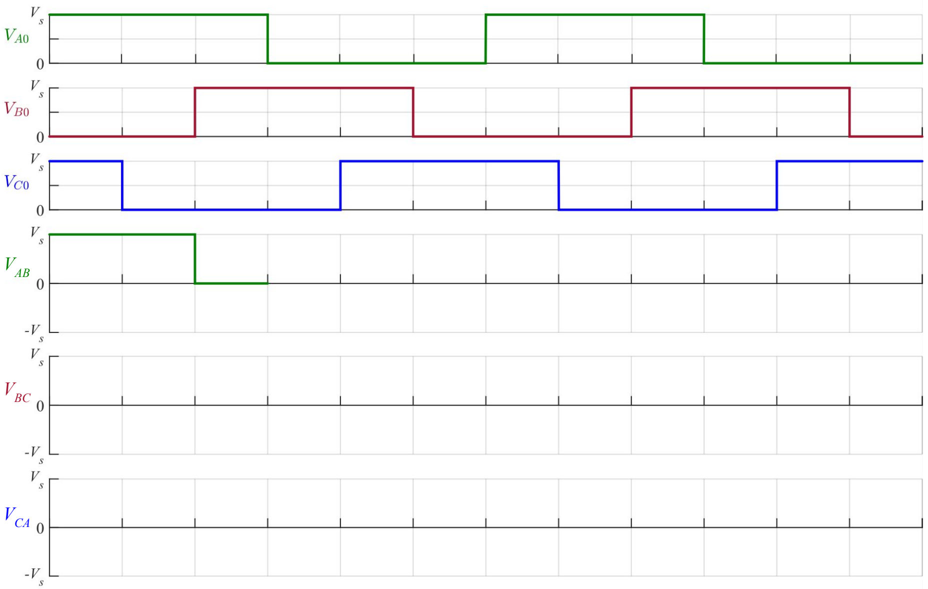
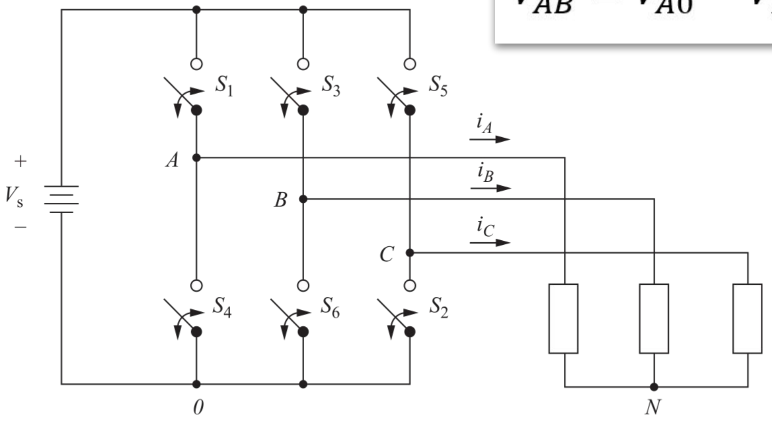


Inversor puente trifásico: formas de onda

Tensiones de línea



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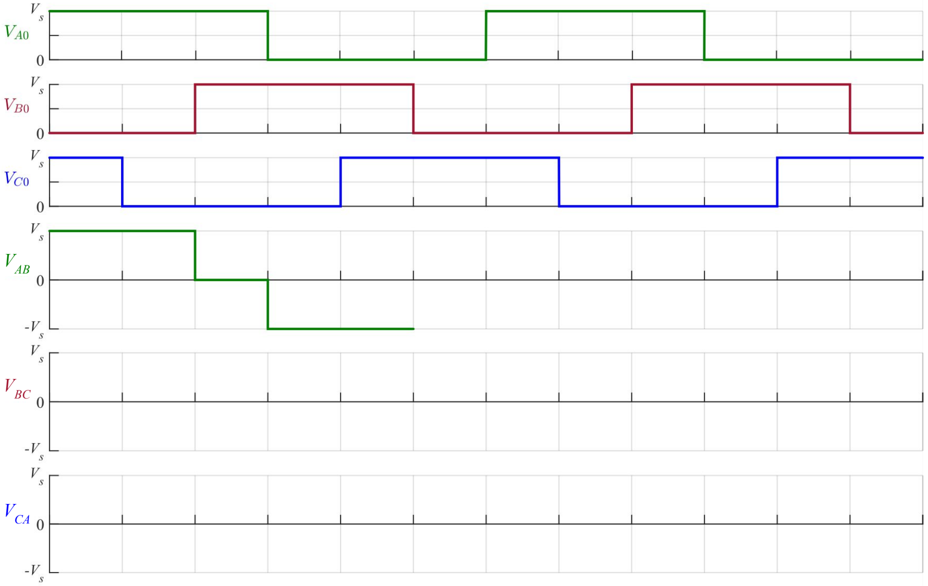

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Tensiones de línea

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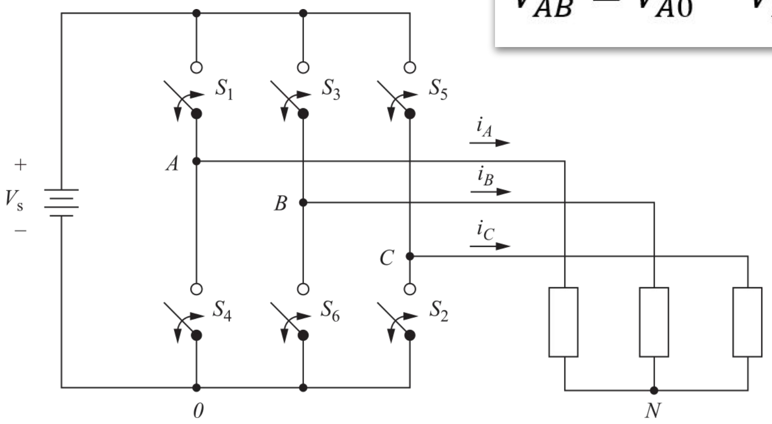

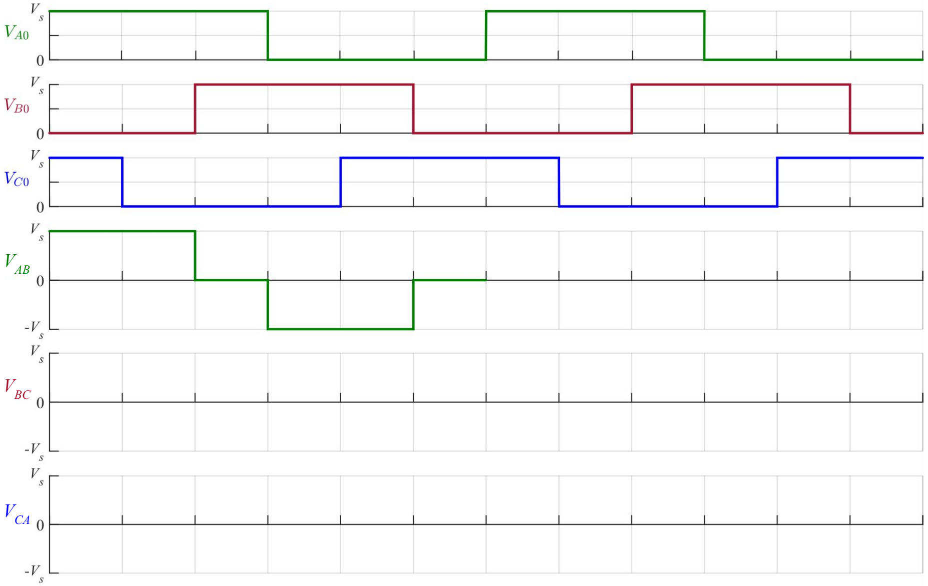

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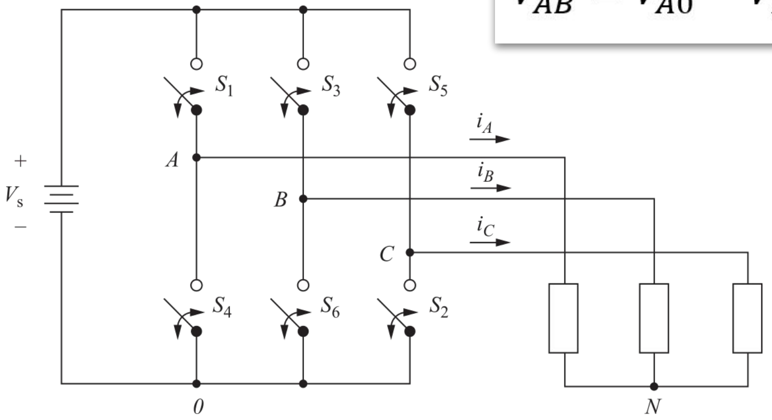
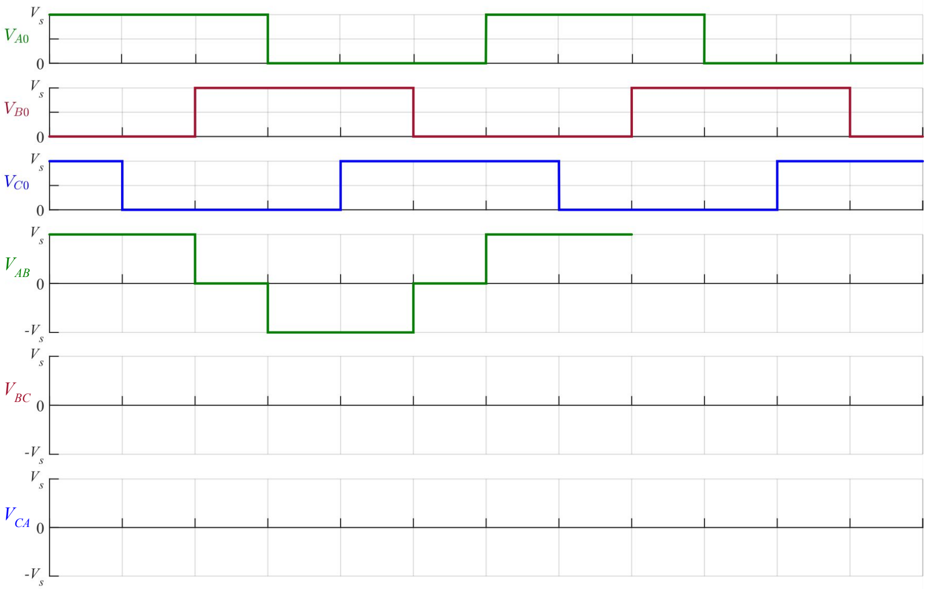
Tensiones de línea



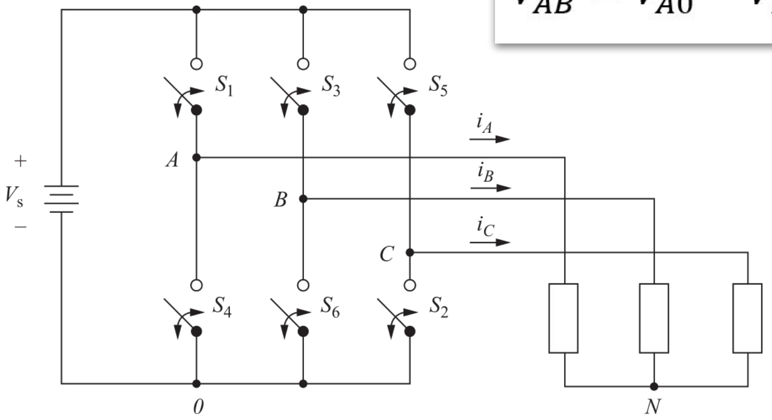
$$V_{AB} = V_{A0} - V_{B0}$$

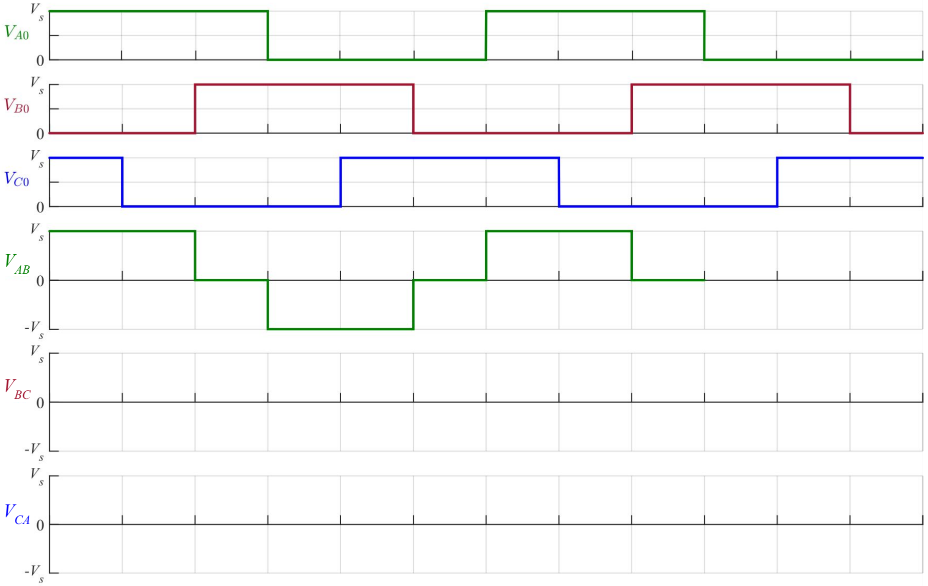


Inversor puente trifásico: formas de onda

Tensiones de línea

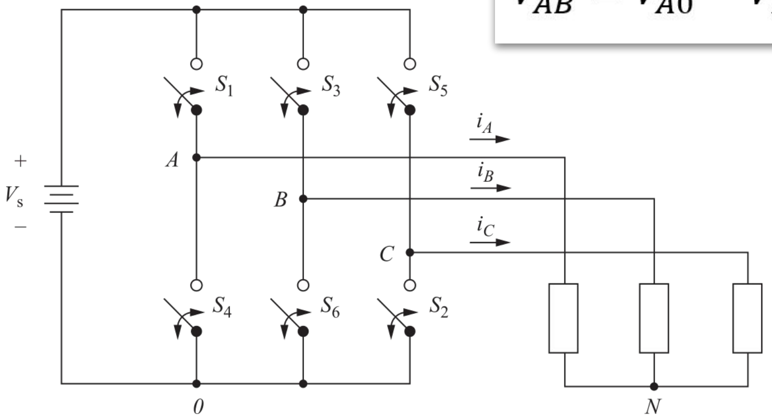
$$V_{AB} = V_{A0} - V_{B0}$$



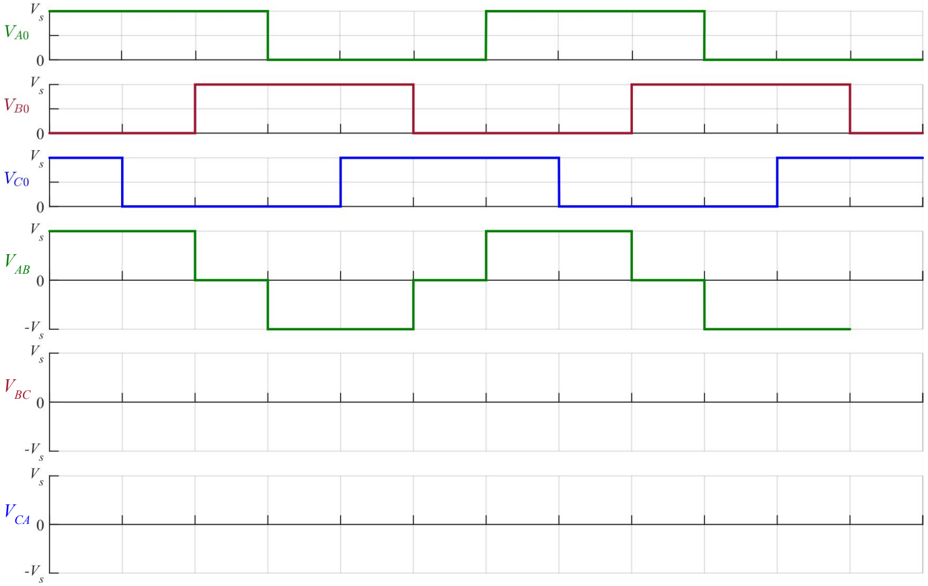


Inversor puente trifásico: formas de onda

Tensiones de línea



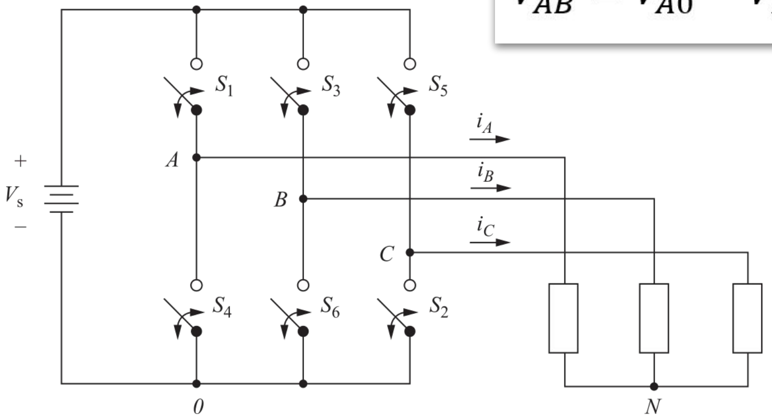
$$V_{AB} = V_{A0} - V_{B0}$$

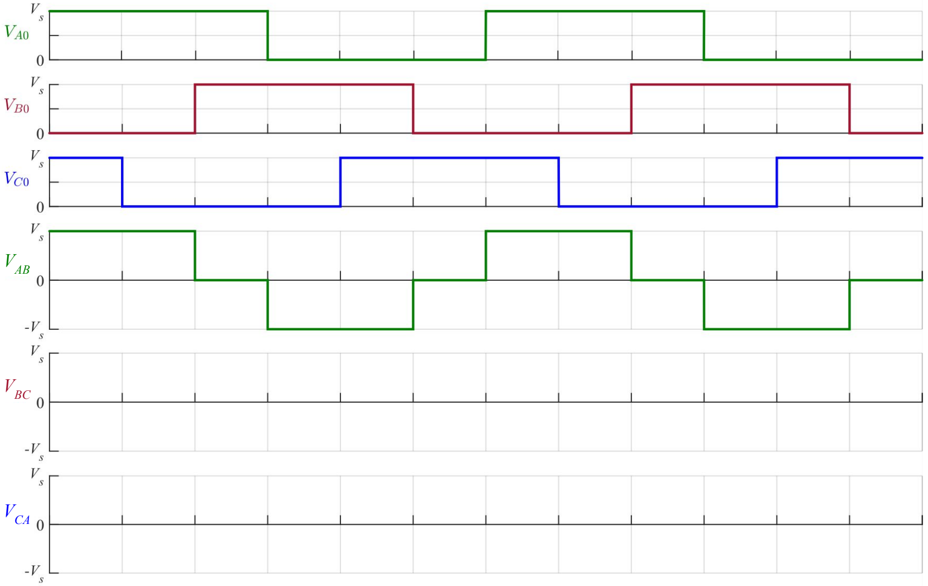


Inversor puente trifásico: formas de onda

Tensiones de línea

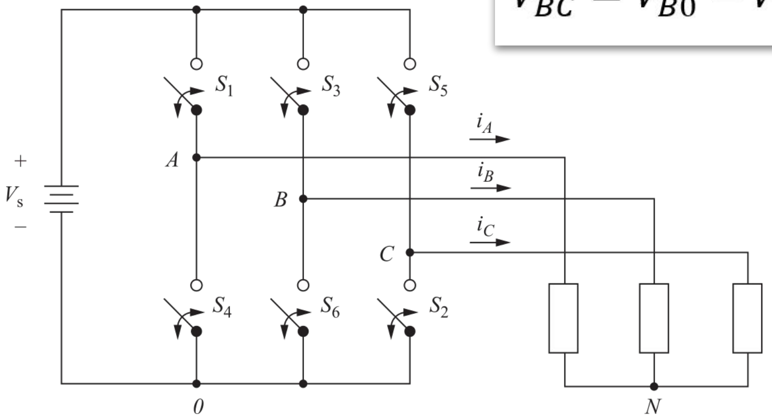
$$V_{AB} = V_{A0} - V_{B0}$$



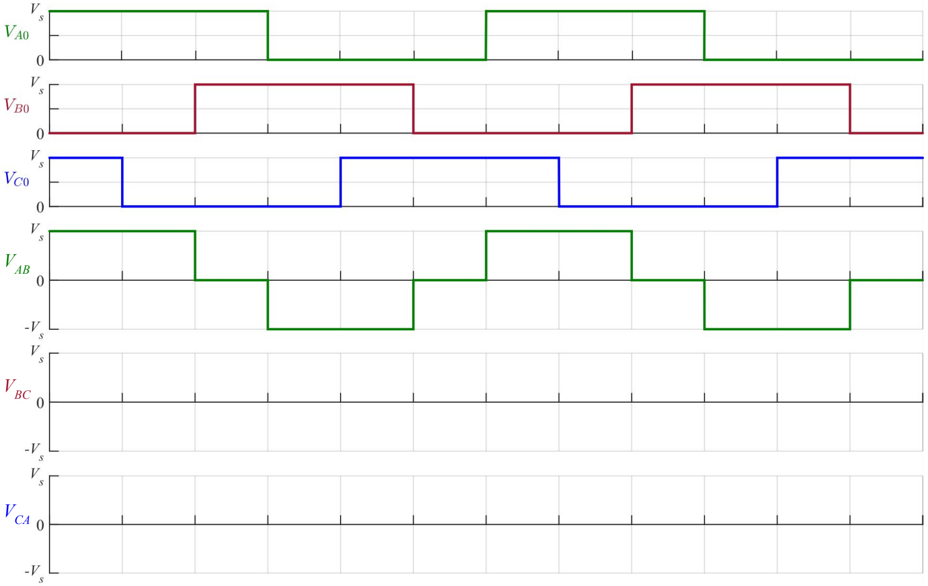


Inversor puente trifásico: formas de onda

Tensiones de línea

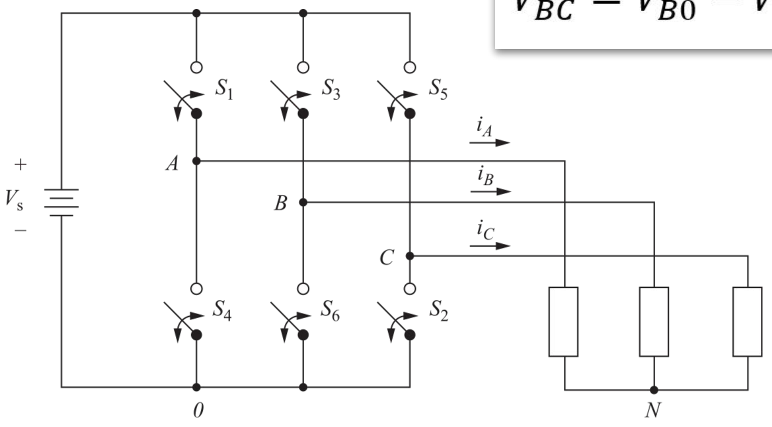


$$V_{BC} = V_{B0} - V_{C0}$$

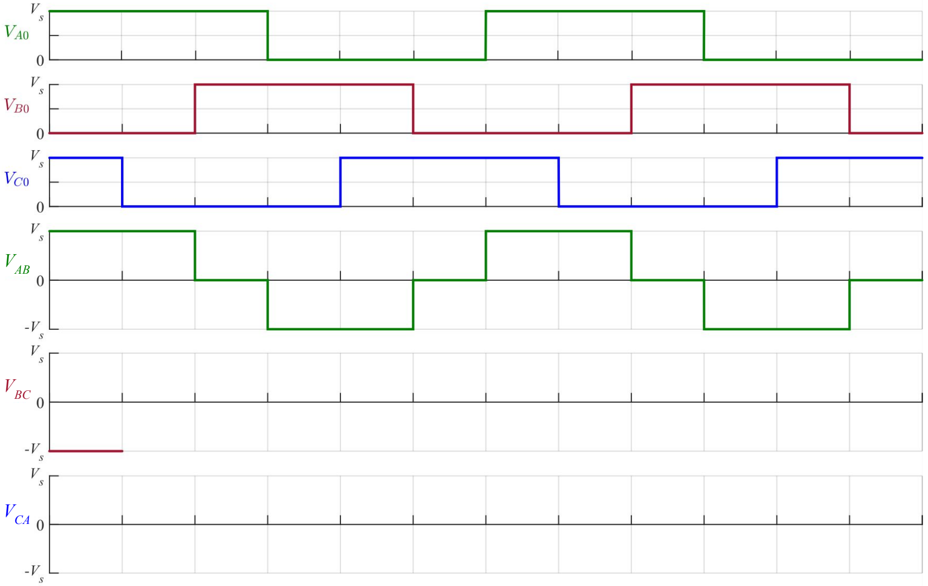


Inversor puente trifásico: formas de onda

Tensiones de línea

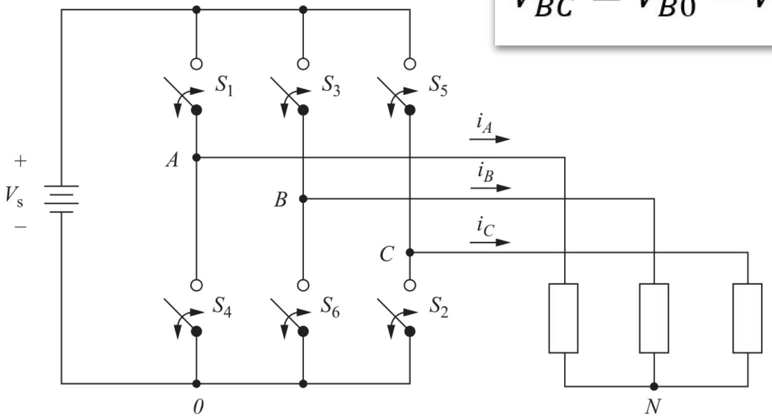


$$V_{BC} = V_{B0} - V_{C0}$$

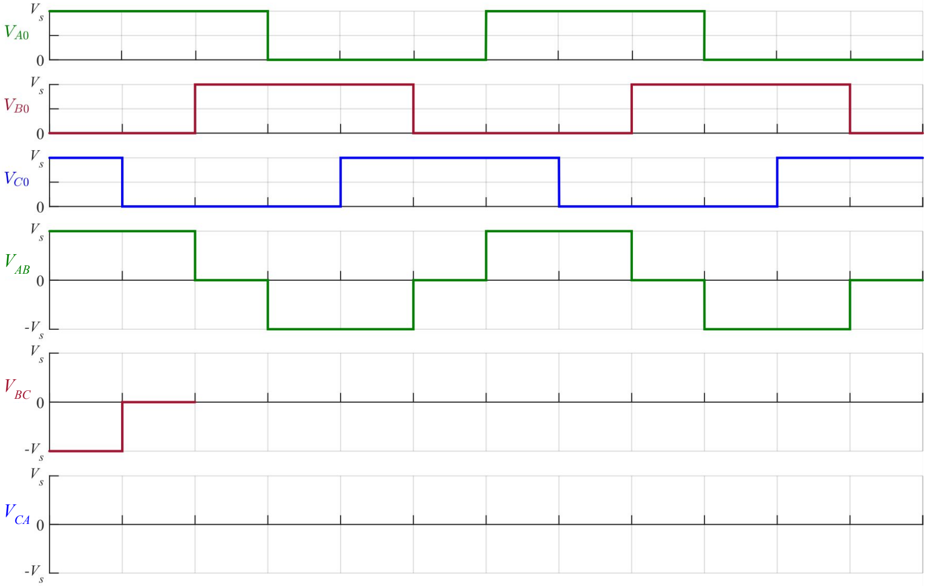


Inversor puente trifásico: formas de onda

Tensiones de línea



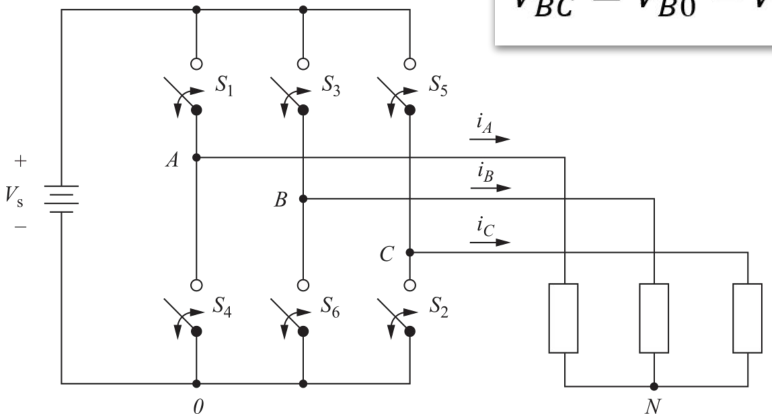
$$V_{BC} = V_{B0} - V_{C0}$$

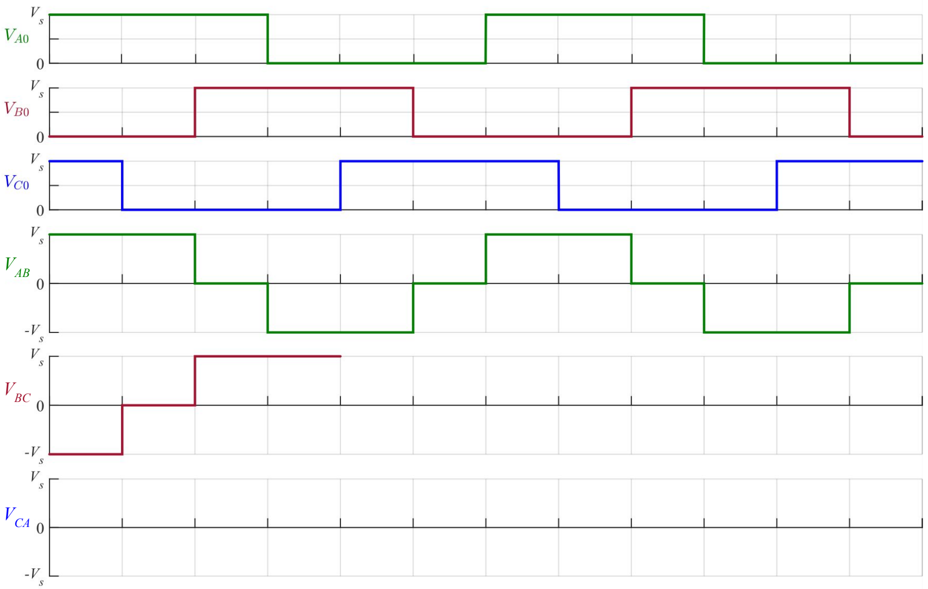


Inversor puente trifásico: formas de onda

Tensiones de línea

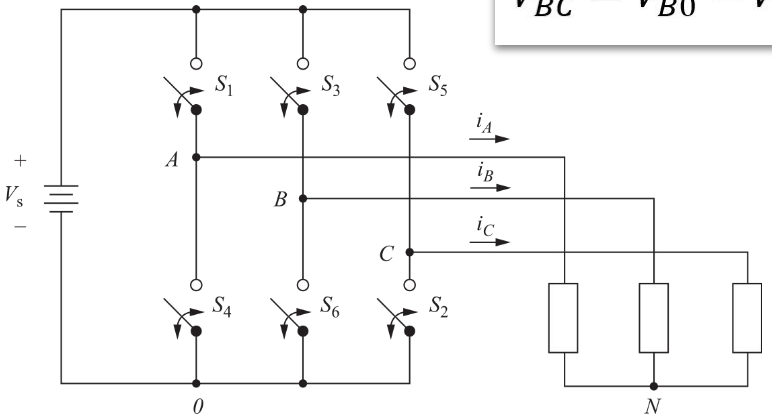
$$V_{BC} = V_{B0} - V_{C0}$$



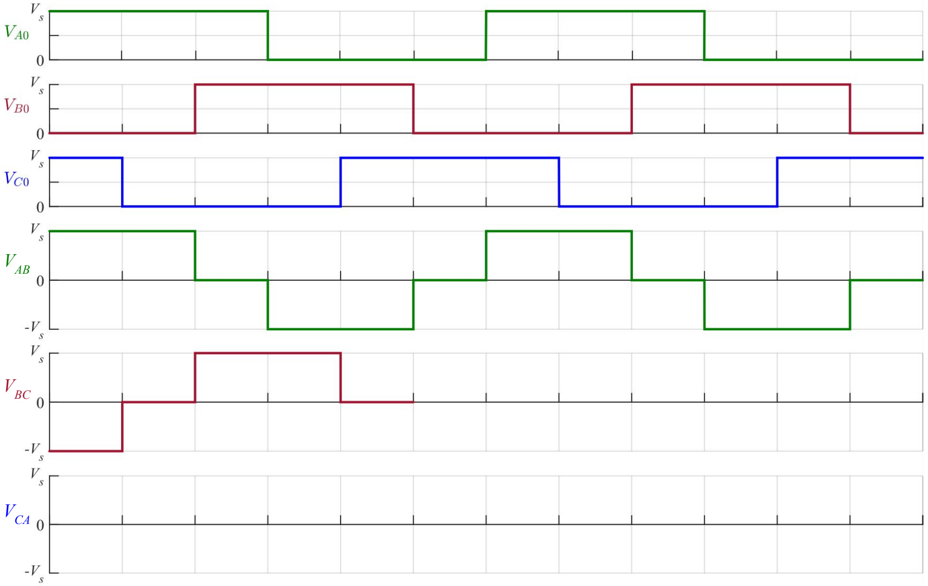


Inversor puente trifásico: formas de onda

Tensiones de línea

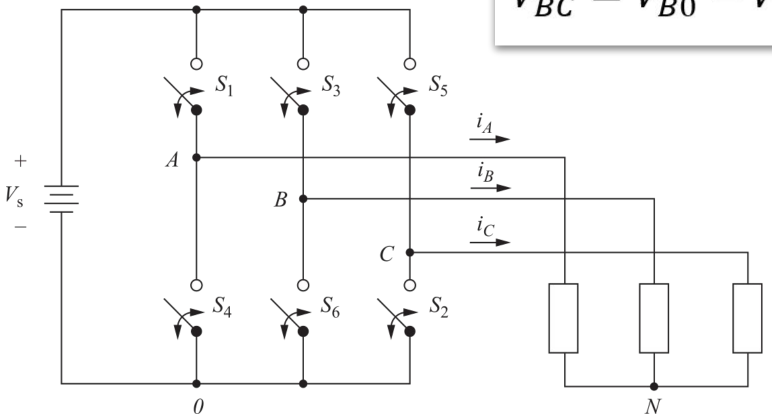


$$V_{BC} = V_{B0} - V_{C0}$$

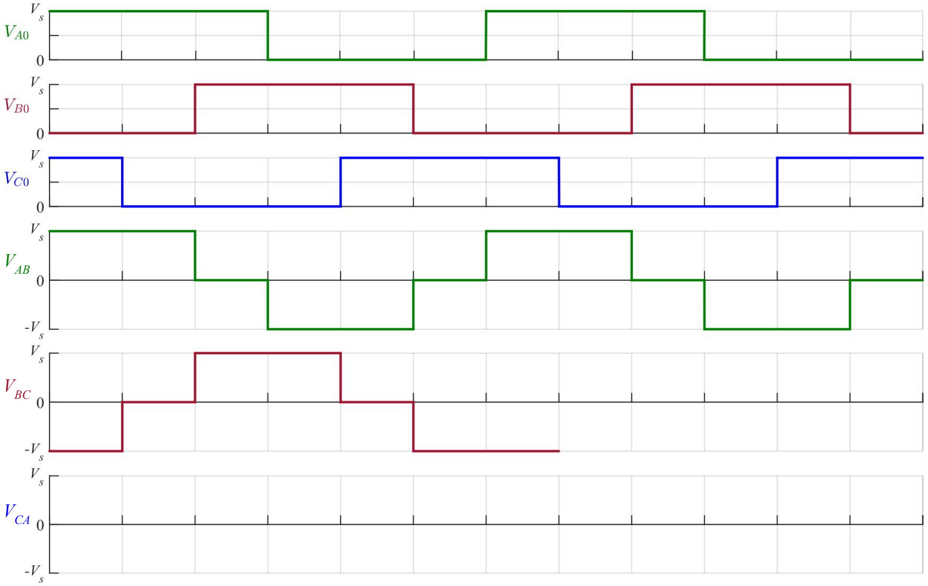


Inversor puente trifásico: formas de onda

Tensiones de línea



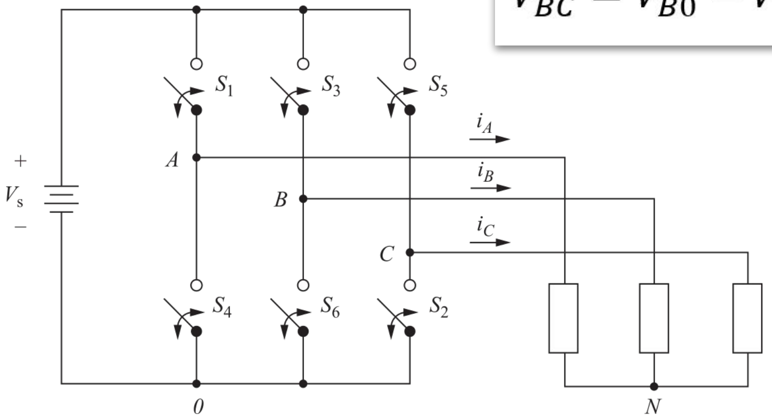
$$V_{BC} = V_{B0} - V_{C0}$$

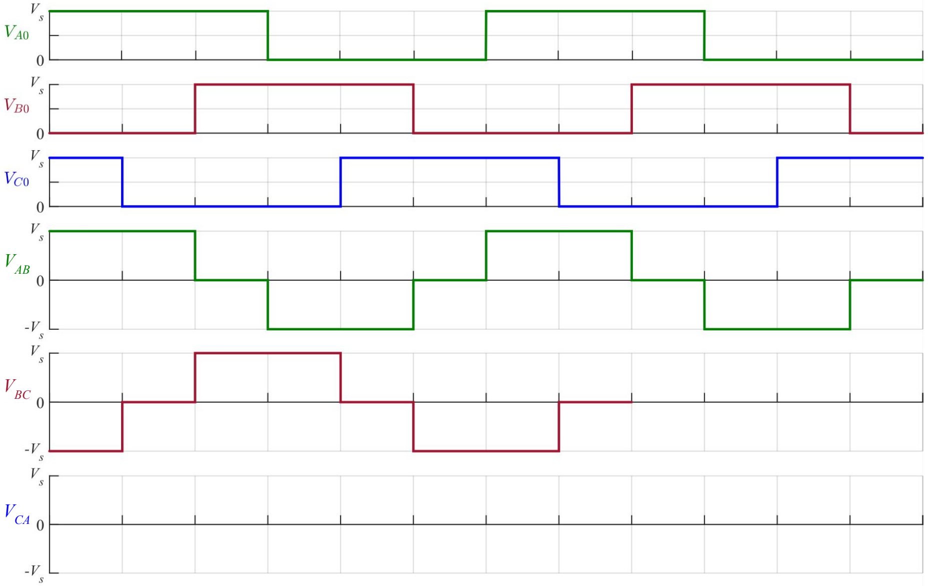


Inversor puente trifásico: formas de onda

Tensiones de línea

$$V_{BC} = V_{B0} - V_{C0}$$

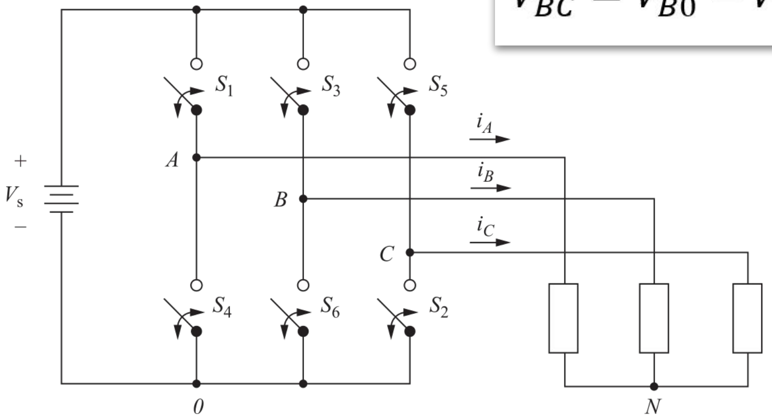


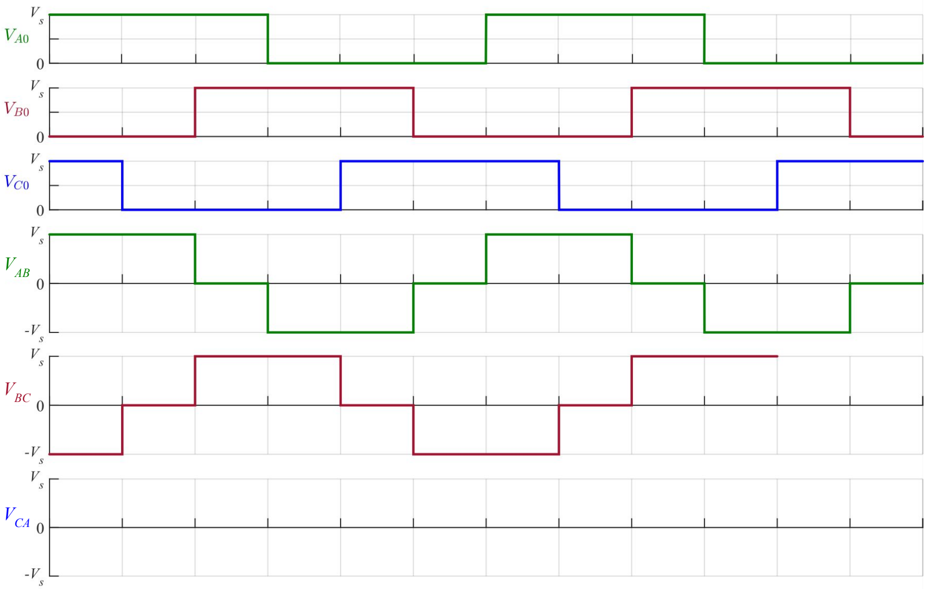


Inversor puente trifásico: formas de onda

Tensiones de línea

$$V_{BC} = V_{B0} - V_{C0}$$

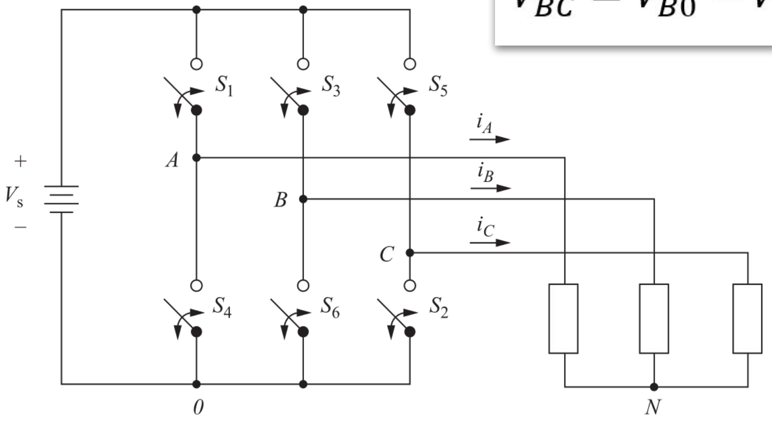


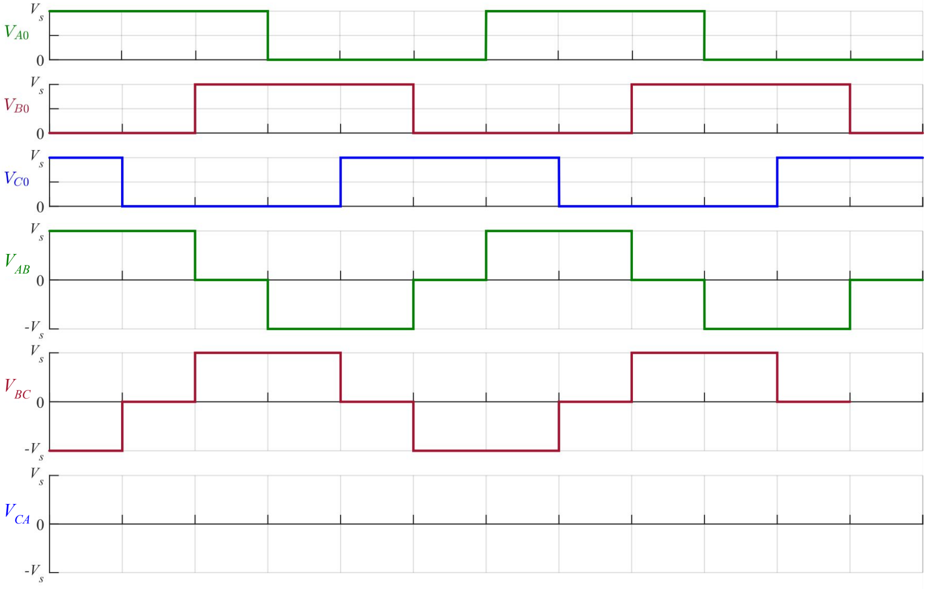


Inversor puente trifásico: formas de onda

Tensiones de línea

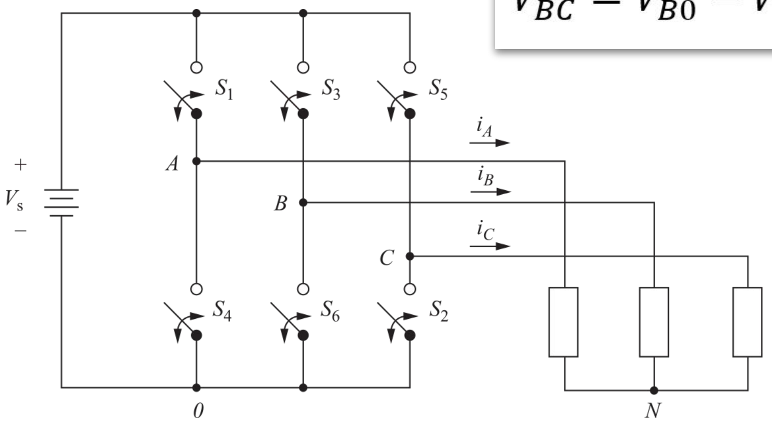
$$V_{BC} = V_{B0} - V_{C0}$$



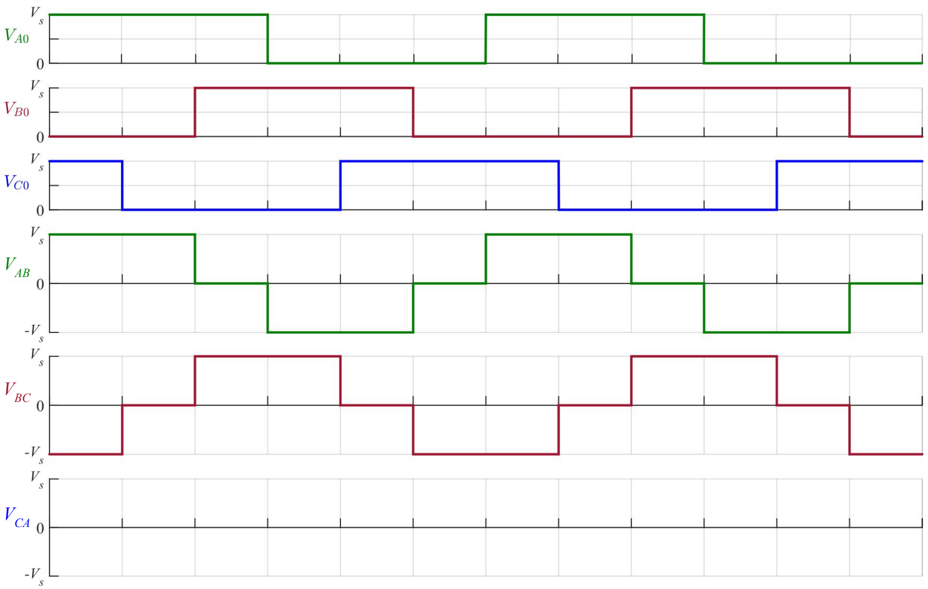


Inversor puente trifásico: formas de onda

Tensiones de línea



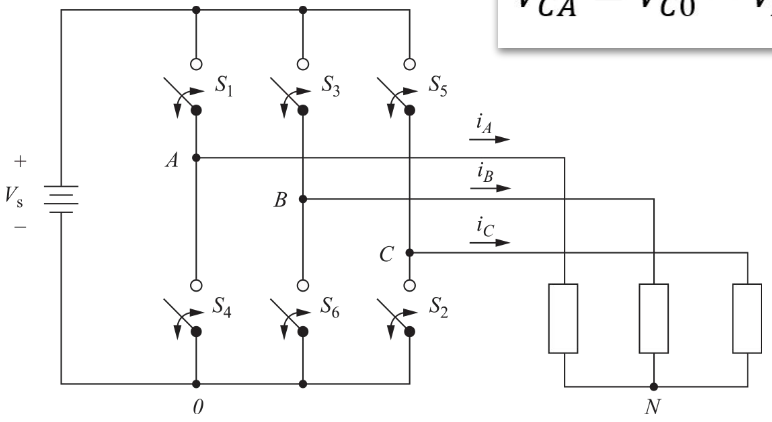
$$V_{BC} = V_{B0} - V_{C0}$$

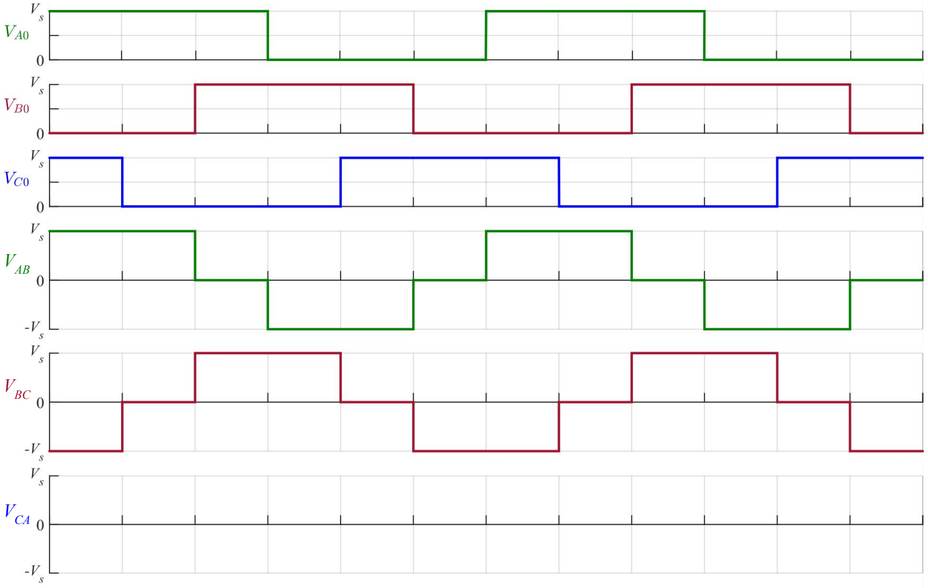


Inversor puente trifásico: formas de onda

Tensiones de línea

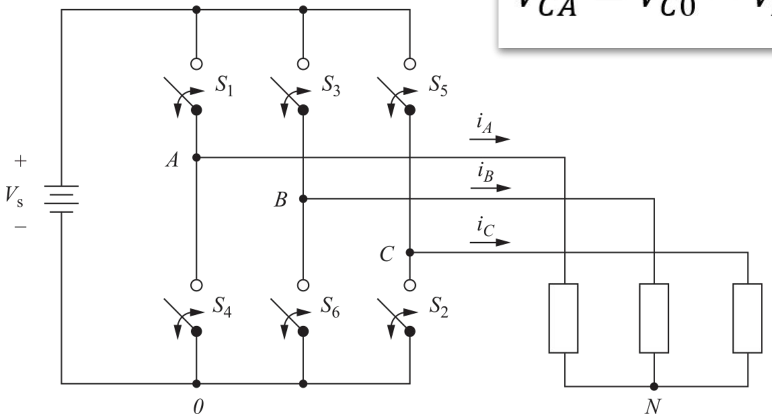
$$V_{CA} = V_{C0} - V_{A0}$$



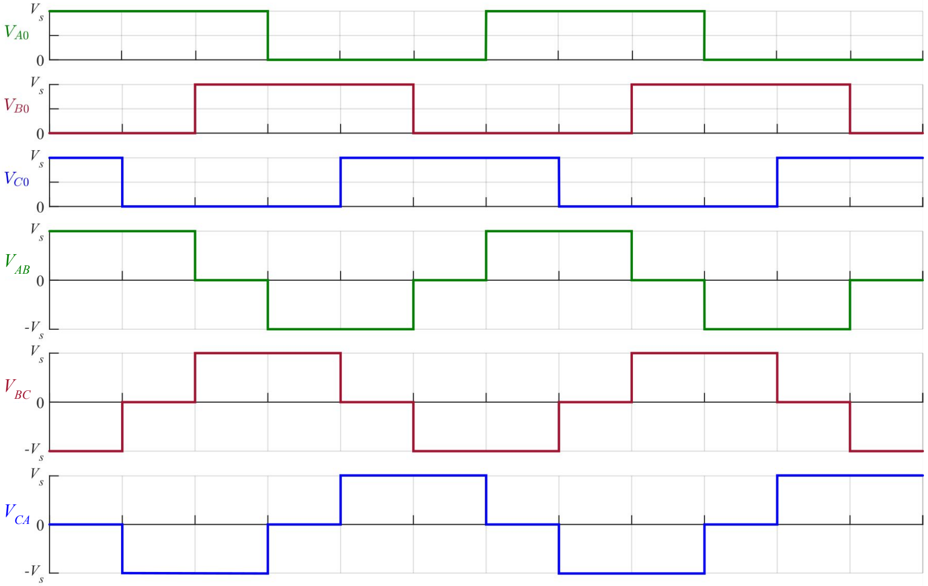


Inversor puente trifásico: formas de onda

Tensiones de línea

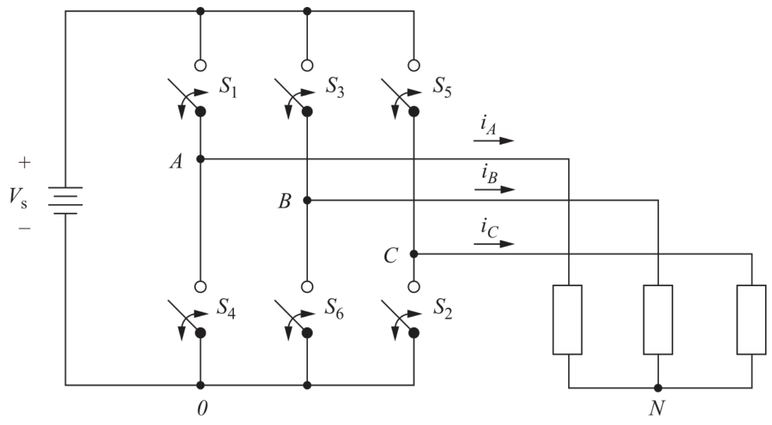


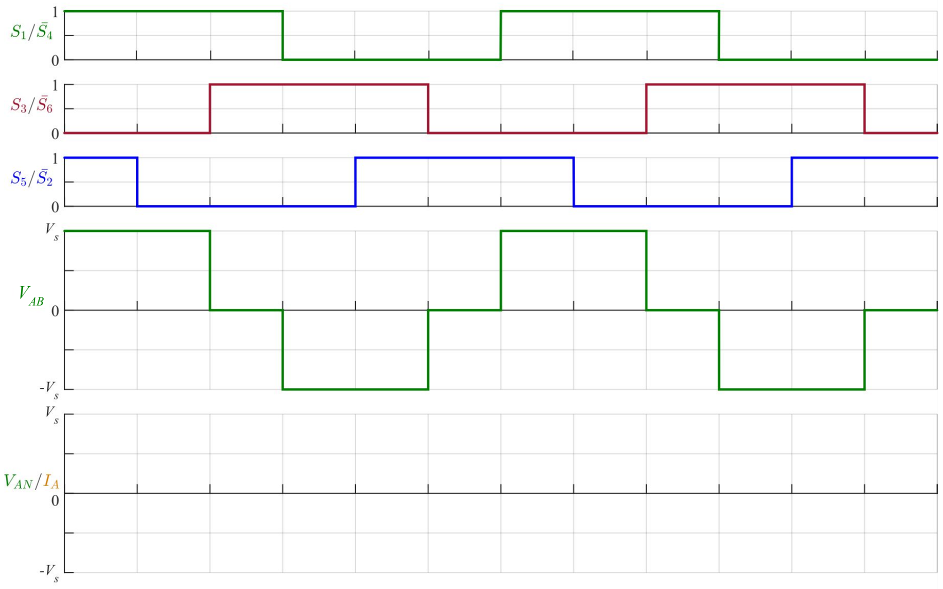
$$V_{CA} = V_{C0} - V_{A0}$$



Inversor puente trifásico: formas de onda

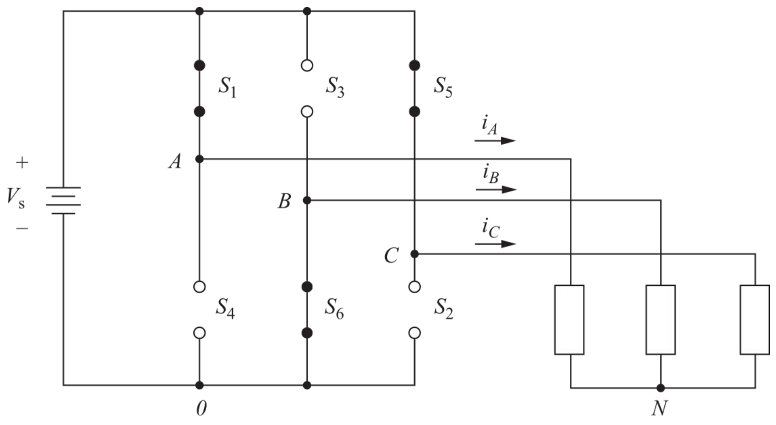
Tensión y corriente de fase

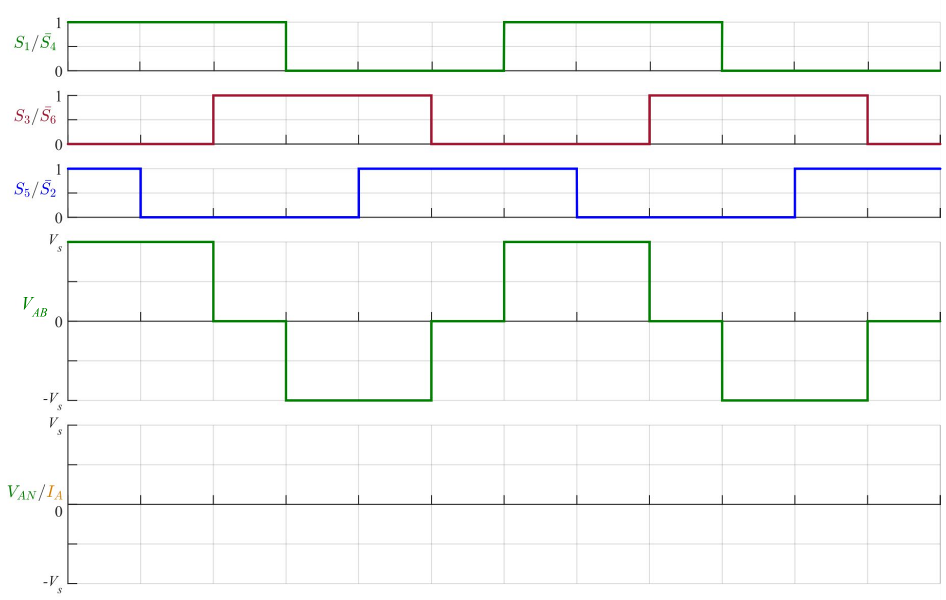




Inversor puente trifásico: formas de onda

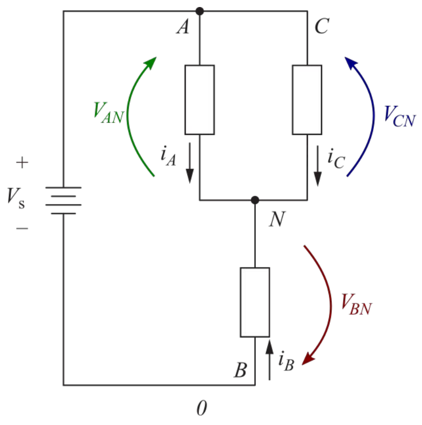
Tensión y corriente de fase

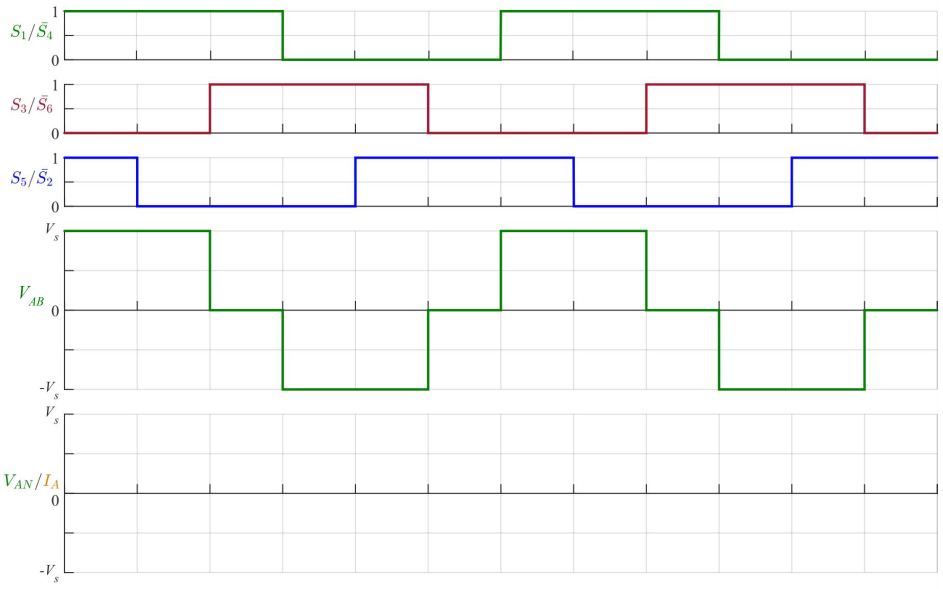




Inversor puente trifásico: formas de onda

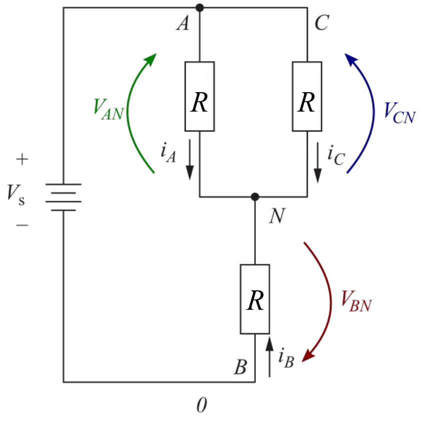
Tensión y corriente de fase

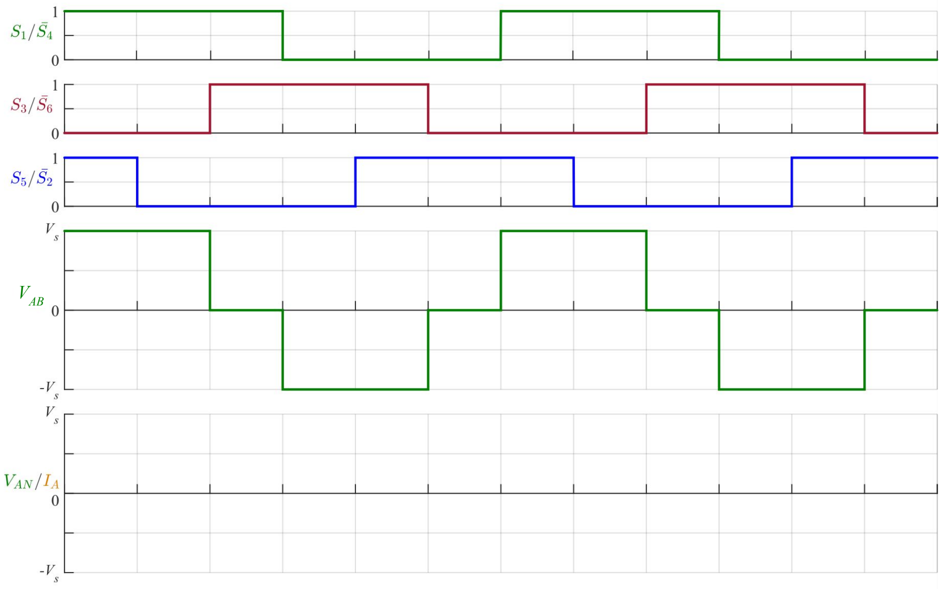




Inversor puente trifásico: formas de onda

Tensión y corriente de fase

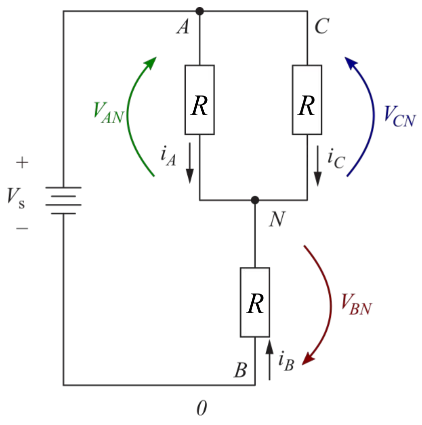


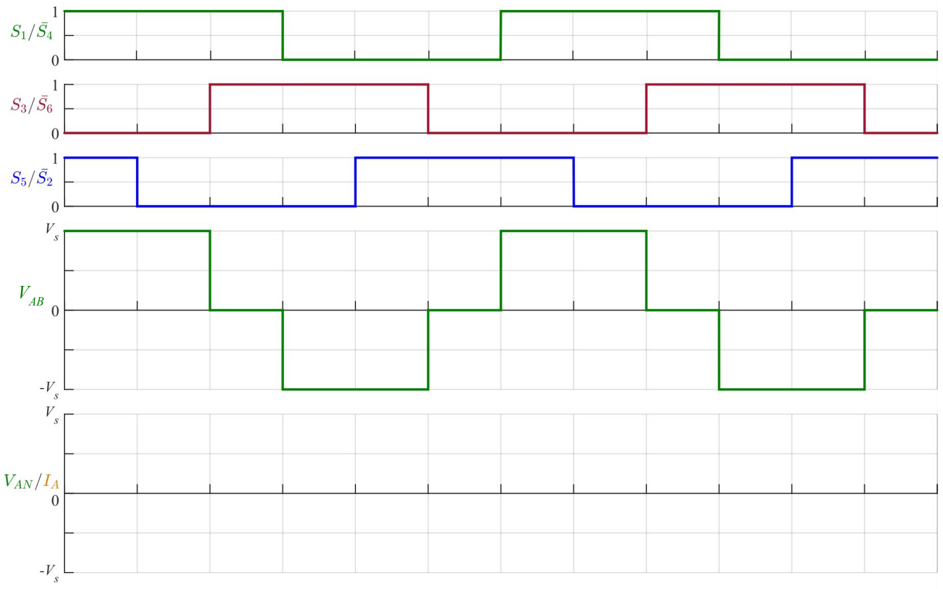


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$

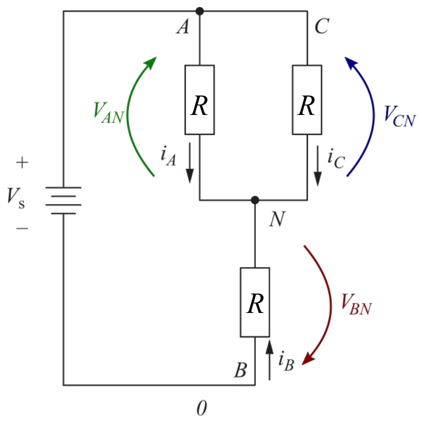




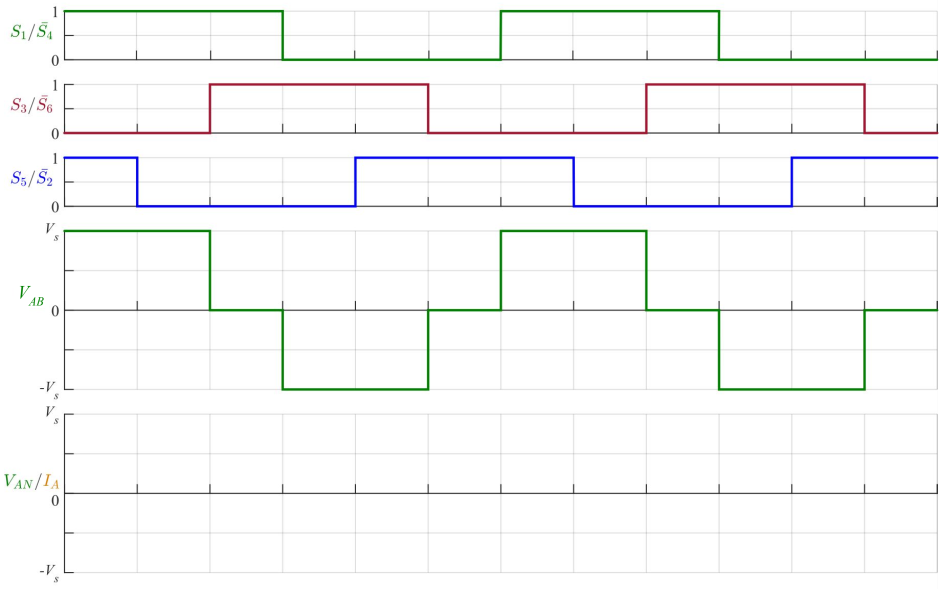
Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



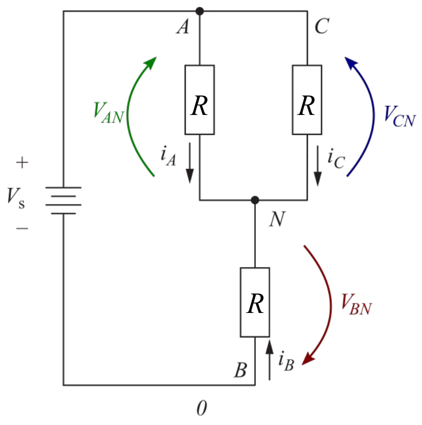
$$i_s = \frac{V_s}{3R/_2} = \frac{2V_s}{3R}$$



Inversor puente trifásico: formas de onda

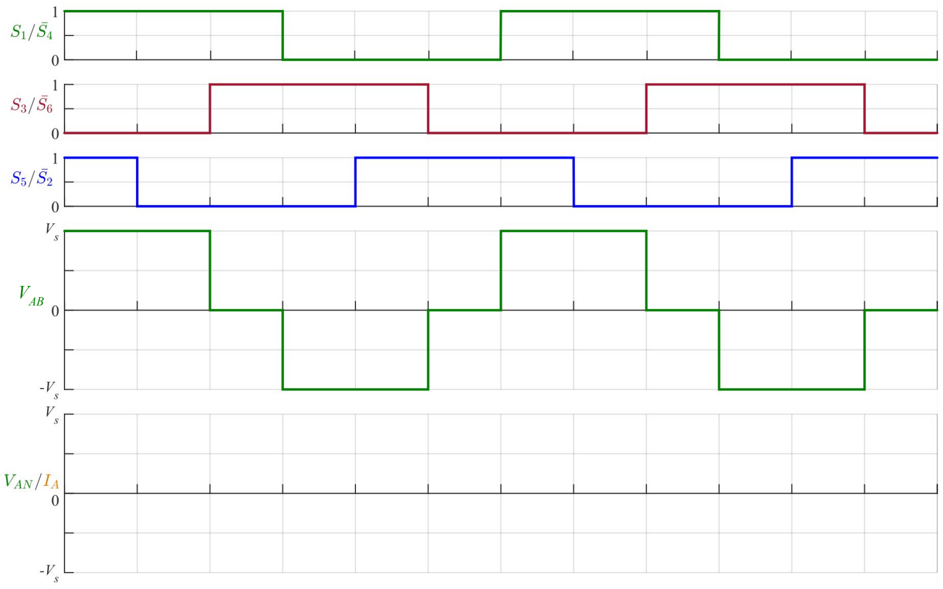
Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{{}^{3R}/_{2}} = \frac{2V_s}{3R}$$

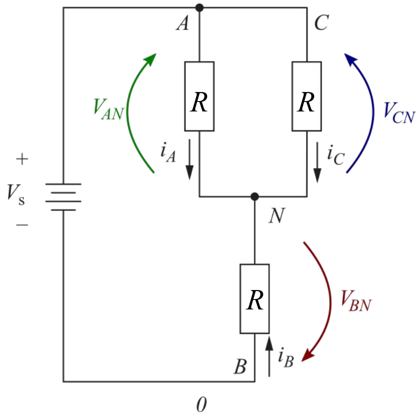
$$i_A = {}^{i_s}/_{2} = \frac{V_s}{3R}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

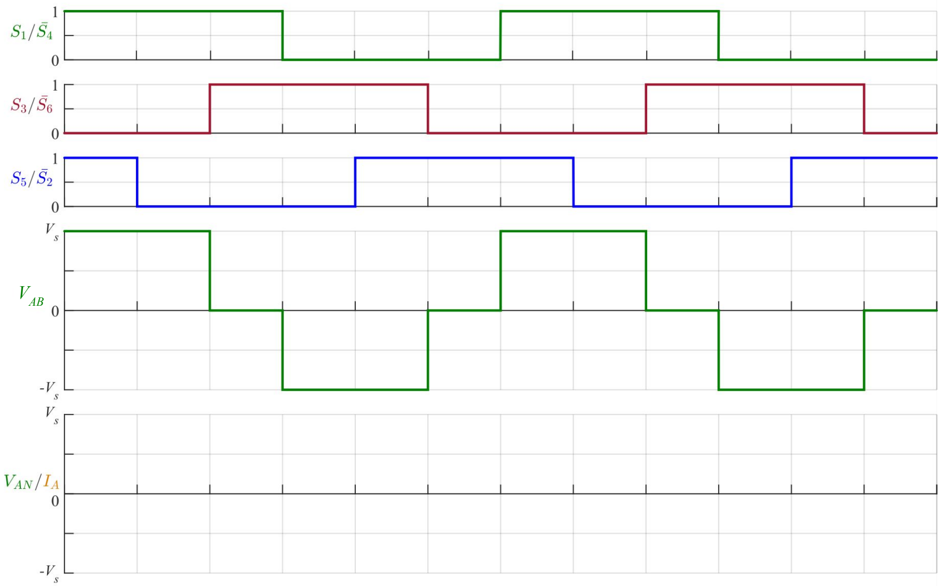
$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{{}^{3R}/_{2}} = \frac{2V_s}{3R}$$

$$i_A = {}^{i_s}/_{2} = \frac{V_s}{3R}$$

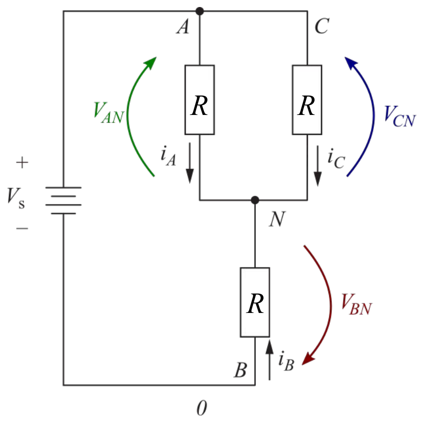
$$\boxed{V_{AN} = R \cdot i_A = \frac{V_s}{3}}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

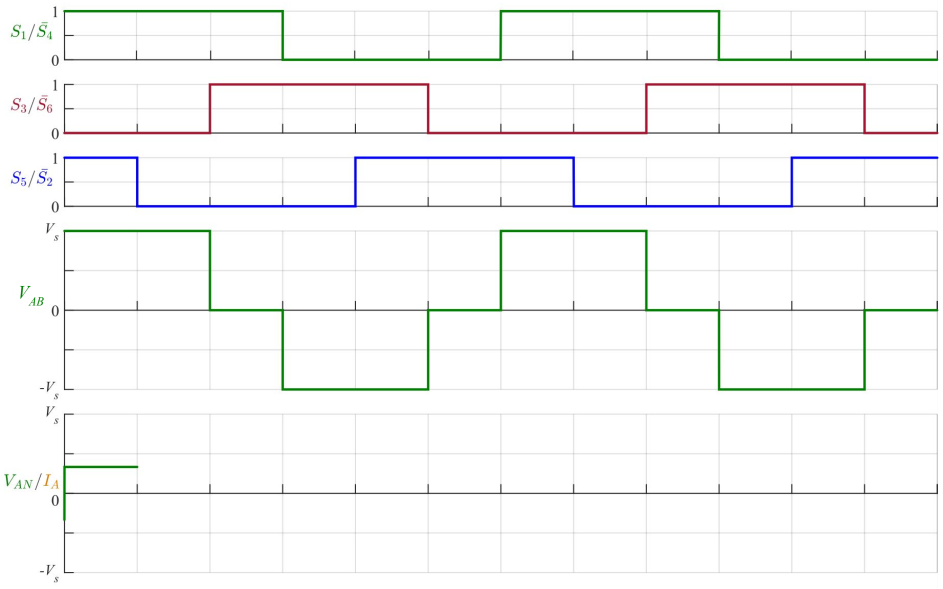
$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{3R/_{2}} = \frac{2V_s}{3R}$$

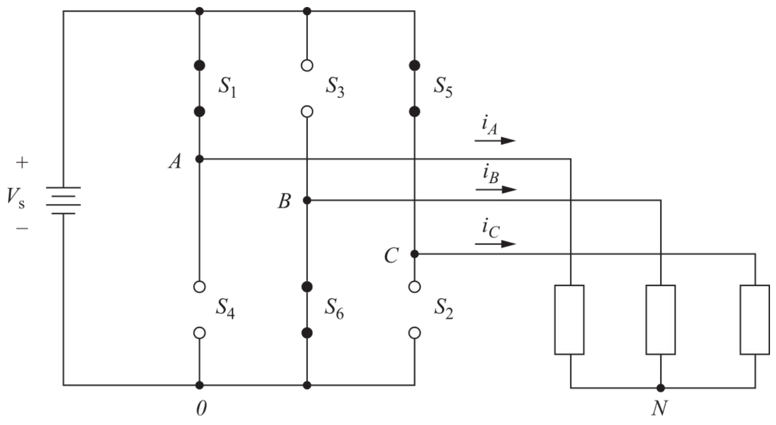
$$i_A = {}^{i_s}/_{2} = \frac{V_s}{3R}$$

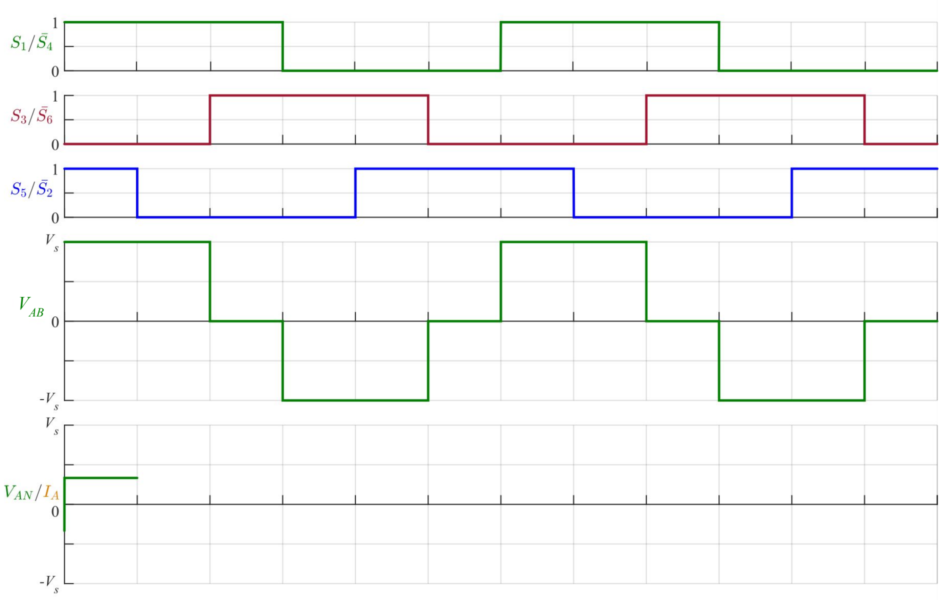
$$\boxed{V_{AN} = R \cdot i_A = \frac{V_s}{3}}$$



Inversor puente trifásico: formas de onda

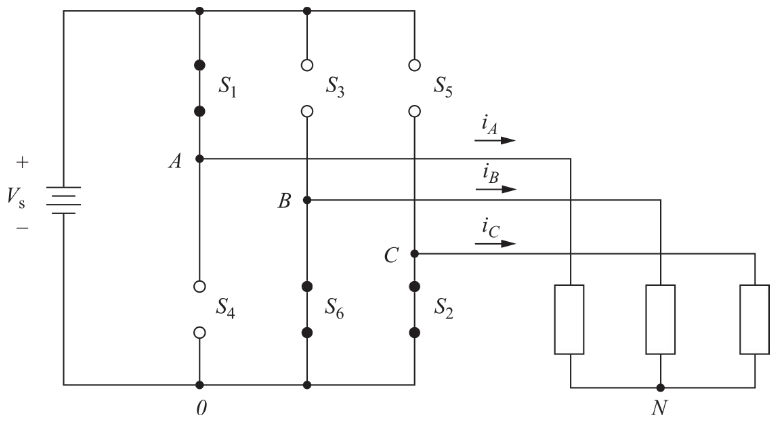
Tensión y corriente de fase

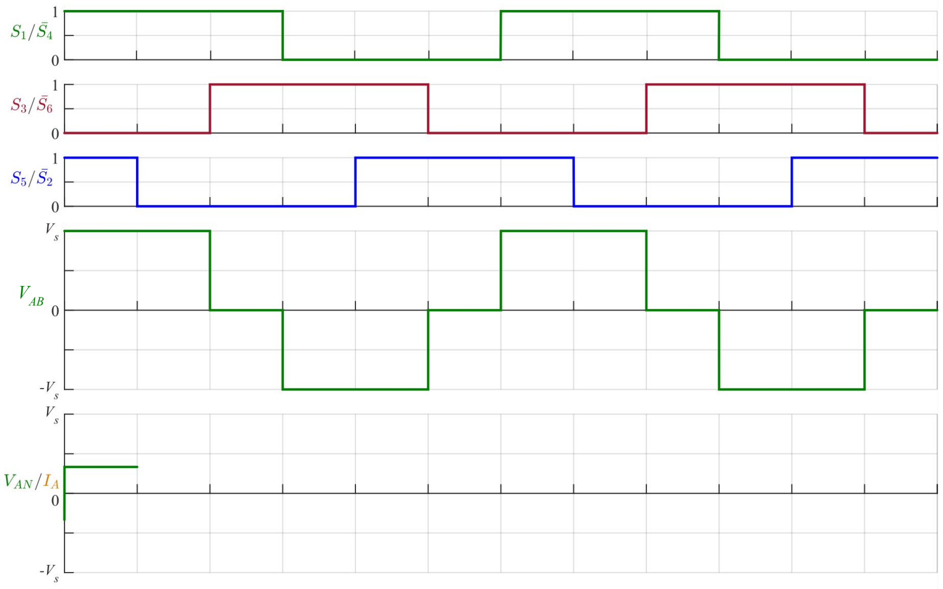




Inversor puente trifásico: formas de onda

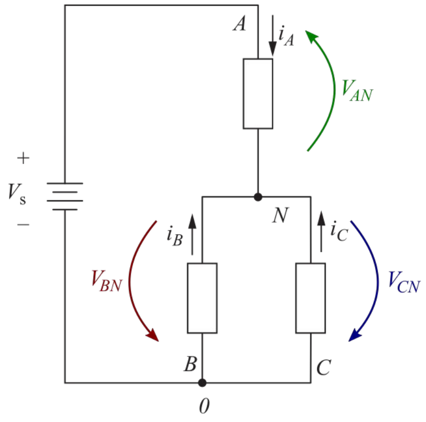
Tensión y corriente de fase

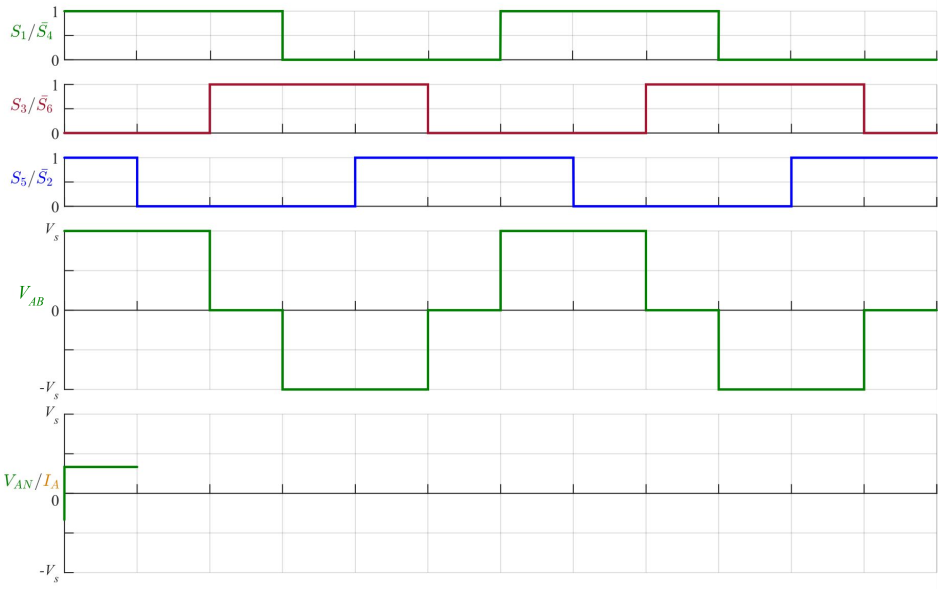




Inversor puente trifásico: formas de onda

Tensión y corriente de fase

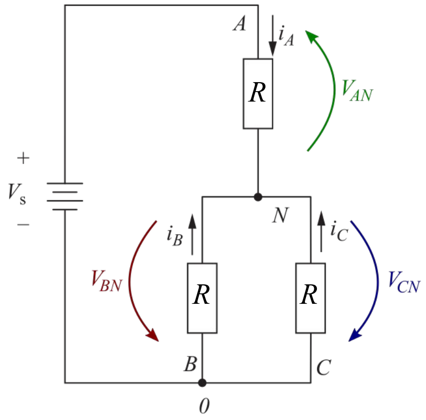


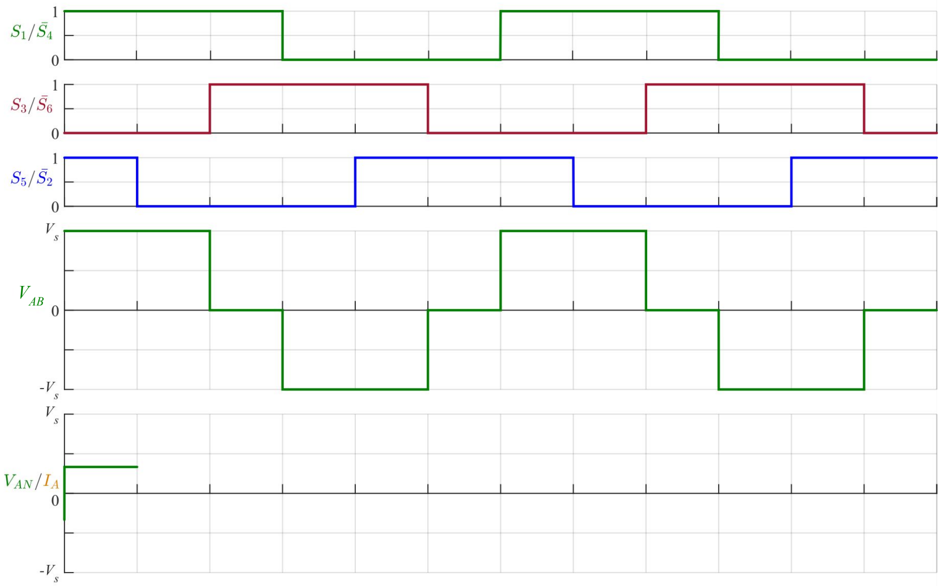


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



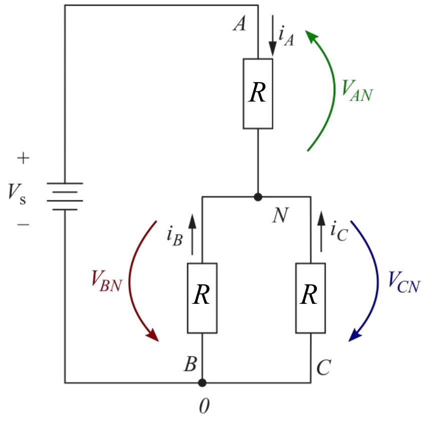


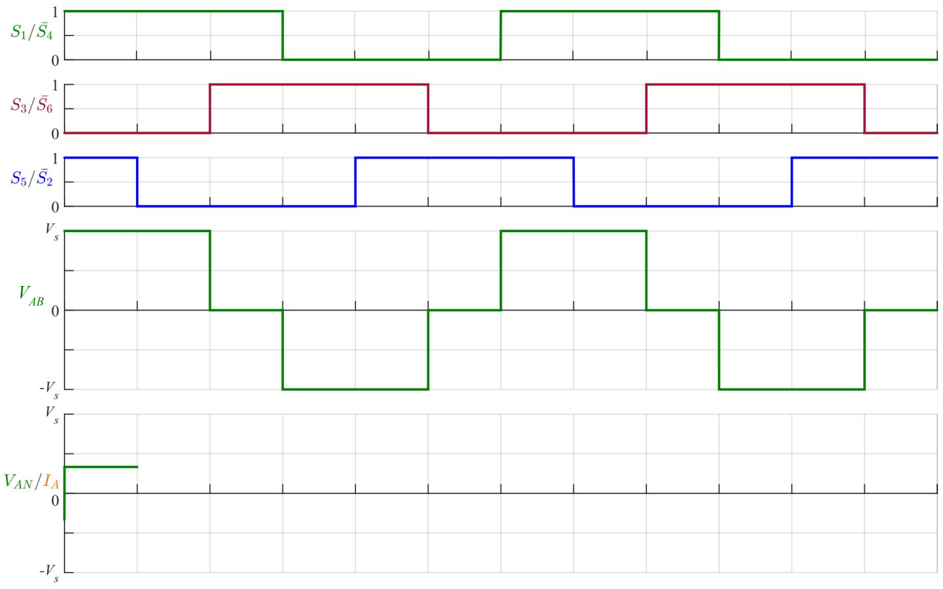
Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$

$$i_s = \frac{V_s}{{}^{3R}/_{2}} = \frac{2V_s}{3R}$$

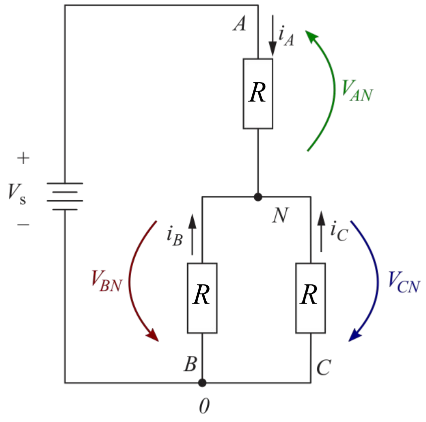




Inversor puente trifásico: formas de onda

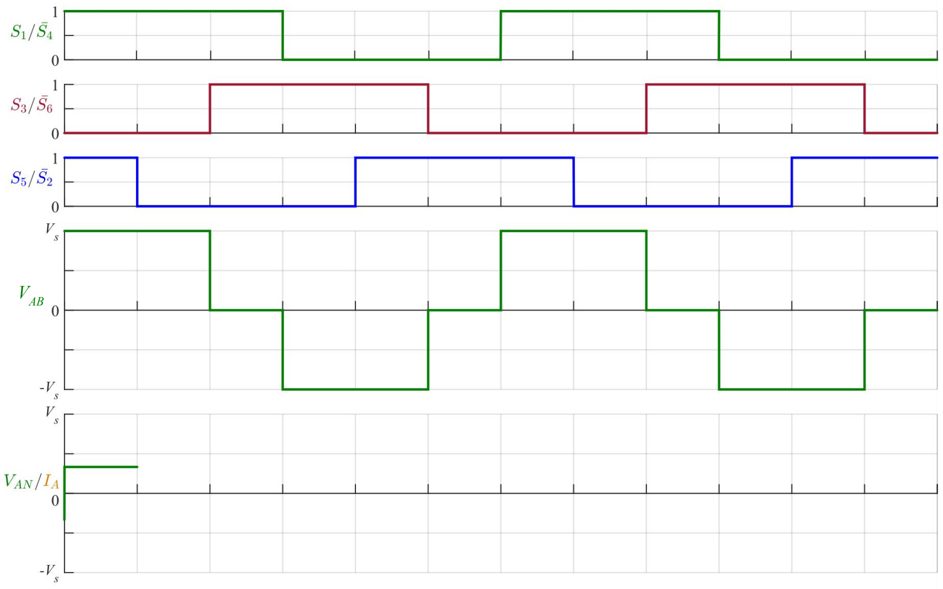
Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{{}^{3R}/_{2}} = \frac{2V_s}{3R}$$

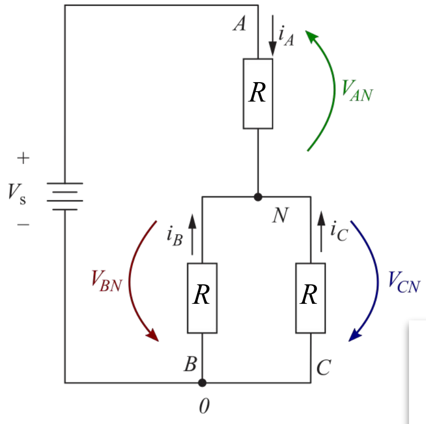
$$i_A = i_s = \frac{2V_s}{3R}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

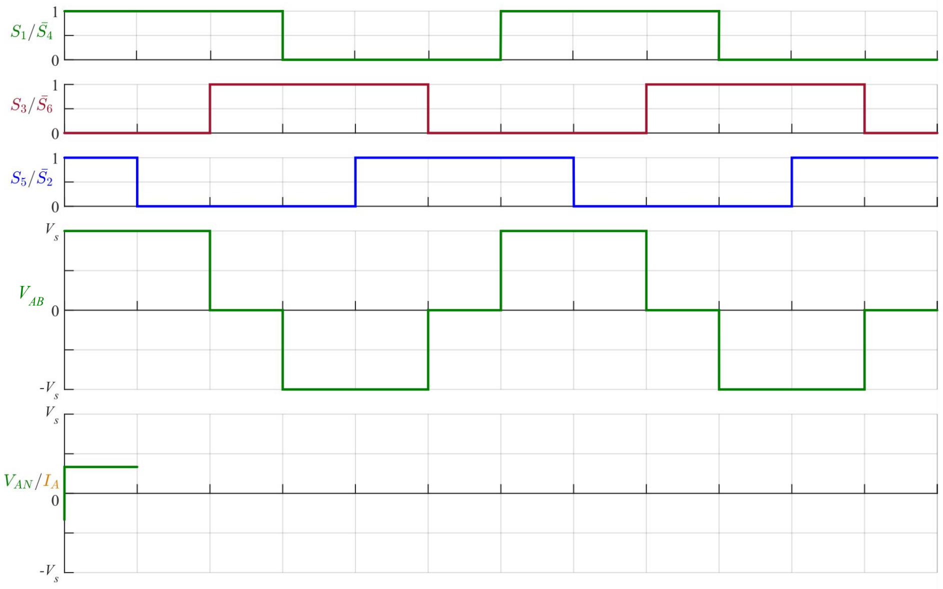
$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{3R/_2} = \frac{2V_s}{3R}$$

$$i_A = i_s = \frac{2V_s}{3R}$$

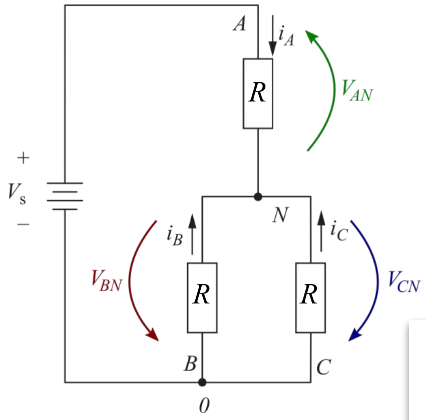
$$V_{AN} = R \cdot i_A = \frac{2V_s}{3}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$



$$i_s = \frac{V_s}{3R/_2} = \frac{2V_s}{3R}$$

$$i_A = i_s = \frac{2V_s}{3R}$$

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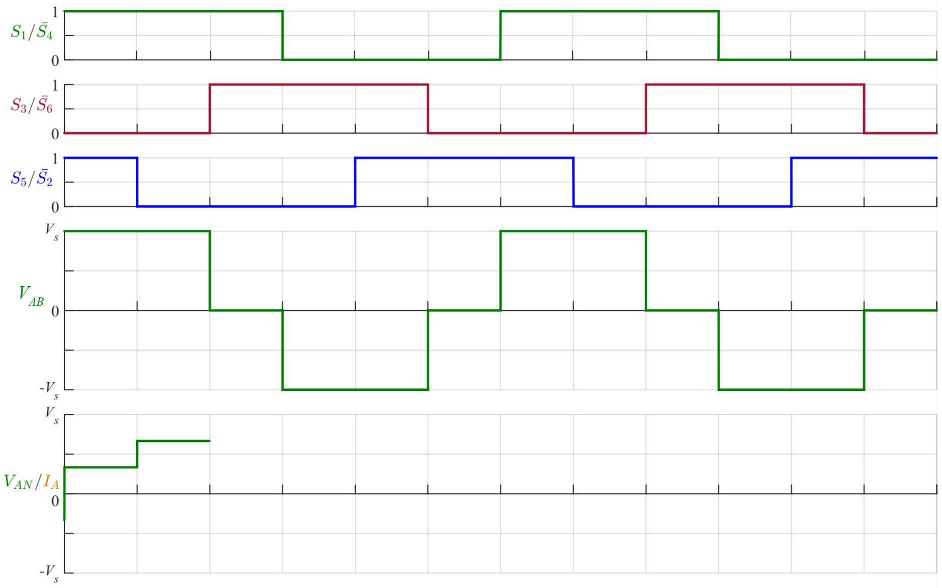
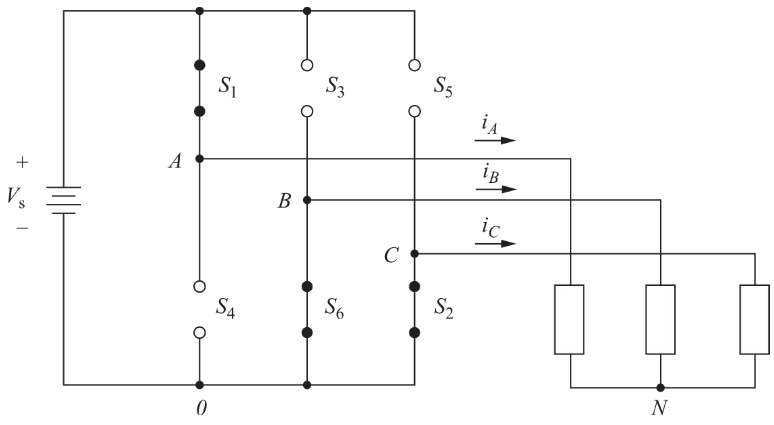
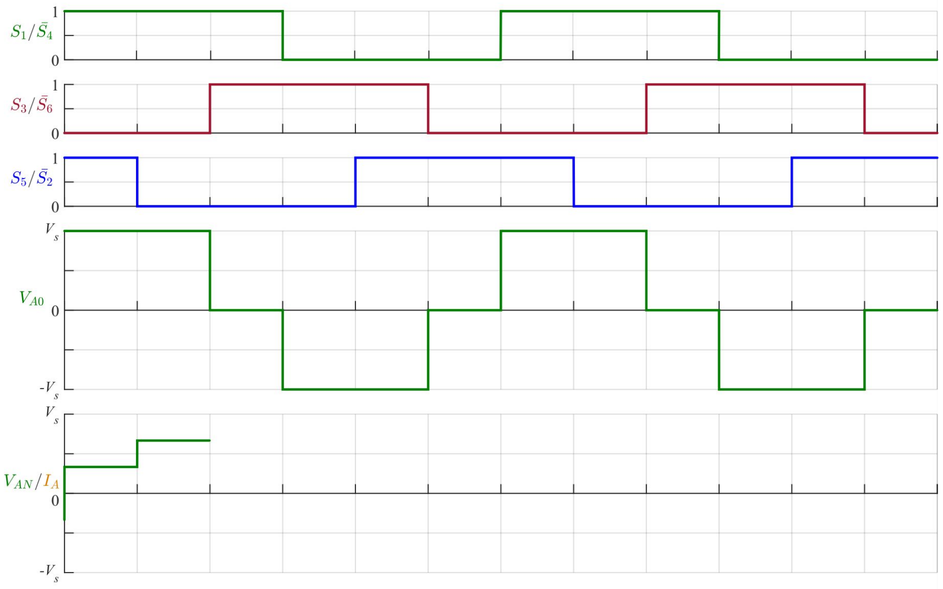


Inversor puente trifásico: formas de onda

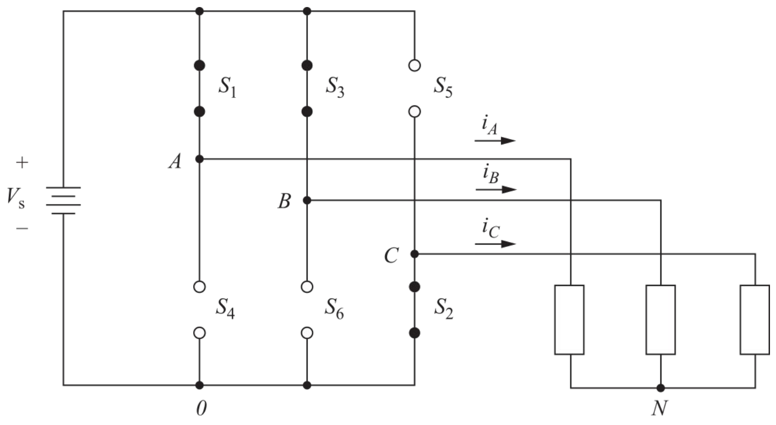
Tensión y corriente de fase

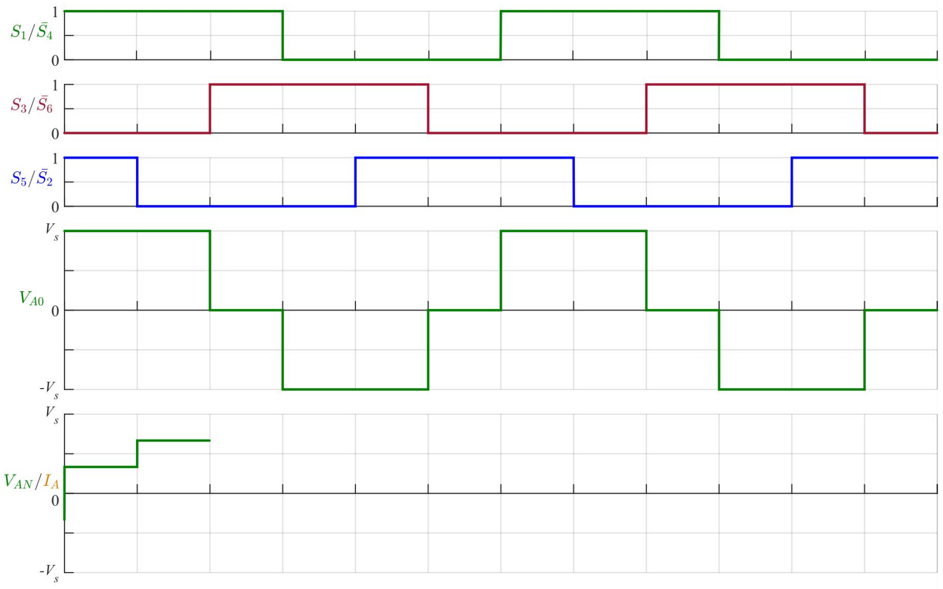




Inversor puente trifásico: formas de onda

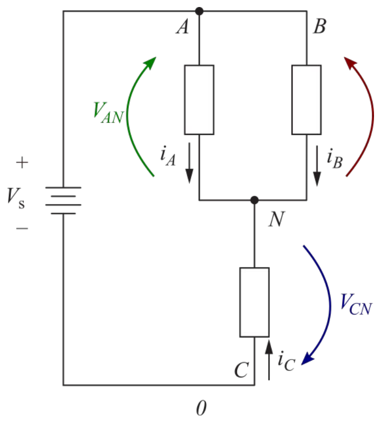
Tensión y corriente de fase

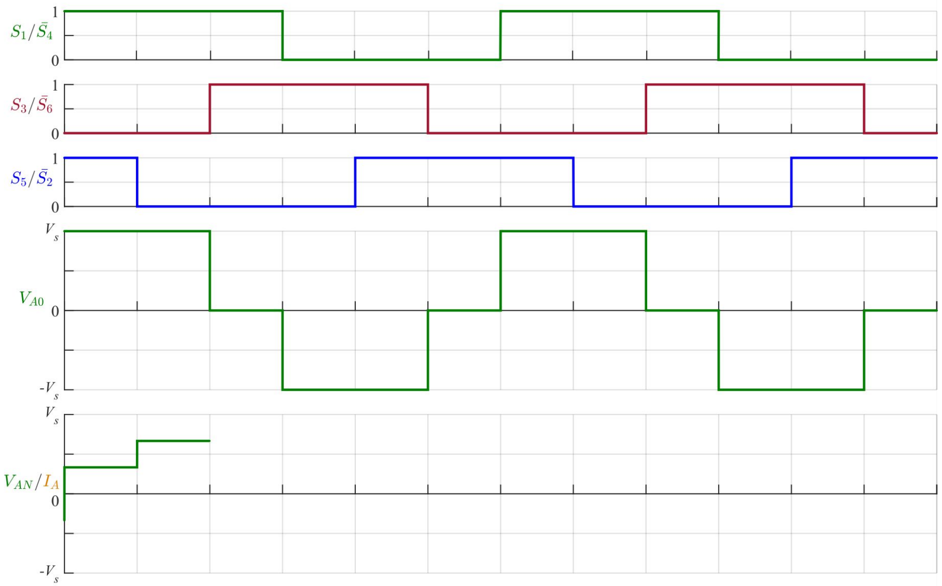




Inversor puente trifásico: formas de onda

Tensión y corriente de fase

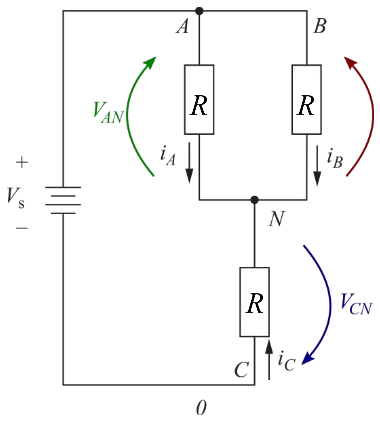


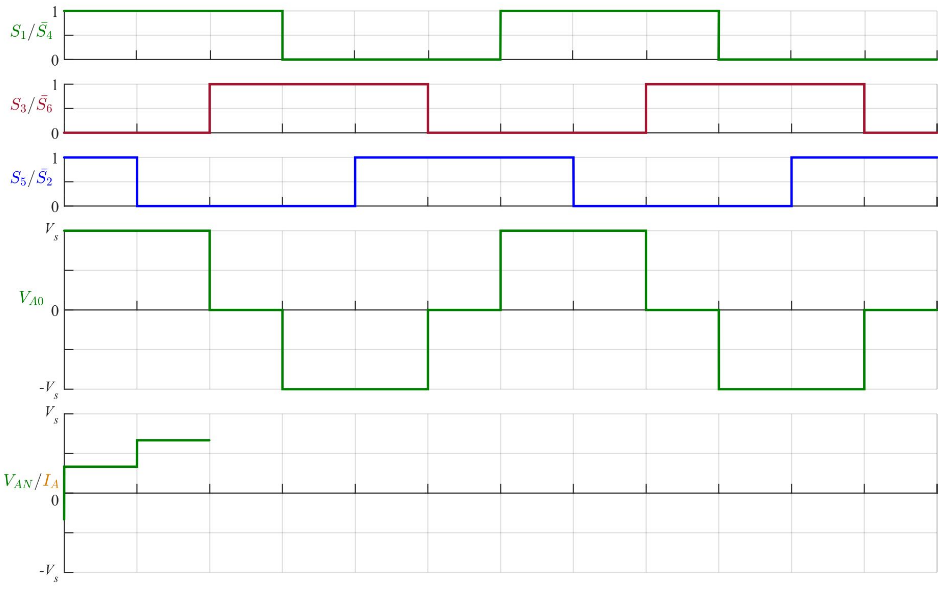


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}/_{2} = \frac{3R}{2}$$





Inversor puente trifásico: formas de onda

Tensión y corriente de fase

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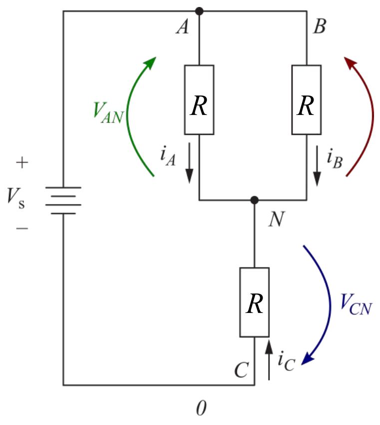
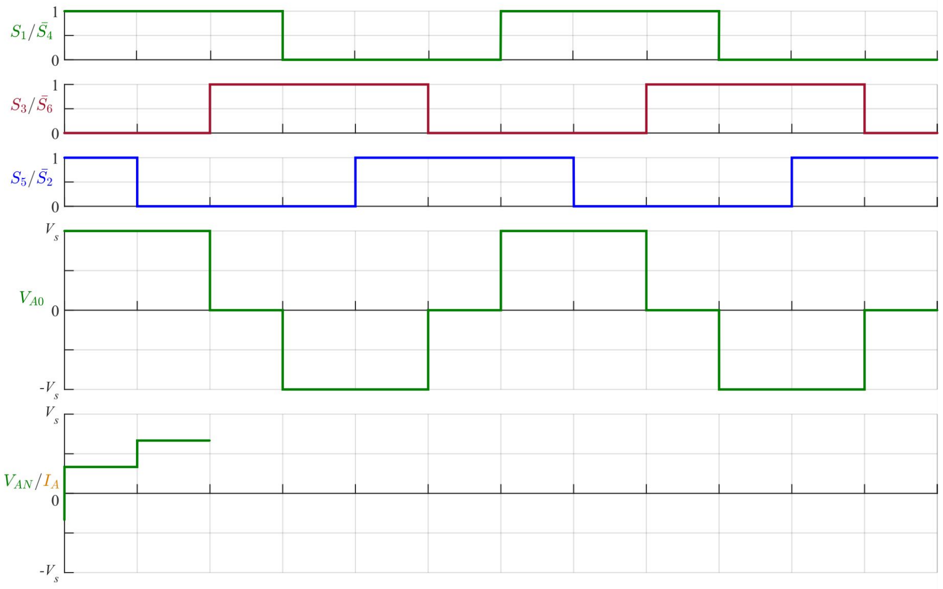


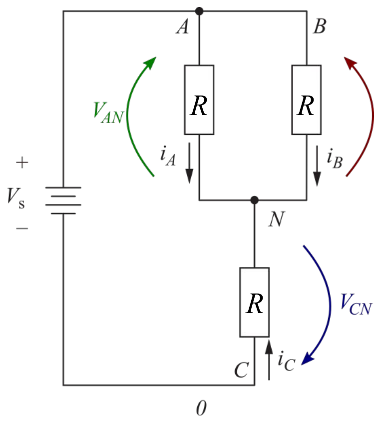
$$i_s = \frac{V_s}{{}^{3R}/_{2}} = \frac{2V_s}{3R}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

$$R_{eq} = R + R//R = R + {}^{R}\!/_{2} = \frac{3R}{2}$$



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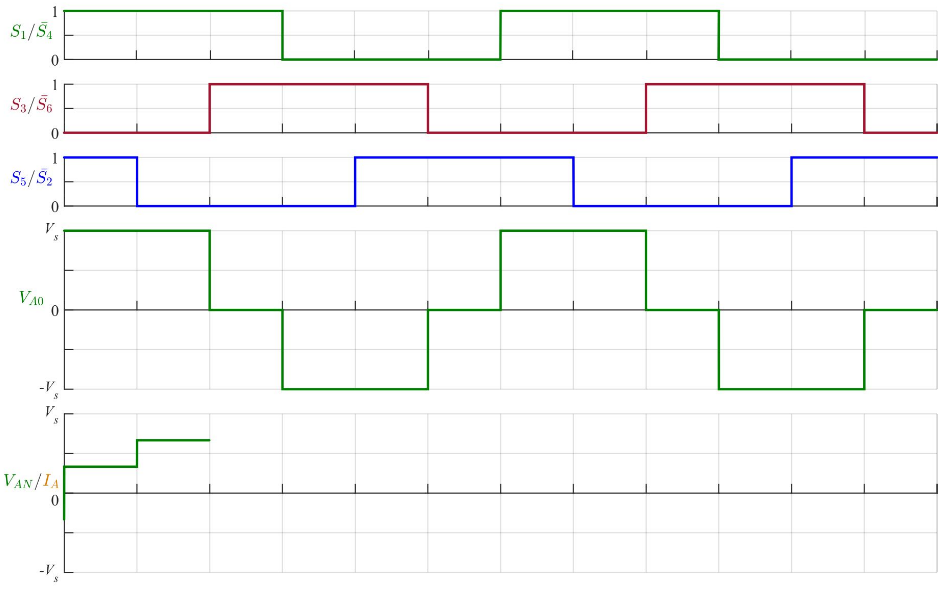

Inversor puente trifásico: formas de onda

Tensión y corriente de fase

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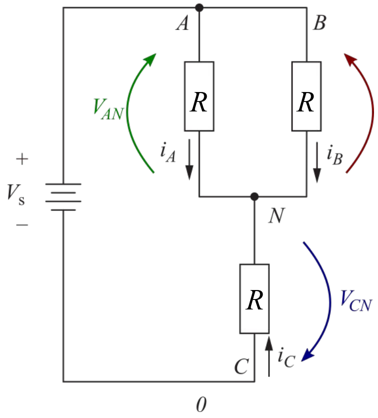
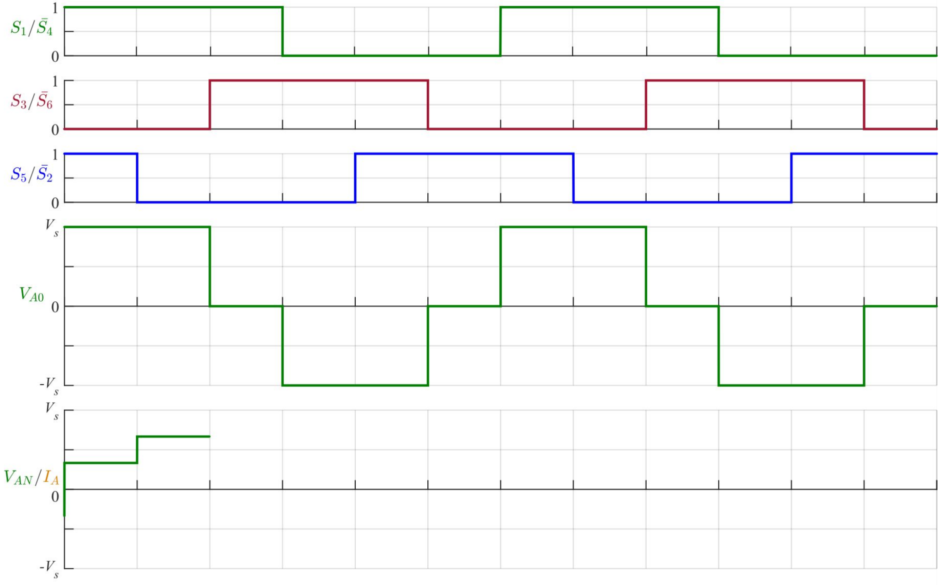


$$i_s = \frac{V_s}{3R/_2} = \frac{2V_s}{3R}$$

$$i_A = {}^{i_s}/_{2} = \frac{V_s}{3R}$$

$$\boxed{V_{AN} = R \cdot i_A = \frac{V_s}{3}}$$



Inversor puente trifásico: formas de onda

Tensión y corriente de fase

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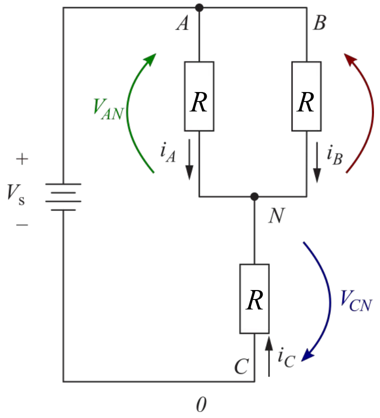

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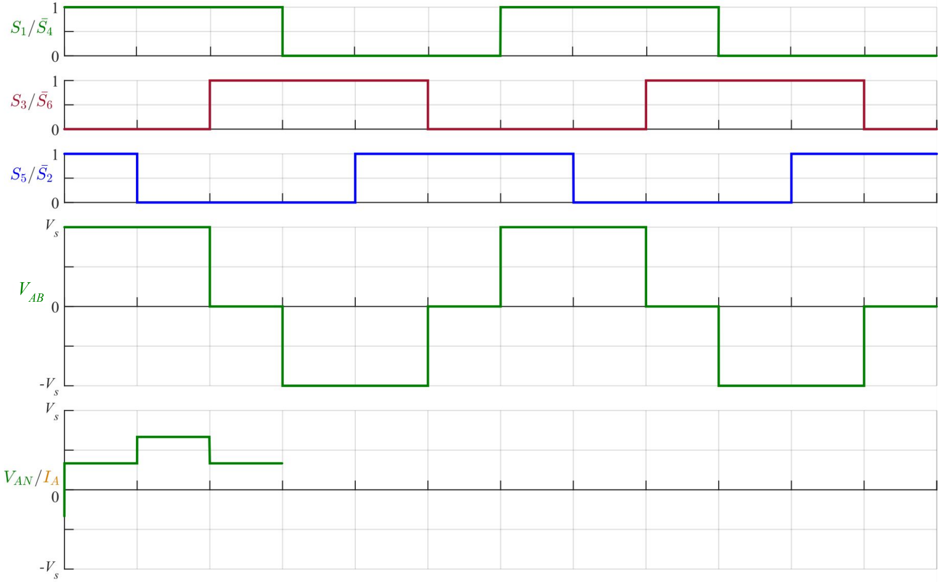
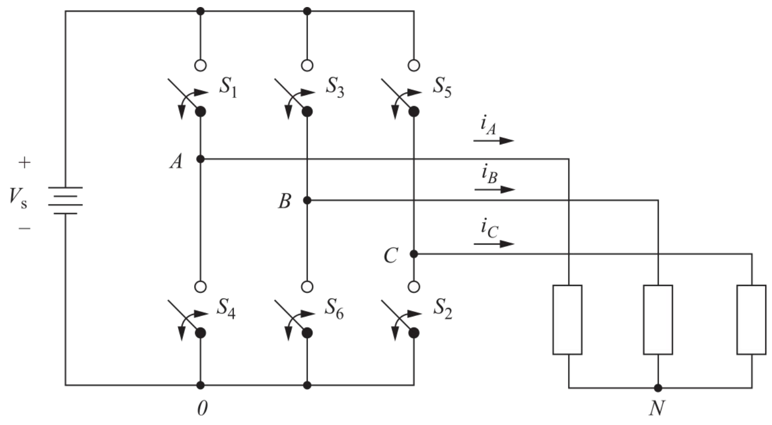
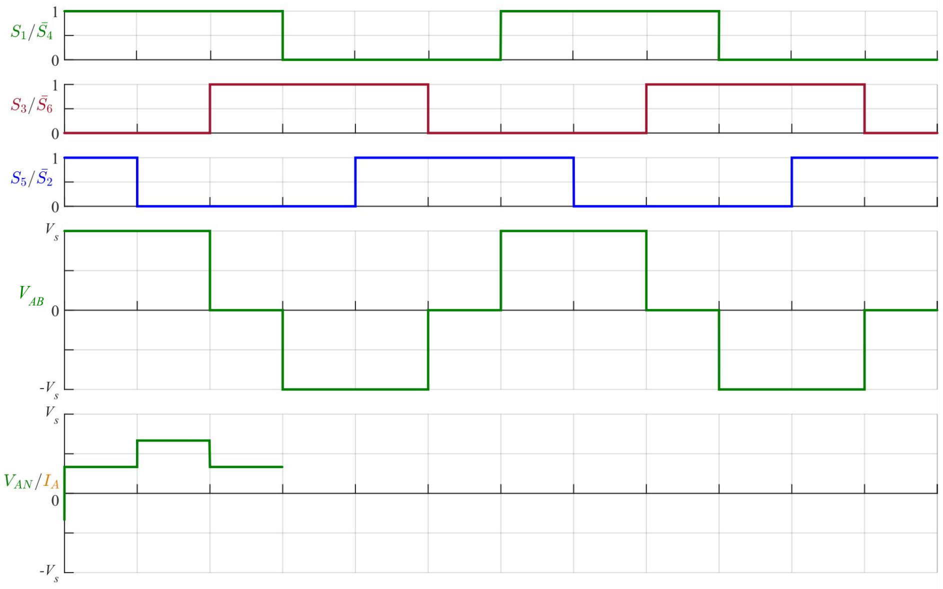


Inversor puente trifásico: formas de onda

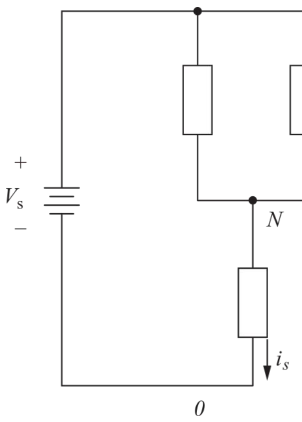
Tensión y corriente de fase

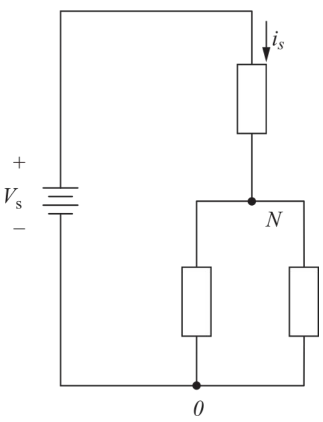


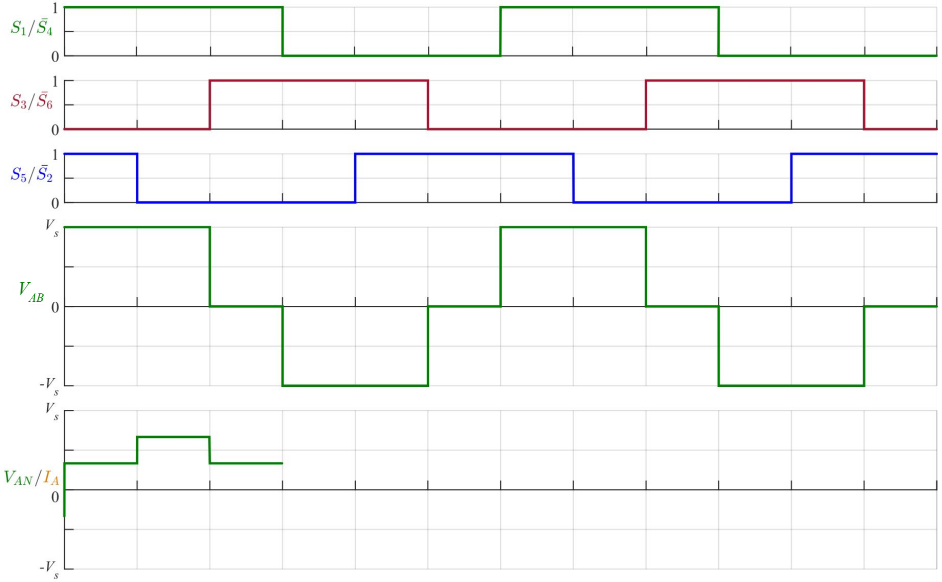


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

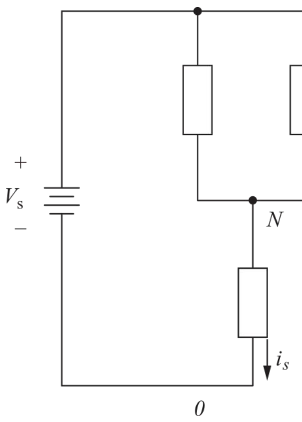


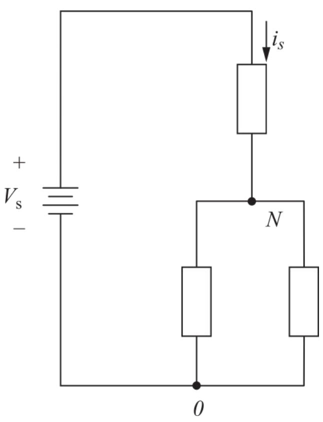


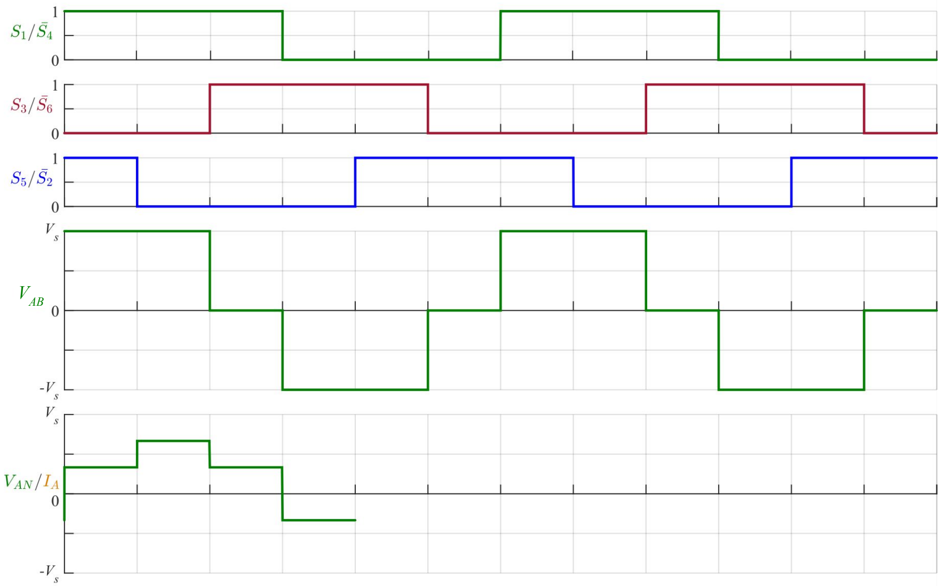


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

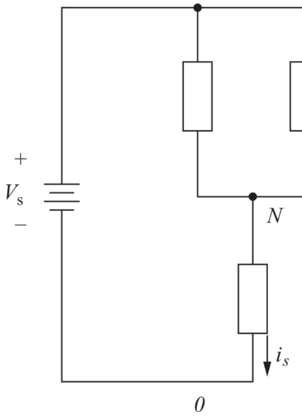


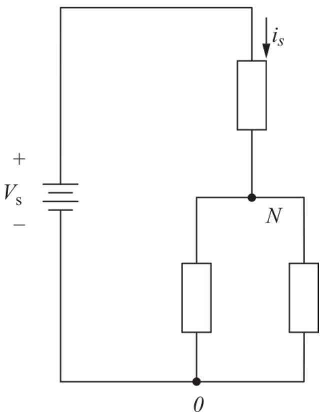


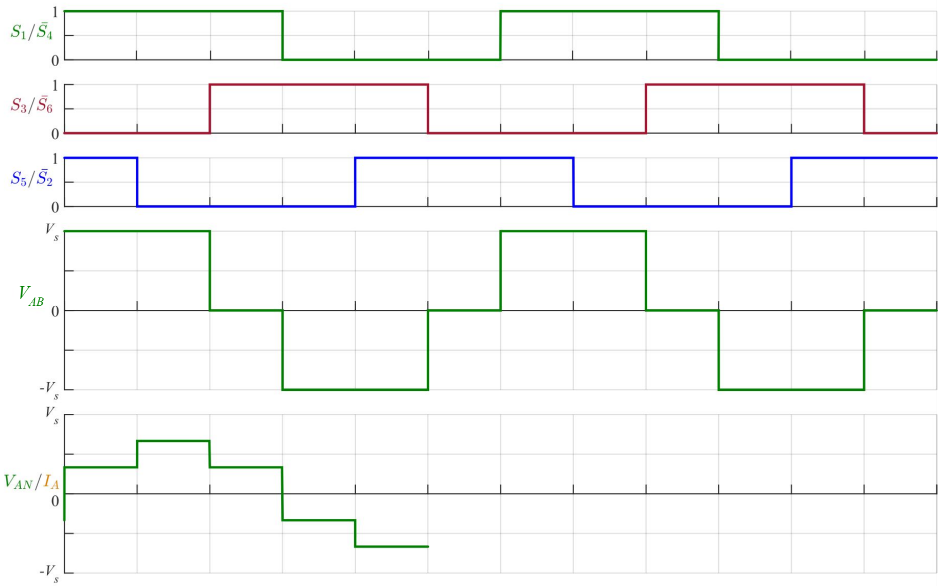


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

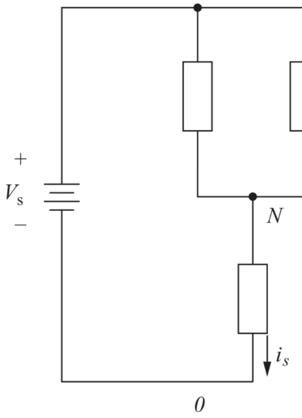


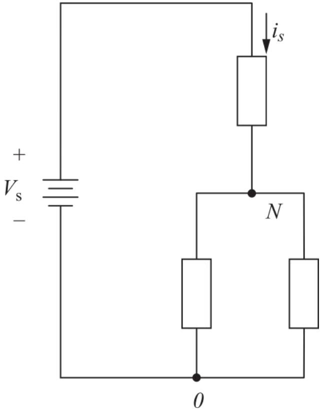


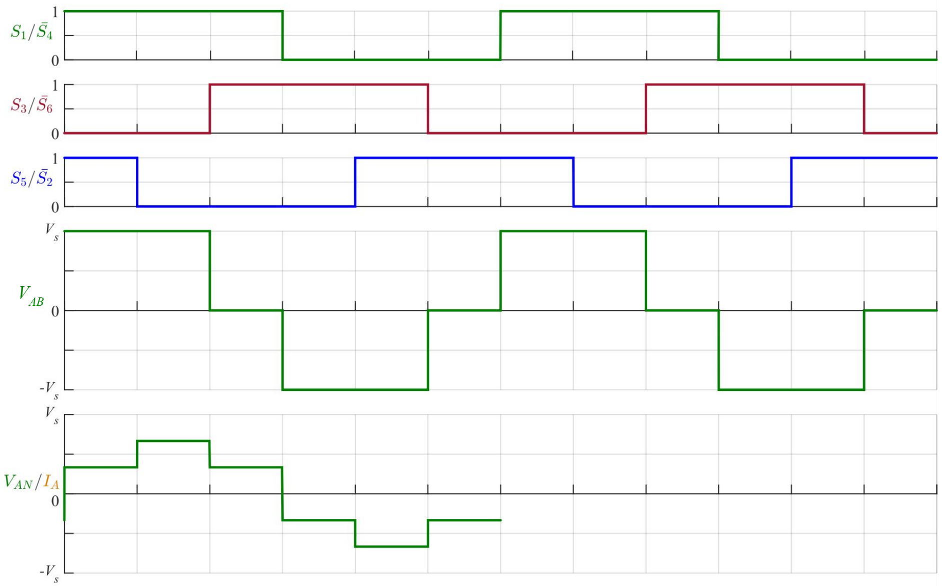


Inversor puente trifásico: formas de onda

Tensión y corriente de fase

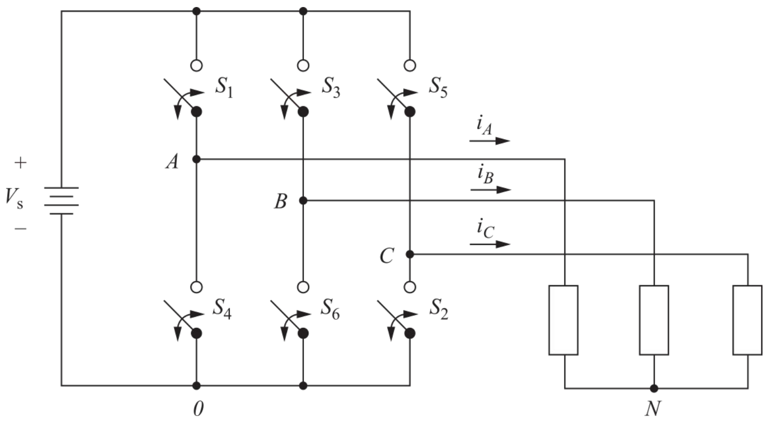


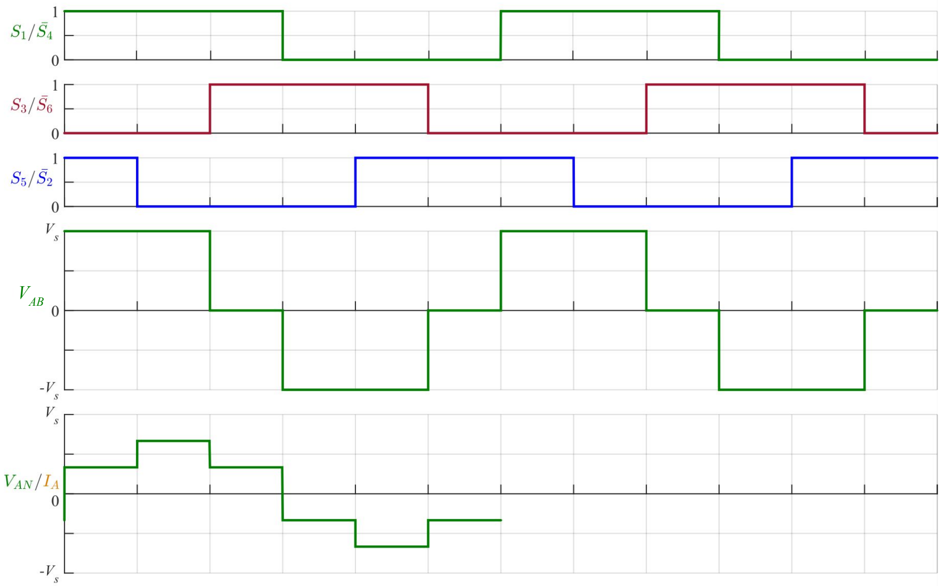




Inversor puente trifásico: formas de onda

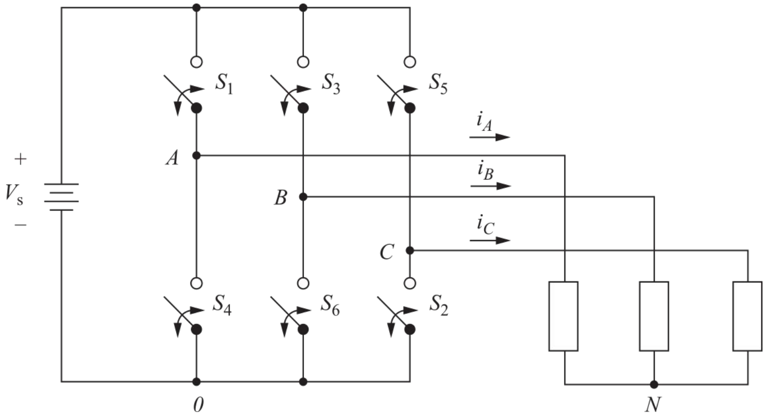
Tensión y corriente de fase

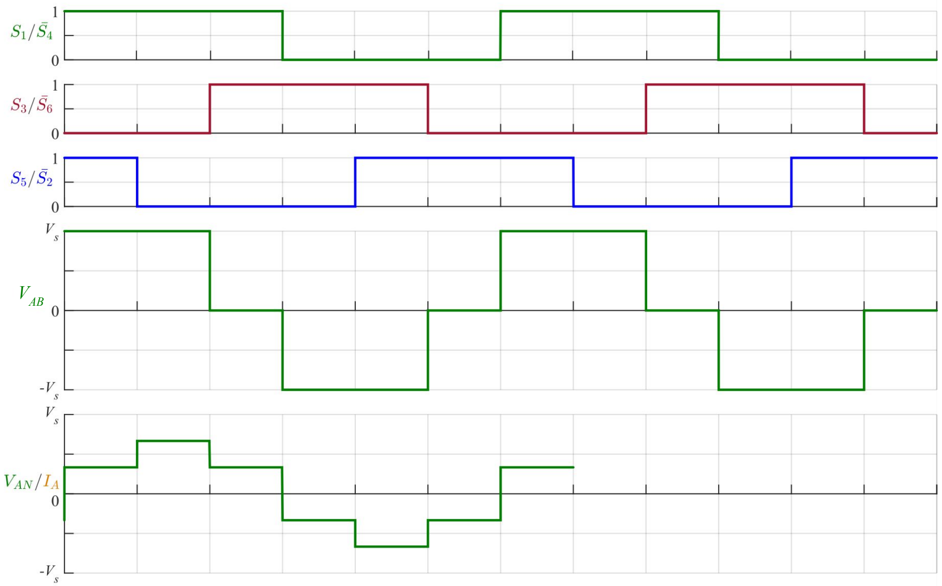




Inversor puente trifásico: formas de onda

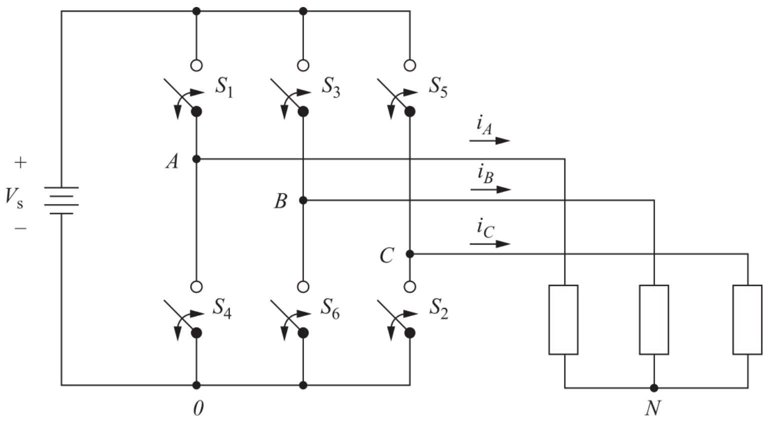
Tensión y corriente de fase

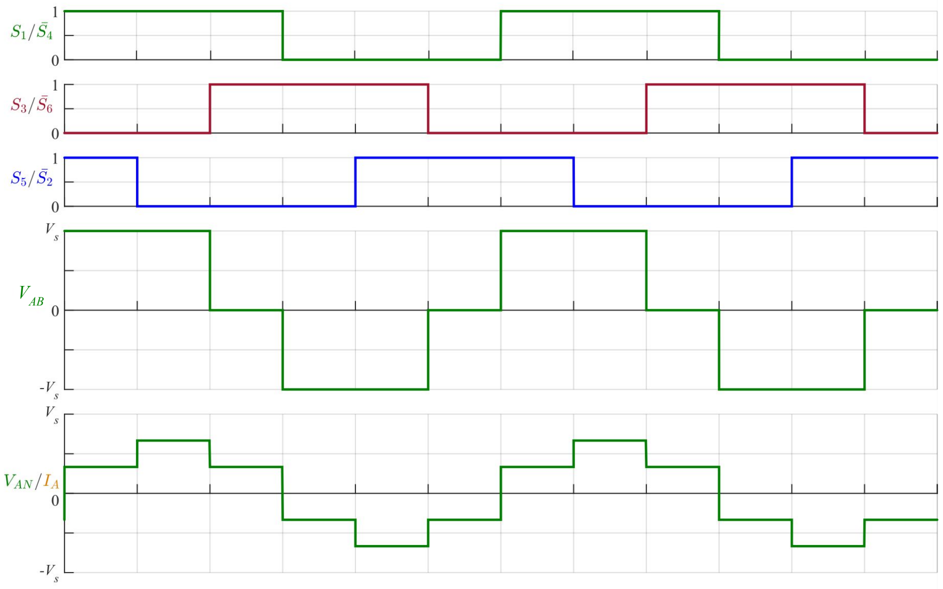




Inversor puente trifásico: formas de onda

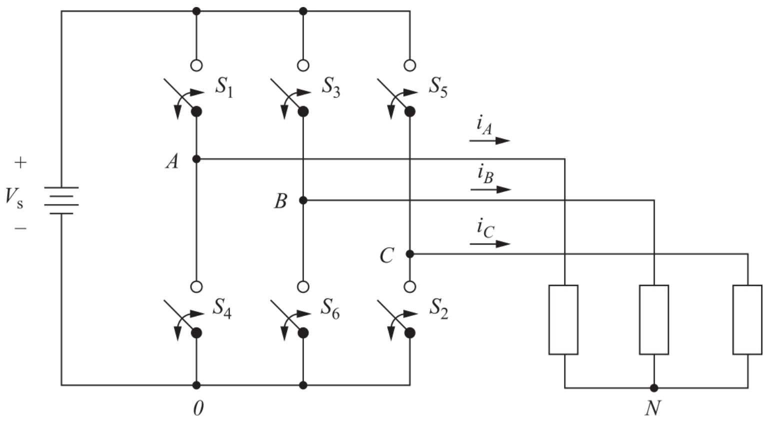
Tensión y corriente de fase

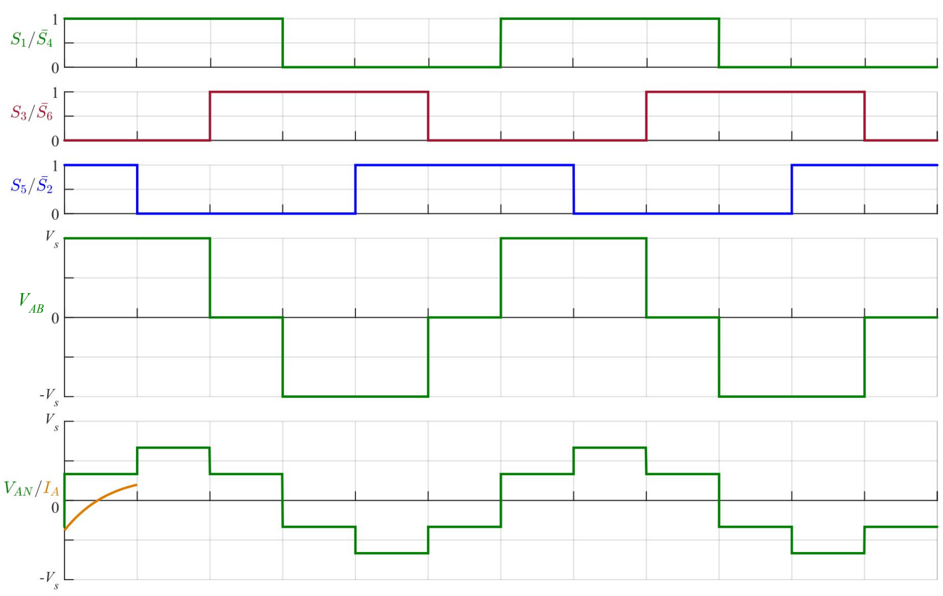




Inversor puente trifásico: formas de onda

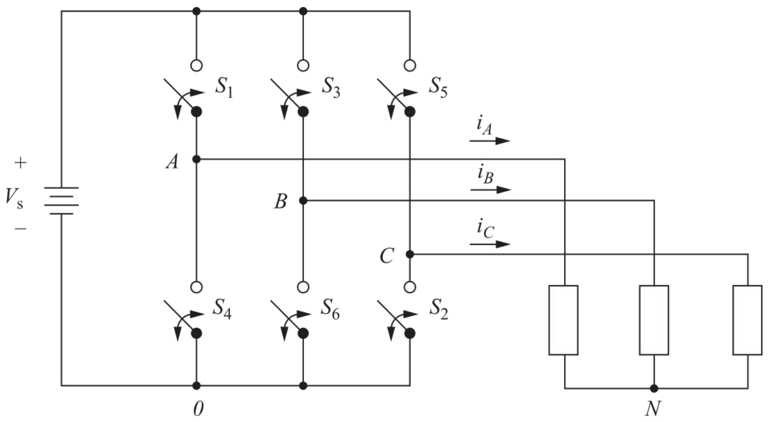
Tensión y corriente de fase

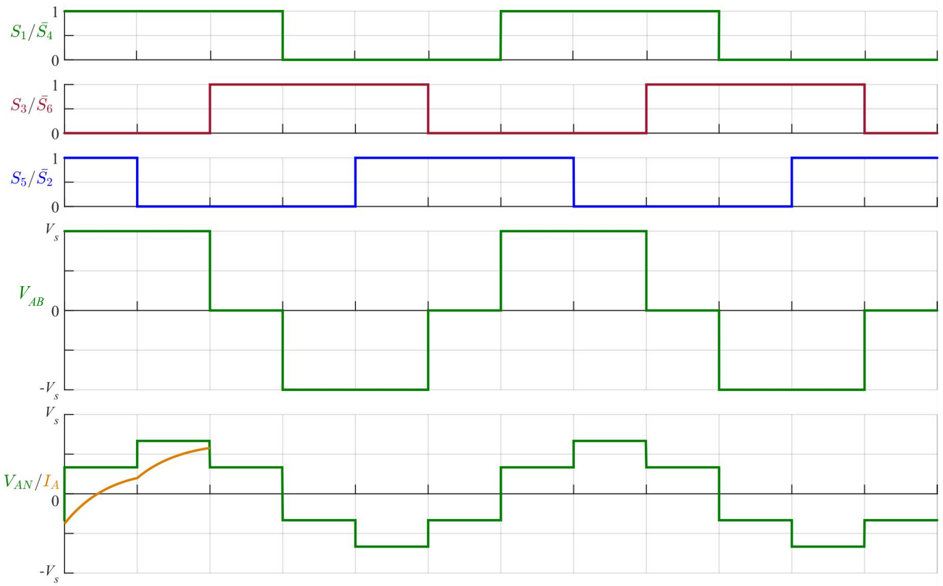




Inversor puente trifásico: formas de onda

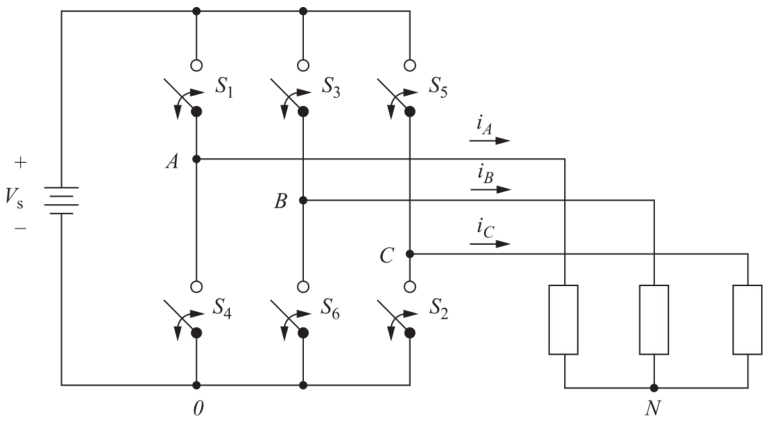
Tensión y corriente de fase

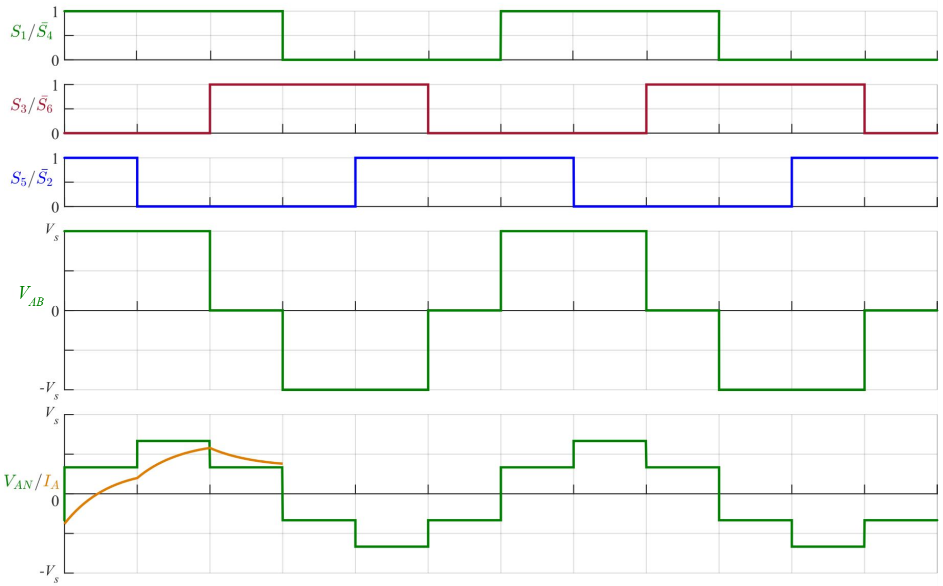




Inversor puente trifásico: formas de onda

Tensión y corriente de fase





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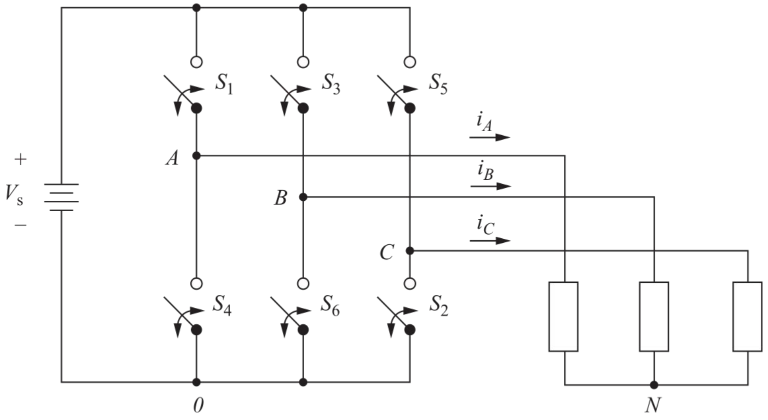
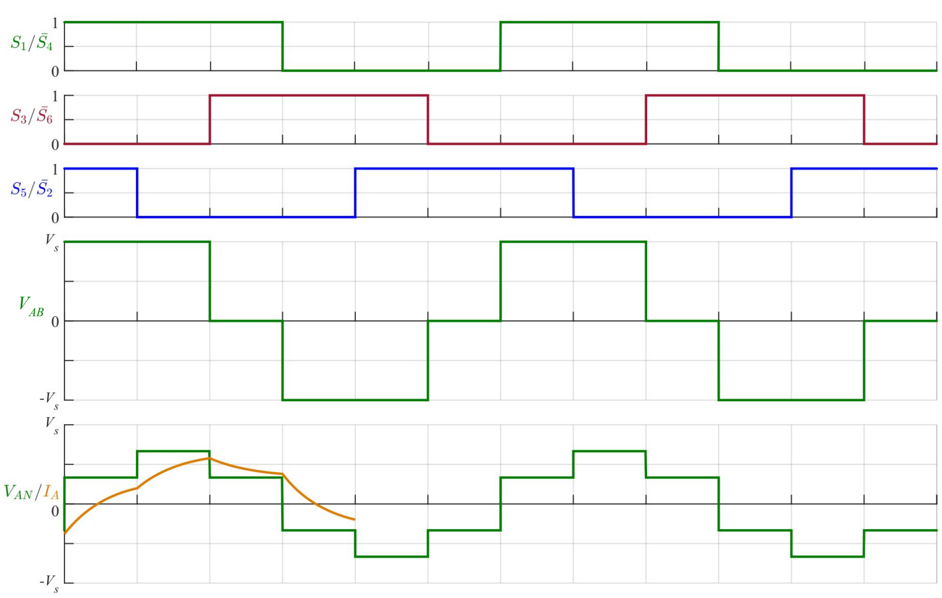
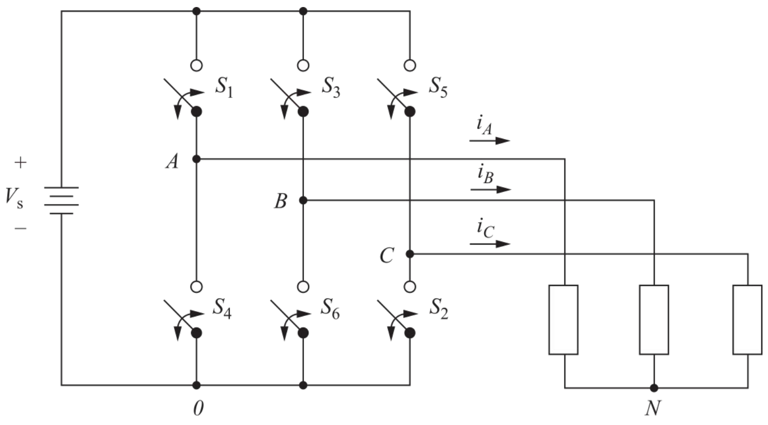
Tensión y corriente de fase

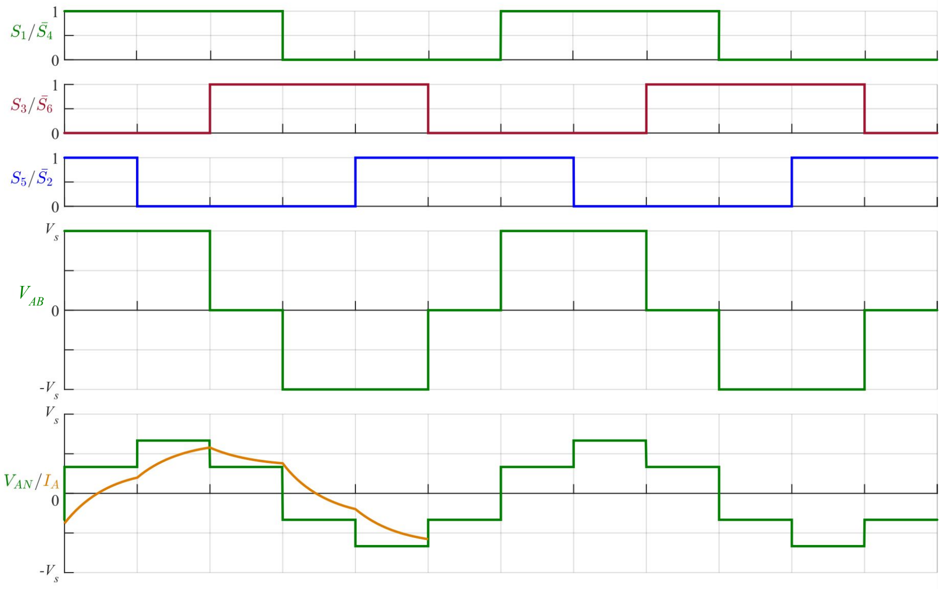




Inversor puente trifásico: formas de onda

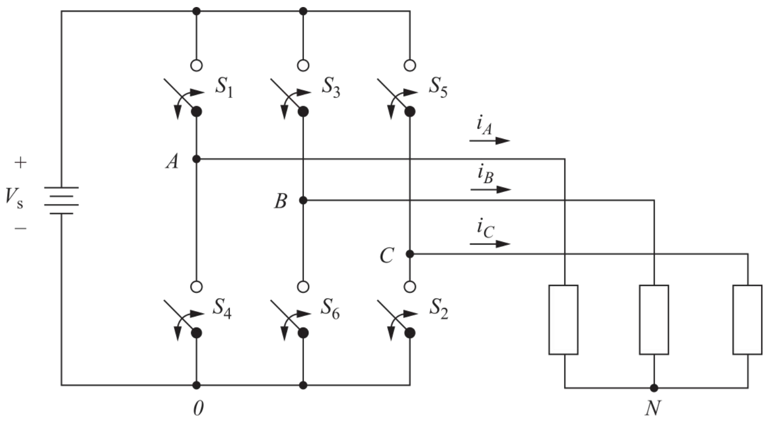
Tensión y corriente de fase

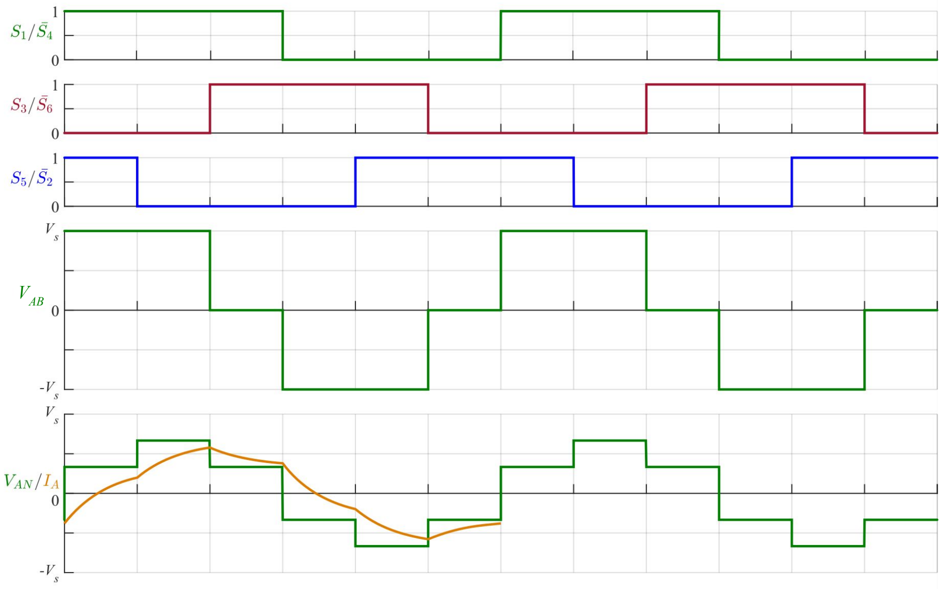




Inversor puente trifásico: formas de onda

Tensión y corriente de fase





Inversor puente trifásico: formas de onda

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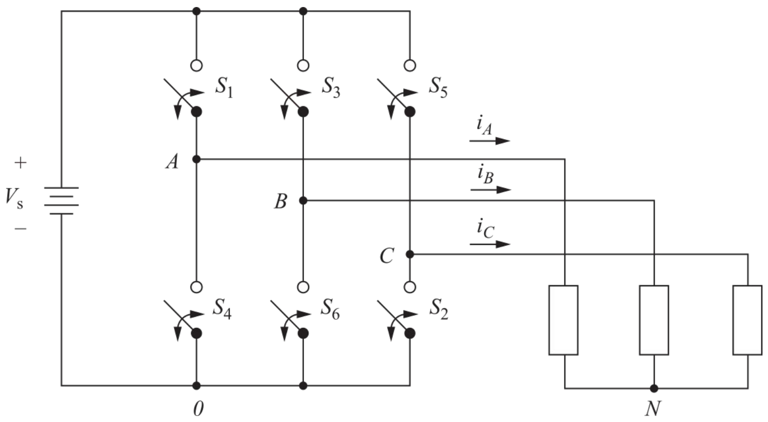

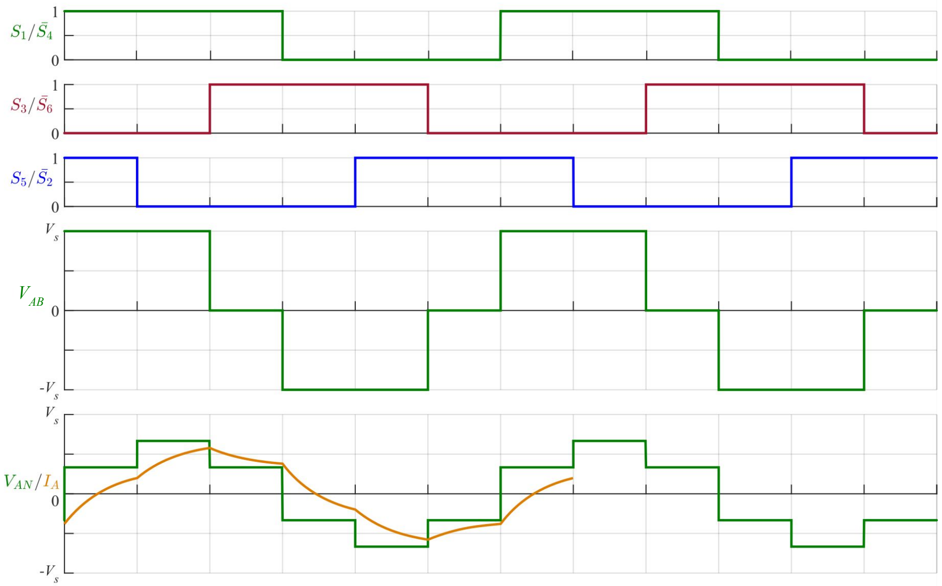

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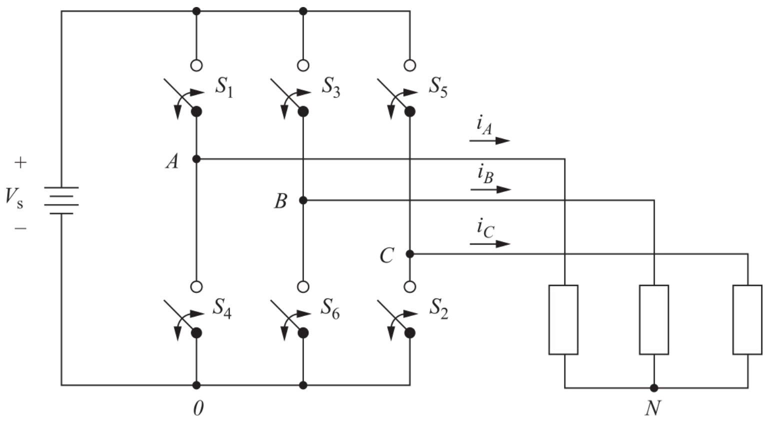

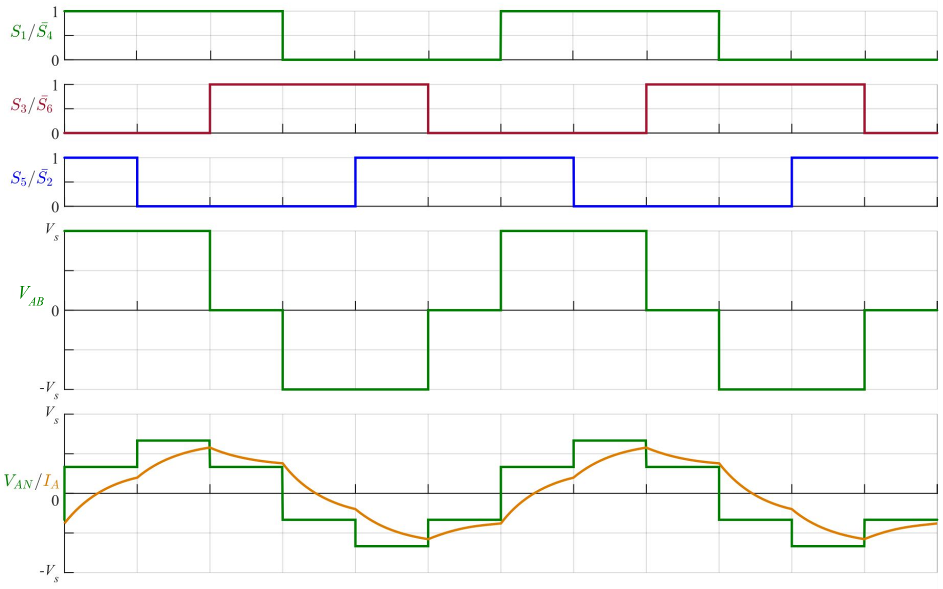

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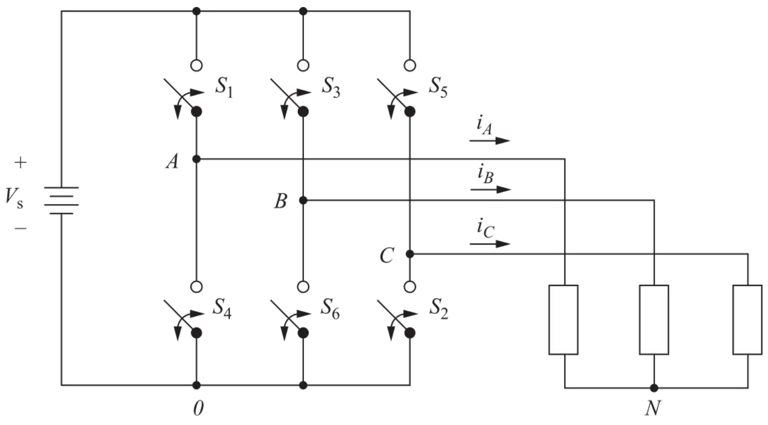
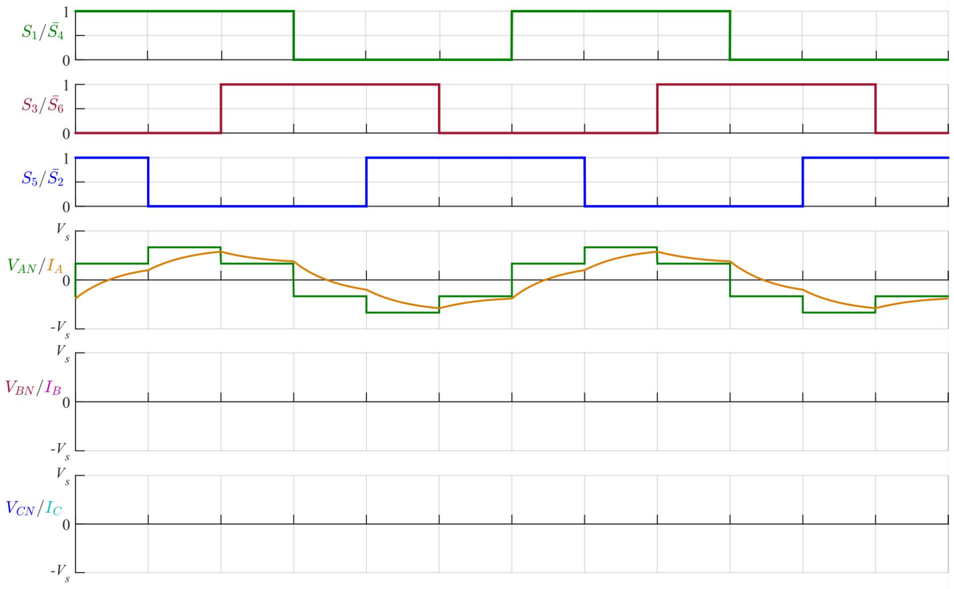
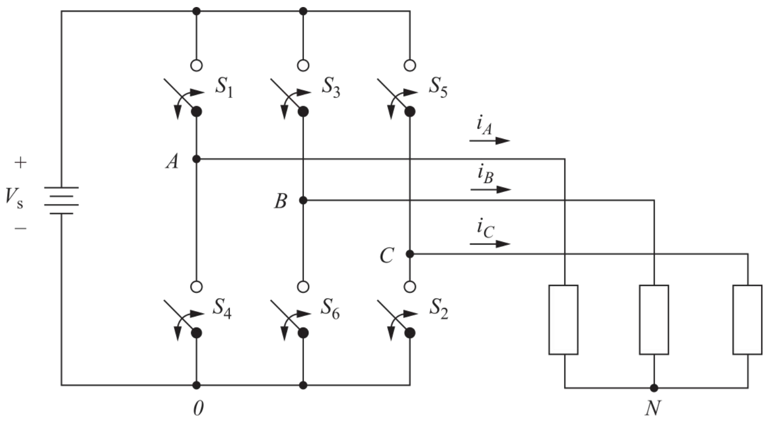
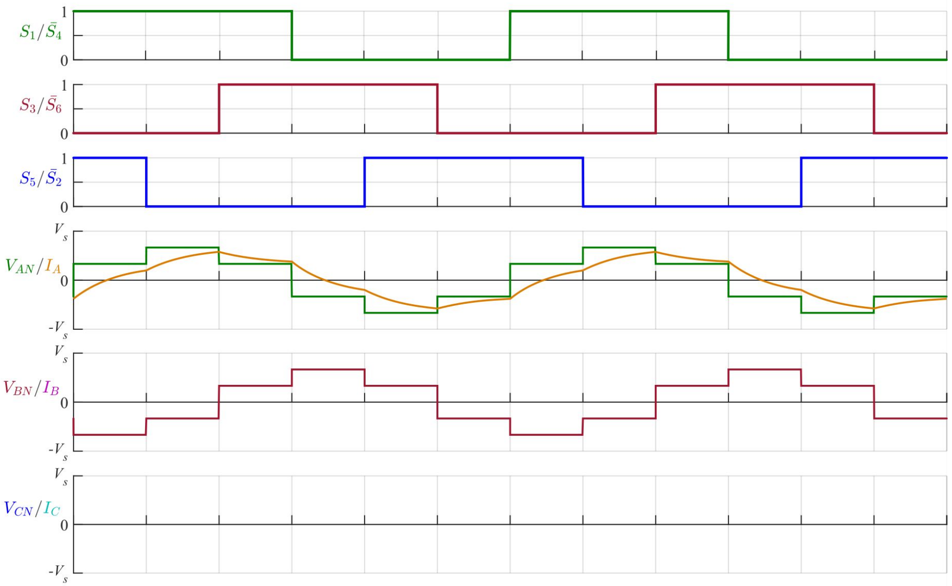




Inversor puente trifásico: formas de onda

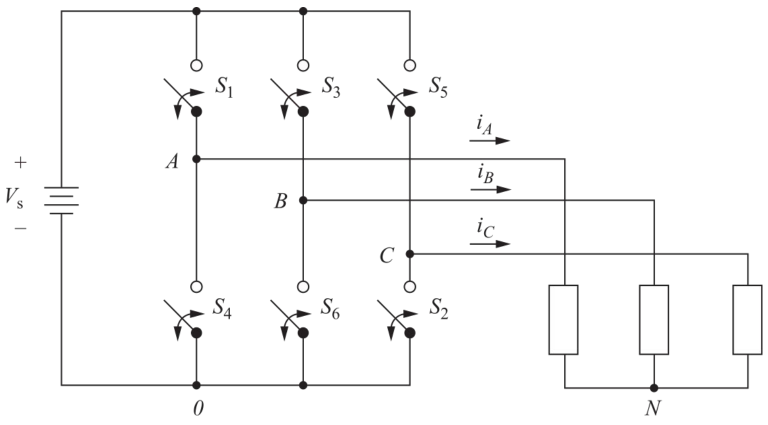
Tensión y corriente de fase

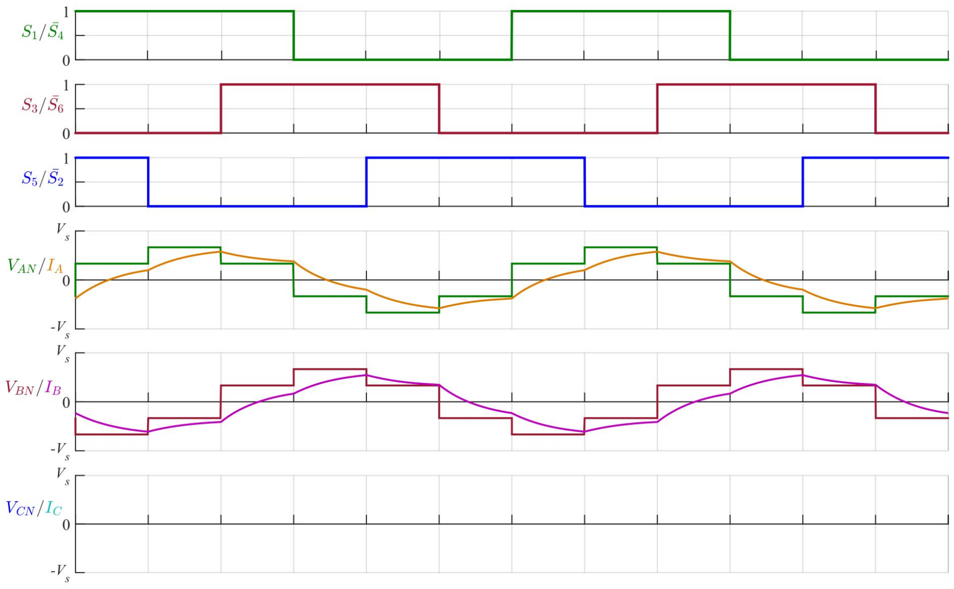




Inversor puente trifásico: formas de onda

Tensión y corriente de fase





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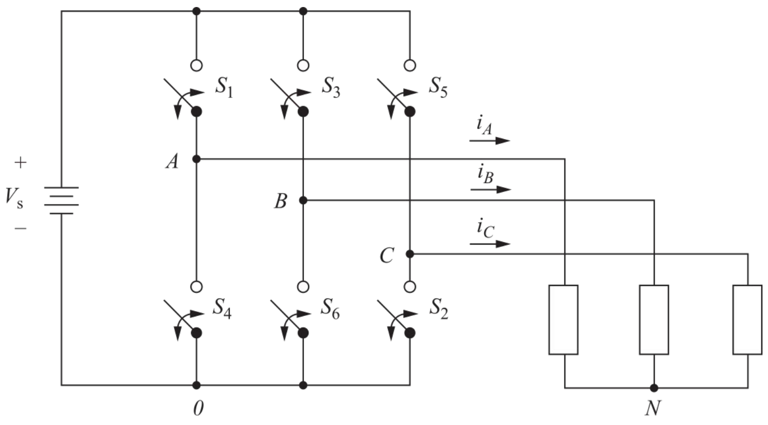
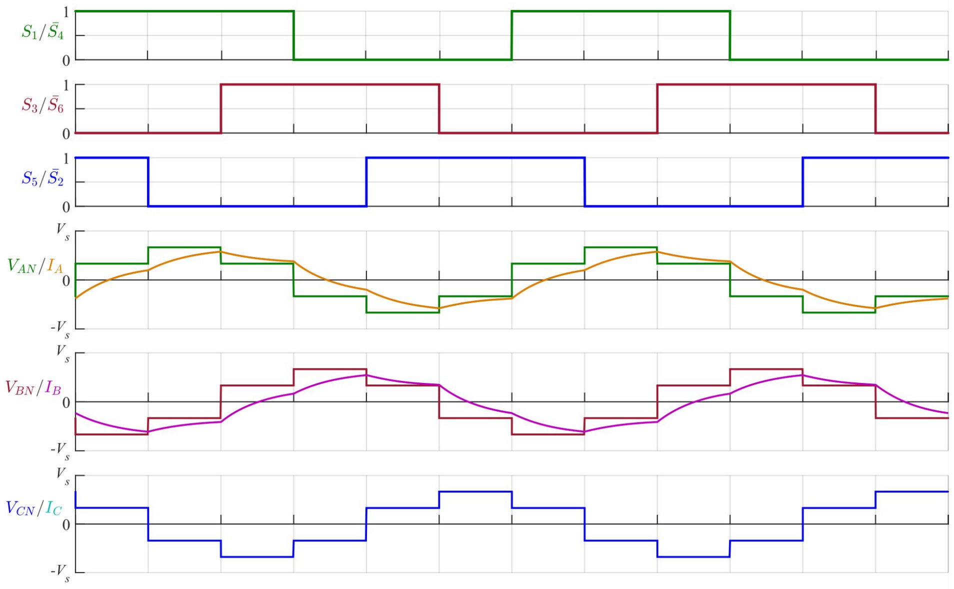
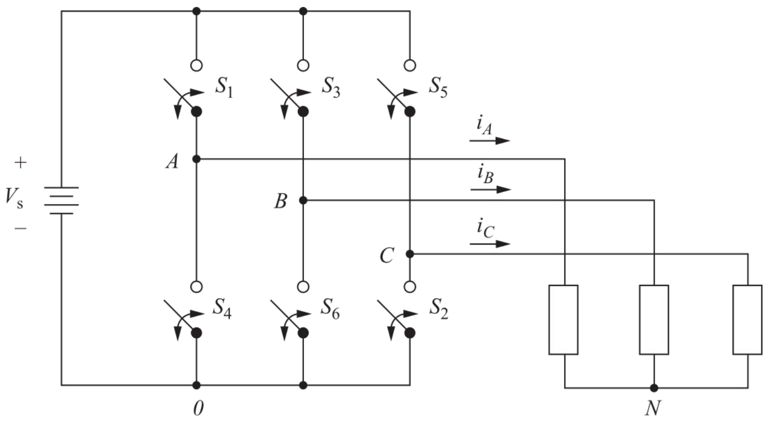
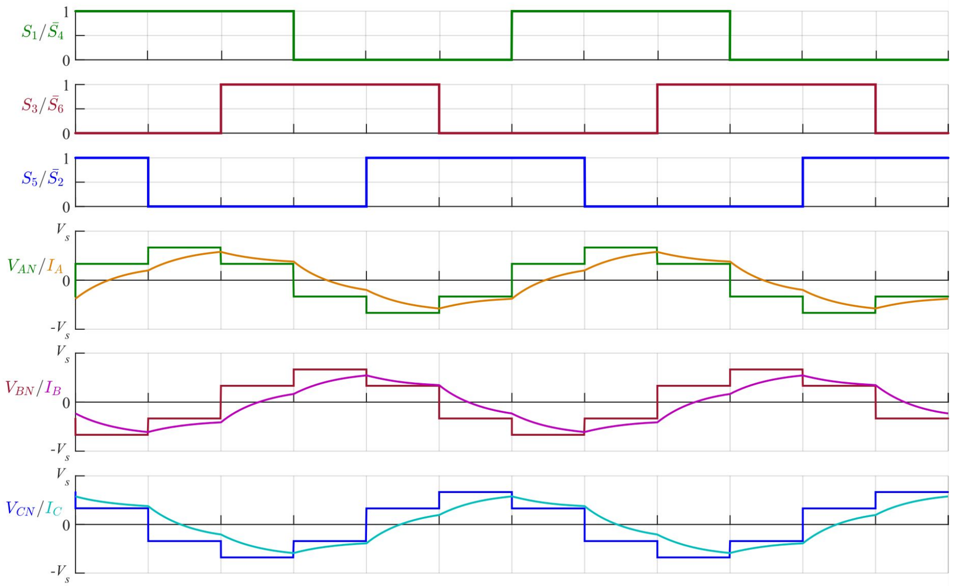




Inversor puente trifásico: formas de onda

Tensión y corriente de fase





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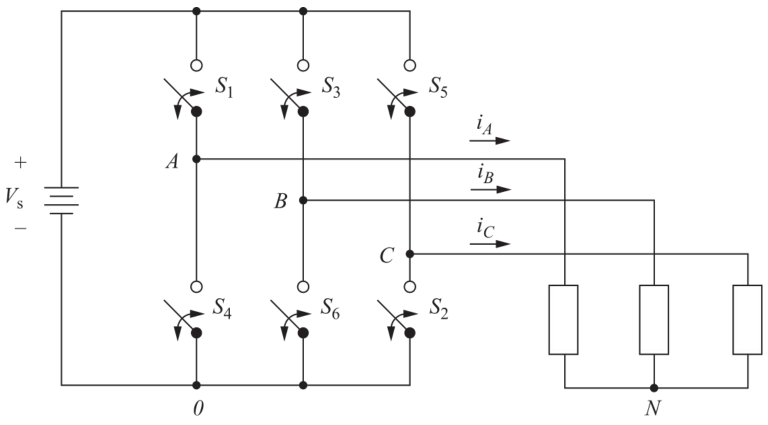
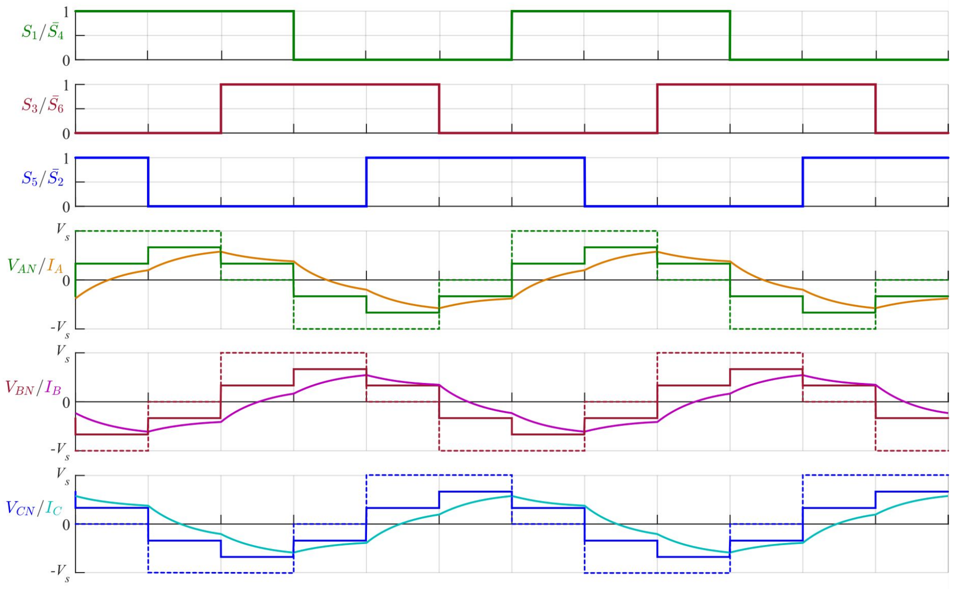
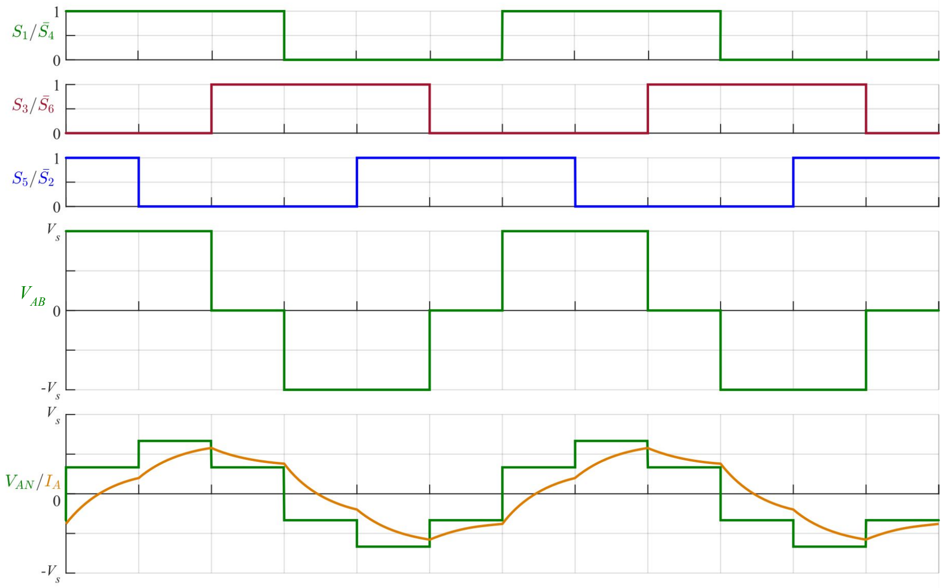




Inversor puente trifásico: análisis espectral

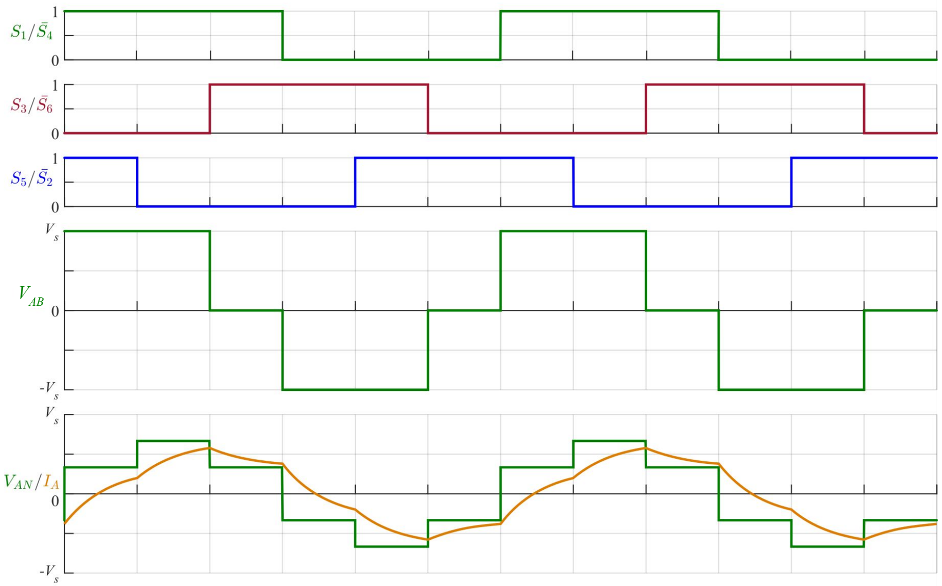
La tensión de línea tiene la forma del caso PWM Monopulso Monofásico, con α = 30°



Inversor puente trifásico: análisis espectral

La tensión de línea tiene la forma del caso PWM Monopulso Monofásico, con α = 30°

$$V_{n,lin} = \frac{4V_s}{n\pi} cos\left(n\frac{\pi}{6}\right)$$



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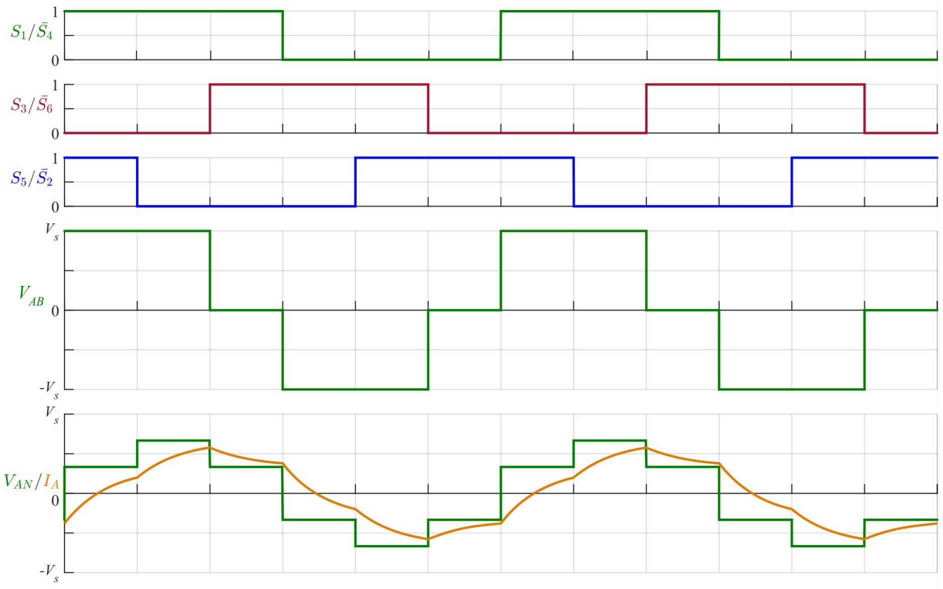
La tensión de fase, algo más compleja, comparte algunas características del espectro:

$$V_{n,fase} = \frac{2V_s}{3n\pi}\left[2 + cos\left(n\frac{\pi}{3}\right) - cos\left(n\frac{2\pi}{3}\right)\right]$$



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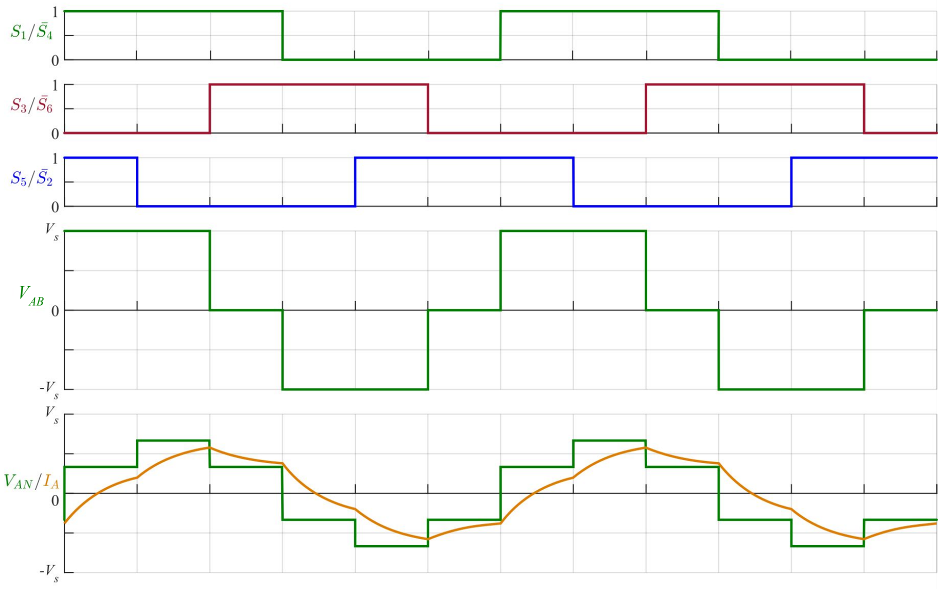
$$n = 1, 5, 7, 11, 13, ...$$



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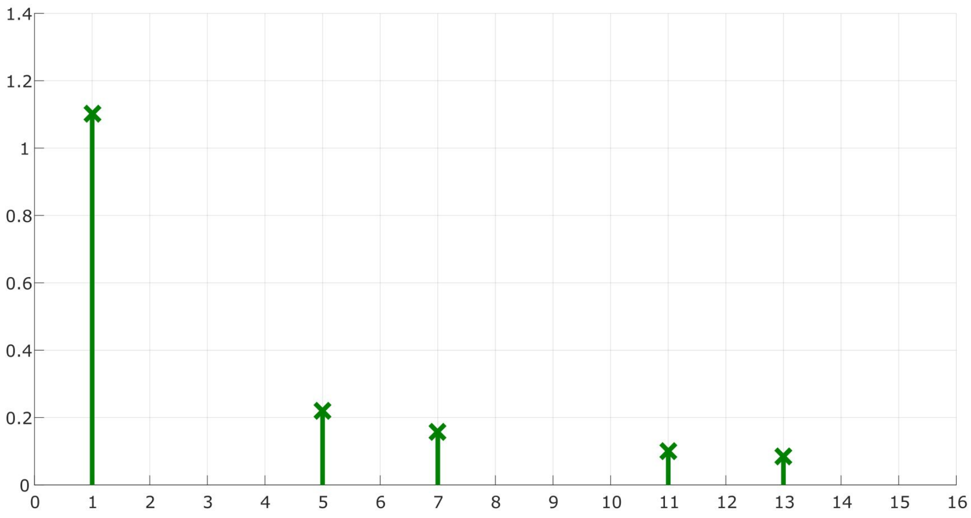

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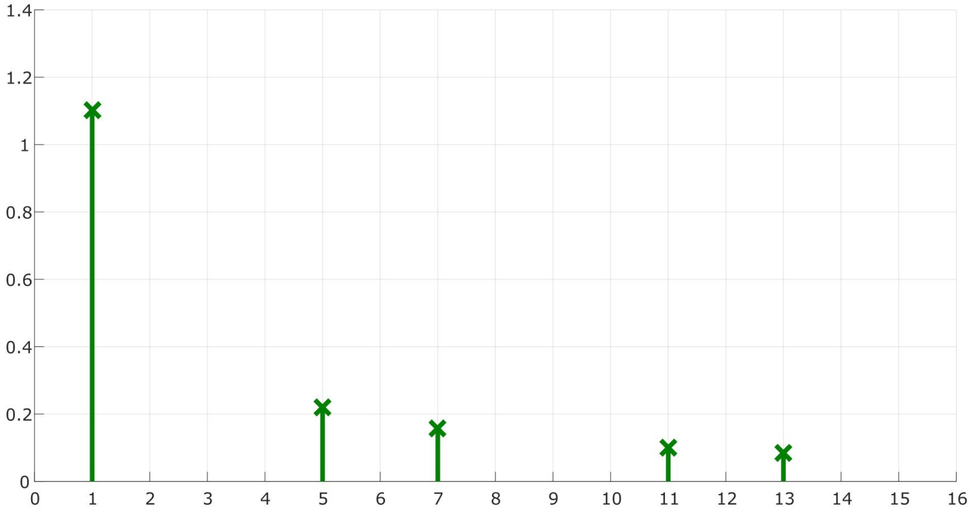

$$THD\% \approx 31\%$$

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